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Fundamentals of Index Coding

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Fundamentals of Index Coding

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ABSTRACT

Index coding is a canonical problem in network information theory that studies the fundamental limit and optimal coding schemes for broadcasting multiple messages to receivers with different side information. The index coding problem provides a simple yet rich model for several important engineering tasks such as satellite communication, content broadcasting, distributed caching, device-to-device relaying, and interference management. This monograph aims to provide a broad overview of this fascinating subject, focusing on the simplest form of multiple-unicast index coding. A unified treatment on coding schemes based on graph-theoretic, algebraic, and information-theoretic approaches is presented. Although the problem of characterizing the optimal communication rate is open in general, several bounds and structural properties are established. The relationship to other problems such as network coding and distributed storage is also discussed.

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1

Introduction

1.1 Motivation

We open our discussion with a simple example. Consider the wireless communication system consisting of one server and three receivers, as depicted in Figure 1.1. The server has three distinct messages x_1, x_2, and x_3. Receiver $i \in \{1, 2, 3\}$ is interested in message x_i and has some of the other messages as side information. In particular, receiver 1 has message x_2 as side information, receiver 2 has x_1 and x_3, and receiver 3 has x_1. The server wishes to communicate all the messages to their designated receivers using the minimum possible number of broadcast transmissions.

The most naive strategy for the server to achieve this goal is to send one message at a time, which takes overall three transmissions. Alternatively, if the server transmits two coded messages $x_1 + x_2$ and x_3 (assuming that the messages can be represented in an alphabet with well-defined addition), then each receiver can recover its desired message using the received coded messages and its side information. Indeed, receiver 1 can recover x_1 from the received message $x_1 + x_2$ and its side information x_2. Similarly, receiver 2 can recover x_2 from $x_1 + x_2$ and x_1. Receiver 3 can trivially recover x_3. This simple example

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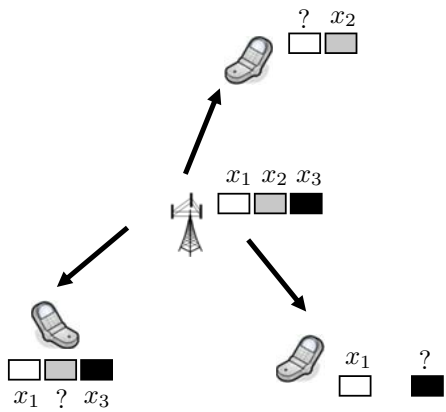


Figure 1.1: An index coding example with three receivers.

illustrates that sending *coded messages* may decrease the number of broadcast transmissions.

Generalizing our initial example, we study the communication problem depicted in Figure 1.2, which is commonly referred to as *index coding*. In this canonical problem in network information theory, a server has a tuple of n messages

$$x^n := (x_1, \ldots, x_n)$$

and broadcasts a coded message y generated from x^n to n receivers. Each receiver $i \in [n] := \{1, 2, \ldots, n\}$ wishes to recover the message x_i from y and a set of other messages

$$x(A_i) := (x_j, j \in A_i), \quad A_i \subseteq [n] \setminus \{i\}$$

it already has as side information. Assuming that the side information sets $A_1, \ldots, A_n$ are known prior to the communication, we are interested in devising a coding scheme that exploits the side information at the receivers and broadcasts the messages reliably with the minimum amount of transmissions.

The index coding problem was introduced by Birk and Kol [26, 27] in the context of satellite communication. Related formulations were studied earlier by Celebiler and Stette [40], Wyner, Wolf, and

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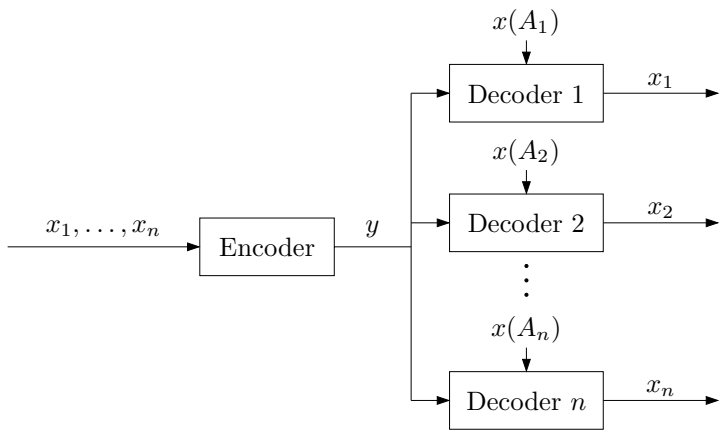


Figure 1.2: The index coding problem.

Willems [157, 161], and Yeung [163]. The term "index coding" is due to Bar-Yossef, Birk, Jayram, and Kol [22], who compared the index coding problem to the problem of zero-error source coding with side information studied by Witsenhausen [158] and contrasted the fact that in the index coding problem the receiver wishes to recover a single component of the source, the *index* of which is unknown to the sender. Hence, as for compound channels [28, 54, 159], the sender can proactively encode its transmission and *broadcast* to the receiver in all possible configurations [49]. In addition to satellite communication, index coding has applications in diverse areas such as multimedia distribution [117], interference management [83], coded caching [106, 84], and distributed computing [96]. This problem is also closely related to many other important problems such as network coding [125, 62, 60], locally recoverable distributed storage [111, 132, 13], guessing games on directed graphs [125, 167, 13], matroid theory [62], and zero-error capacity of channels [135]. Due to this significance, the index coding problem has been broadly studied over the past two decades. Tools from various disciplines including graph theory, coding theory, and information theory have been utilized to propose numerous nontrivial coding schemes [26, 104, 23, 44, 117, 30, 8, 107, 83, 134, 7, 9, 153, 119, 82, 145, 150, 167], as well as performance bounds [165, 23, 56, 29, 18, 145].

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1.2 Formal Definition of the Problem

We formulate the problem more precisely. A $(t_1, \ldots, t_n, r)$ *index code* is defined by

- n messages, where the i-th message $x_i = (x_{i1}, \ldots, x_{it_i})$ takes values from $\mathcal{X}^{t_i}$ for some common finite alphabet $\mathcal{X}$,
- an encoder $\phi : \prod_{i=1}^{n} \mathcal{X}^{t_i} \to \mathcal{X}^r$ that maps the message n-tuple x^n to an index $y = (y_1, \ldots, y_r) \in \mathcal{X}^r$, and
- n decoders, where the decoder at receiver $i \in [n]$, $\psi_i : \mathcal{X}^r \times \prod_{j \in A_i} \mathcal{X}^{t_j} \to \mathcal{X}^{t_i}$, maps the received index $y = \phi(x^n)$ and the side information $x(A_i)$ back to x_i.

Thus, for every $x^n \in \prod_{i=1}^{n} \mathcal{X}^{t_i}$,

$$\psi_i(\phi(x^n), x(A_i)) = x_i, \quad i \in [n].$$

A $(t, \ldots, t, r)$ code will be written simply as a (t, r) code.

Suppose that $\mathcal{X}$ is an alphabet over which linear operations are well-defined, for example, a finite field $\mathbb{F}_q$ or a ring (see, for example, [48]). If the encoder of a code is a linear function of $x^n = (x_1^{t_1}, \ldots, x_n^{t_n}) = (x_{11}, \ldots, x_{1t_1}, \ldots, x_{n1}, \ldots, x_{nt_n})$ and the decoders are also linear functions of x^n (and $y = (y_1, \ldots, y_r)$), then the code is referred to as a *linear* index code. If $t_i = 1$ for all $i \in [n]$, then the linear index code is said to be a *scalar* linear index code. Otherwise, the code is a *vector* linear index code.

As an example, for the 3-message index coding problem in Figure 1.1, a $(1, 1, 1, 2)$ index code with $\mathcal{X} = \mathbb{F}_2$, the encoder defined by

$$y_1 = x_1 + x_2 \quad \text{and} \quad y_2 = x_3,$$

and the decoders defined by

$$x_1 = y_1 + x_2, \quad x_2 = y_1 + x_3, \quad \text{and} \quad x_3 = y_2,$$

where in both cases the addition operations are in $\mathbb{F}_2$, is a scalar linear code.

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A tuple $(R_1, \ldots, R_n) \in \mathbb{R}^n_{\geq 0}$ of nonnegative real numbers is said to be an *achievable* rate tuple for the index coding problem if there exists a $(t_1, \ldots, t_n, r)$ index code such that

$$R_i \leq \frac{t_i}{r}, \quad i \in [n].$$

The *capacity region* $\mathscr{C}$ of the index coding problem is defined as the closure of the set of all achievable rate tuples. The ultimate goal of studying the index coding problem is to characterize the capacity region of a general index coding problem and develop a simple coding scheme that achieves or approximates the capacity region.

Remark 1.1. Our definition of capacity region, with the stringent requirement of perfect, zero-error recovery of the messages, should be distinguished from the more common definition of *vanishing-error capacity region* in network information theory that allows for arbitrarily small probability of error. It can be shown, however, that these two capacity regions coincide; see Appendix 1.A for details.

The definition of the capacity region depends on the alphabet $\mathcal{X}$ on which the messages are defined and one may well denote it by $\mathscr{C}_{\mathcal{X}}$ to emphasize this dependence. As we will prove in Appendix 1.B, however, the choice of $\mathcal{X}$ is irrelevant to the actual capacity region itself.

Lemma 1.1. For any two finite alphabets $\mathcal{X}$ and $\mathcal{X}'$,

$$\mathscr{C}_{\mathcal{X}} = \mathscr{C}_{\mathcal{X}'}.$$

Consequently, we assume without loss of generality that $\mathcal{X} = \mathbb{F}_2$ for a general index code and consequently that the base of logarithm is 2 throughout, unless specified otherwise.

By limiting our attention to *linear* codes, we can similarly define *linearly achievable* rate tuple and *linear capacity region* $\mathscr{L}$. In contrast to the capacity region, the linear capacity region of the index coding problem may depend on the chosen alphabet $\mathcal{X}$ and indeed does so [104] (see Section 6.7).

The capacity region, linear or nonlinear, is closed by definition. Based on the standard time-sharing argument (see, for example, [61, Sec. 4.4]), it can be readily checked to be *convex*.

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In many cases, it is convenient to focus on a single performance metric instead of a multidimensional region. Let $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_n) \in \mathbb{R}^n_{\geq 0}$ be a tuple of nonnegative real numbers. The $\boldsymbol{\mu}$-*directed capacity* of the index coding problem is defined as

$$C(\boldsymbol{\mu}) = \max\{R\colon R\boldsymbol{\mu} \in \mathscr{C}\}.$$

Remark 1.2. The capacity region can be written in terms of $\boldsymbol{\mu}$-directed capacities as

$$\mathscr{C} = \bigcup_{\boldsymbol{\mu}} \{(R_1, \ldots, R_n)\colon R_i \leq C(\boldsymbol{\mu})\mu_i,\ i \in [n]\}. \tag{1.1}$$

Note that if $\boldsymbol{\mu} = c\boldsymbol{\mu}'$ for some constant c, then $C(\boldsymbol{\mu})\boldsymbol{\mu} = C(\boldsymbol{\mu}')\boldsymbol{\mu}'$ and thus in (1.1), it suffices to take the union only over normalized vectors, e.g., over $\boldsymbol{\mu}$ such that $\sum_{i=1}^n \mu_i = n$.

The $\mathbf{1}$-directed capacity of the index coding problem is referred to as the *symmetric capacity* (or the *capacity* in short), that is,

$$C_{\text{sym}} = C(\mathbf{1}) = \max\{R\colon (R, \ldots, R) \in \mathscr{C}\}.$$

The symmetric capacity can be equivalently defined as

$$C_{\text{sym}} = \sup_r \sup_{(t,r)\ \text{codes}} \frac{t}{r} = \lim_{r\to\infty} \sup_{(t,r)\ \text{codes}} \frac{t}{r},$$

where the equality between the supremum and the limit follows by Fekete's lemma [69] and the superadditivity

$$\sup_{(t,r_1+r_2)\ \text{codes}} t \geq \sup_{(t_1,r_1)\ \text{codes}} t_1 + \sup_{(t_2,r_2)\ \text{codes}} t_2.$$

The reciprocal of the symmetric capacity,

$$\beta = \frac{1}{C_{\text{sym}}},$$

is referred to as the *broadcast rate*, which can be alternatively defined as

$$\beta = \inf_t \inf_{(t,r)\ \text{codes}} \frac{r}{t} = \lim_{t\to\infty} \inf_{(t,r)\ \text{codes}} \frac{r}{t}. \tag{1.2}$$

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The *linear broadcast rate* is similarly defined as

$$\lambda = \inf_{t} \inf_{(t,r) \text{ linear codes}} \frac{r}{t} = \lim_{t\to\infty} \inf_{(t,r) \text{ linear codes}} \frac{r}{t}.$$

As with the linear capacity region $\mathscr{L}$, the linear broadcast rate depends on the underlying alphabet $\mathcal{X}$ and will be sometimes denoted $\lambda_{\mathcal{X}}$ to emphasize this dependence. For any $\mathcal{X}$, $\beta \leq \lambda_{\mathcal{X}}$.

Note that the capacity region $\mathscr{C}$ of the index coding problem includes the simplex $\{(R_1, \ldots, R_n)\colon R_1 + \cdots + R_n = 1, R_i \geq 0,\ i \in [n]\}$ and is included in the hypercube $\{(R_1, \ldots, R_n)\colon 0 \leq R_i \leq 1,\ i \in [n]\}$. Consequently, the capacity and the broadcast rate are bounded as $1/n \leq C_{\text{sym}} \leq 1$ and $1 \leq \beta \leq n$, respectively. Similar bounds hold for $\mathscr{L}$ and λ as well.

Any instance of the index coding problem is fully determined by the side information sets $A_1, \ldots, A_n$, and is represented compactly by a sequence $(i|A_i), i \in [n]$. For example, the 3-message index coding problem with $A_1 = \{2\}, A_2 = \{1, 3\}$, and $A_3 = \{1\}$ in Figure 1.1 is represented as

$$(1|2), (2|1, 3), (3|1).$$

Each instance of the index coding problem can be equivalently specified by a directed graph $G = (V, E)$ with n vertices, referred to as the *side information graph* [27, 44]. Each vertex of G corresponds to a receiver (and its associated message) and there is a directed edge $j \to i$ if and only if (iff) receiver i knows message x_j as side information, i.e., $j \in A_i$ (see Figure 1.3). The reader is cautioned that in the literature the opposite convention is sometimes used to describe the availability of side information, in which there is a directed edge $i \to j$ if $j \in A_i$.

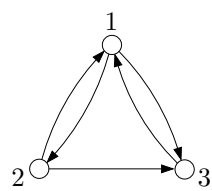


Figure 1.3: The graph representation for the index coding problem with $A_1 = \{2, 3\}, A_2 = \{1\}$, and $A_3 = \{1, 2\}$.

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Either way, the number of index coding problems with n messages is $2^{n(n-1)}$, which blows up quickly with n. Even when we remove *isomorphic* (i.e., symmetric up to vertex relabeling) instances and concentrate on nonisomorphic instances, the number of such problems is equal to that of nonisomorphic directed graphs with n vertices [151, Seq. A000273], which grows superexponentially. Throughout the monograph, we identify an instance of the index coding problem with its side information graph G and often write "index coding problem G." We also denote the broadcast rate and the capacity region of problem G with $\beta(G)$ and $\mathscr{C}(G)$, respectively, when this dependence is to be emphasized.

The index coding problem can be also formulated as a special case of the multiple-unicast network coding problem [2, 164]; see Section 10 for a self-contained description of the latter. For example, the index coding problem with $A_1 = \{2\}, A_2 = \{1, 3\}$, and $A_3 = \{1\}$ can be represented as a network coding problem depicted in Figure 1.4. In this network coding graph, each edge (solid line) represents a link of unit capacity and each vertex represents a node. There are three messages communicated from source nodes on the left to destination nodes on the right (depicted by dashed lines). Each source node is connected to the top left node, which encodes the messages into an index (coded message) and communicates it to the top right node (under the capacity constraint). The top right node is connected to each destination node and forwards (broadcasts) the index to all of them. The remaining edges connect source nodes and destination nodes directly according to the availability pattern of side information. The essence of each problem

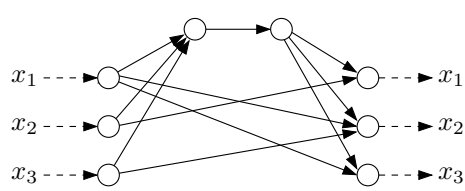


Figure 1.4: The network coding representation of the index coding problem with $A_1 = \{2\}, A_2 = \{1, 3\}$, and $A_3 = \{1\}$.

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instance is captured by such direct connections between the source and destination nodes. If we consider the subgraph consisting of only source and destination nodes and overlap each source–destination pair, then we recover the side information graph in Figure 1.3.

1.3 Objectives and the Organization

As mentioned earlier, our main objectives in studying the index coding problem are to characterize the capacity region for a general index coding instance in a computable expression and to develop the coding scheme that can achieve it. Despite their simplicity, these two closely related questions are extremely difficult and precise answers to them, after twenty years of vigorous investigation, are still in terra incognita. There are, nonetheless, many elegant results that shed light on the fundamental challenges in multiple-unicast network communication and expose intriguing interplay between coding theory, graph theory, and information theory. This monograph thus aims to provide a concise survey of these results in a unified framework.

To facilitate the development of this framework, the rest of the monograph is organized as follows. Section 2 reviews some known results in graph theory that will be recalled frequently throughout. Our main story starts with Section 3, which presents a few noncomputable characterizations of the capacity of a general index coding problem in graph-theoretic and information-theoretic expressions. As a main application of these characterizations, we present basic structural properties of index coding capacity in Section 4. The next two sections develop upper and lower bounds on the capacity. In Section 5, we establish a few capacity upper bounds and the relationships among them. In Section 6, we develop several coding schemes based on algebraic, graph-theoretic, and information-theoretic tools along with the corresponding lower bounds on the capacity. Section 7 is devoted to the notion of criticality, namely, whether the side information graph cannot be reduced without lowering the capacity, and presents necessary and sufficient conditions for an index coding problem to be critical. In Section 8, we combine the results in Sections 4 through 7 to characterize the capacity of several classes of the index coding problem. The capacity

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approximation results beyond these classes of problems are presented in Section 9. The next two sections explore the connection between index coding and other related problems. In Section 10, we relate index coding to the well-known multiple-unicast network coding problem. In Section 11, we present the intriguing duality between index coding, locally recoverable distributed storage, and guessing games. We conclude our discussion in Section 12 by pointing out numerous variations and extensions of the basic index coding problem presented thus far. Some technical proofs are relegated to the end of each section.

1.A Capacity Region Under Average Error Probability Criterion

Let X_i and $\hat{X}_i$ be random variables representing the i-th message and its estimate, respectively. Suppose that $(X_1, \ldots, X_n)$ is uniformly distributed over $\mathcal{X}^{t_1} \times \cdots \times \mathcal{X}^{t_n}$, i.e., the messages are uniformly distributed and independent of each other. A rate tuple $(R_1, \ldots, R_n)$ is said to be *vanishing-error achievable* if for every $\epsilon > 0$, there exists a $(t_1, \ldots, t_n, r)$ code with $R_i \leq t_i/r$, $i \in [n]$, such that the average probability of error

$$P_e = \mathsf{P}\{(\hat{X}_1, \ldots, \hat{X}_n) \neq (X_1, \ldots, X_n)\} \leq \epsilon. \tag{1.3}$$

Equivalently, a rate tuple $(R_1, \ldots, R_n)$ is vanishing-error achievable if there exists a sequence of $(\lceil rR_1 \rceil, \ldots, \lceil rR_n \rceil, r)$ index codes such that the average probability of error P_e converges to 0 as $r \to \infty$. The *vanishing-error capacity region* $\mathscr{C}^*$ of the index coding problem is the closure of the set of all vanishing-error achievable rate tuples $(R_1, \ldots, R_n)$.

For a general network communication problem, the vanishing-error capacity region and the (zero-error) capacity region are not the same [59, 154]. For a network with a single sender that broadcasts multiple messages, however, these two regions can be shown to be identical [156] (see also [61, Problem 8.11]), which was rediscovered by Chan and Grant [42], and Langberg and Effros [92] in the context of index coding and single-sender network coding problems.

Lemma 1.2. For any index coding problem, $\mathscr{C} = \mathscr{C}^*$.

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The proof is delegated to Appendix 1.C. One can similarly define *vanishing-error linearly achievability*. The *vanishing-error linear capacity region* $\mathscr{L}^*$ is then defined to be the closure of the set of all vanishing-error achievable rate tuples, which is also the same as the (zero-error) linear capacity region.

Lemma 1.3. For any $\mathcal{X} = \mathbb{F}_q$, $\mathscr{L} = \mathscr{L}^*$.

To prove Lemma 1.3, we establish the following stronger result.

Lemma 1.4. For any linear index code, if the probability of error $P_e < 1/2$, then $P_e = 0$.

Proof. Let $t_\Sigma = \sum_{i=1}^n t_i$. Every linear encoder ϕ is specified by a matrix $\Phi \in \mathbb{F}_q^{r \times t_\Sigma}$ such that $y = \phi(x^n) = \Phi x^n$. Assume by contradiction that $0 < P_e < 1/2$. Since the probability of error is nonzero, there exist distinct $\tilde{x}^n, \tilde{z}^n \in \mathbb{F}_q^{t_\Sigma}$ such that $\Phi\tilde{x}^n = \Phi\tilde{z}^n$ and $\tilde{x}(A_i) = \tilde{z}(A_i)$ for some $i \in [n]$, or equivalently, $\Phi e^n = 0$ and $e(A_i) = 0$, where $e^n = \tilde{x}^n - \tilde{z}^n$. Now for every x^n, since the code is linear, there exists $z^n = x^n + e^n$ for which $\Phi x^n = \Phi z^n$ and $x(A_i) = z(A_i)$ for some i. Therefore, at most one of x^n and z^n can be recovered correctly and consequently the average probability of error $P_e \geq 1/2$, which contradicts the assumption. □

We are now ready to prove Lemma 1.3. Clearly $\mathscr{L} \subseteq \mathscr{L}^*$. Thus, it suffices to show that $\mathscr{L}^* \subseteq \mathscr{L}$. Let $(R_1, \ldots, R_n)$ be a vanishing error linearly achievable rate tuple. Then, by definition, there exists a sequence of $(\lceil rR_1 \rceil, \ldots, \lceil rR_n \rceil, r)$ index codes for which (1.3) is satisfied. Therefore, there exists a sufficiently large r such that the error probability of the index code $(\lceil rR_1 \rceil, \ldots, \lceil rR_n \rceil, r)$ is less than $1/2$. By Lemma 1.4, the error probability of this code is zero and thus, $(R_1, \ldots, R_n)$ is also a (zero-error) linearly achievable rate tuple. Hence, we have $\mathscr{L}^* \subseteq \mathscr{L}$, which completes the proof.

1.B Proof of Lemma 1.1

Let I and I' be index coding instances defined over finite alphabets $\mathcal{X}$ and $\mathcal{X}'$, respectively, and let $\mathscr{A}$ and $\mathscr{A}'$ be the associated sets of achievable rate tuples. We consider two cases.

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Case 1. $\log_{|\mathcal{X}|}|\mathcal{X}'|$ is a rational number, i.e., $\log_{|\mathcal{X}|}|\mathcal{X}'| = a/b$ for some $a, b \in \mathbb{N}$. To show that the capacity regions are equal, it suffices to show $\mathscr{A} = \mathscr{A}'$. Suppose that $\mathbf{R} = (R_1, \ldots, R_n) \in \mathscr{A}$. Then, by definition, there exists a $(\mathbf{t}, r) = (t_1, \ldots, t_n, r)$ code for problem I such that $R_i \leq t_i/r$, $i \in [n]$. Repeat the $(\mathbf{t}, r)$ code a times to construct an $(a\mathbf{t}, ar)$ index code for problem I. Since the two instances are both defined on the same set of side information, and $|\mathcal{X}|^a = |\mathcal{X}'|^b$, this leads to a $(b\mathbf{t}, br)$ code for problem I'. Therefore, $\mathbf{R} \in \mathscr{A}'$, and thus $\mathscr{A} \subseteq \mathscr{A}'$. By similar steps we can show $\mathscr{A}' \subseteq \mathscr{A}$, which completes the proof.

Case 2. $\log_{|\mathcal{X}|}|\mathcal{X}'|$ is irrational. First, we show that $\mathscr{A}' \subseteq \mathscr{C}_{\mathcal{X}}$. Suppose that $\mathbf{R} \in \mathscr{A}'$. Then, by definition, there exists a $(\mathbf{t}, r)$ index code for problem I' such that $R_i \leq t_i/r$, $i \in [n]$. For any $b \in \mathbb{N}$ sufficiently large, there exists $a \in \mathbb{N}$ such that $a/b < \log_{|\mathcal{X}|}|\mathcal{X}'| < (a+1)/b$. Construct a $(b\mathbf{t}, br)$ index code for problem I' by repeating the $(\mathbf{t}, r)$ code b times. Since $|\mathcal{X}|^a < |\mathcal{X}'|^b < |\mathcal{X}|^{a+1}$ and the two problems are defined on the same set of side information, a $(a\mathbf{t}, (a+1)r)$ code for problem I can be constructed from the $(b\mathbf{t}, br)$ code for problem I'. Letting $b \to \infty$ (and hence $a \to \infty$) proves that $\mathbf{R} \in \mathscr{C}_{\mathcal{X}}$, and thus $\mathscr{A}' \subseteq \mathscr{C}_{\mathcal{X}}$. Since $\mathscr{C}_{\mathcal{X}}$ is closed, we have $\mathscr{C}_{\mathcal{X}'} \subseteq \mathscr{C}_{\mathcal{X}}$. By similar steps we can show $\mathscr{C}_{\mathcal{X}} \subseteq \mathscr{C}_{\mathcal{X}'}$, which completes the proof.

1.C Proof of Lemma 1.2

We adapt Telatar's simplification of the classical proof by Willems [156] on the invariance of the broadcast channel capacity region under the average and maximal error probability criteria that appeared in [61, Problem 8.11].

It is trivial to see that $\mathscr{C} \subseteq \mathscr{C}^*$. We thus prove the other direction. Let $(R_1, \ldots, R_n) \subseteq \mathscr{C}^*$. Then for every $\epsilon > 0$, there exists a sequence of $(t_1, \ldots, t_n, r)$ codes with $t_i = \lceil rR_i \rceil$, $i \in [n]$, such that the average probability of error $P_e \leq \epsilon$ for r sufficiently large. Assume without loss of generality that $R_i > 0$, $i \in [n]$. (Otherwise, the message x_i of zero rate $R_i = 0$ is fixed and can be ignored.) We will identify the set $\{0,1\}^t$ of all t-bit sequences with the set $[2^t] = \{1, \ldots, 2^t\}$ of integers throughout. Then the set of codewords of the $(t_1, \ldots, t_n, r)$ index code

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can be expressed as

$$\mathcal{C} = \{\phi(x^n) \in [2^r] : x^n \in [2^{t_1}] \times \cdots \times [2^{t_n}]\},$$

which will be referred to as the *codebook*. For each message tuple x^n, we define its probability of error as

$$P_e(x^n) = \mathsf{P}\{X^n \neq \hat{X}^n \,|\, X^n = x^n\}. \tag{1.4}$$

Note that $P_e(x^n)$ is either 0 or 1 for any index code. We say that a codeword $\phi(x^n)$ is said to be "bad" if the corresponding $P_e(x^n) = 1$. Since the average probability of error is

$$\epsilon = \frac{1}{2^{t_\Sigma}} \sum_{x^n} P_e(x^n),$$

there are $2^{t_\Sigma}\epsilon$ "bad" codewords $\phi(x^n)$, where $t_\Sigma = \sum_{i=1}^n t_i$. Randomly and independently permute the messages $x_1, \ldots, x_n$ to generate a new codebook $\overline{\mathcal{C}}$ that consists of codewords $\phi(\pi_1(x_1), \ldots, \pi(x_n))$, where $\pi_1, \ldots, \pi_n$ denote the independent random permutations.

We now proceed with the *multicoding* technique by Marton [108] originally developed for broadcast channels. We partition the codebook $\overline{\mathcal{C}}$ into subcodebooks $\overline{\mathcal{C}}(x'_1, \ldots, x'_n)$ for a new set of message tuples $(x'_1, \ldots, x'_n) \in [2^{t_1}/r^2] \times \cdots \times [2^{t_n}/r^2]$, each subcodebook consisting of $r^2 \times \cdots \times r^2 = r^{2n}$ codewords of length r. We will show that there exists a new encoder $\phi'(x'_1, \ldots, x'_n)$ that maps each message tuple $(x'_1, \ldots, x'_n)$ to some codeword in the corresponding subcodebook $\overline{\mathcal{C}}(x'_1, \ldots, x'_n)$, so that every codeword $\phi'(x'_1, \ldots, x'_n)$ is "good" (= not "bad") and hence distinguishable from the rest with zero error. Since the rate of the new code is $R'_i = (t_i - 2\log r)/r = (\lceil rR_i \rceil - 2\log r)/r$, which converges to the original rate R_i as $r \to \infty$, there is no asymptotic rate loss for achieving the zero error probability. Details on the existence of $\phi'(x'_1, \ldots, x'_n)$ are as follows.

First note that every subcodebook has the same distribution as the set

$$\{\phi(\pi_1(x_1), \ldots, \pi_n(x_n)) \colon x_1, \ldots, x_n \in [r^2]\}.$$

The probability that all r^{2n} codewords in this set are "bad" is upper bounded by the probability that all r^2 "diagonal" codewords, that is,

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all codewords in

$$\{\phi(\pi_1(x), \ldots, \pi_n(x)) \colon x \in [r^2]\},$$

are "bad." Since the permutations are independent and there are $2^{t_\Sigma}\epsilon$ "bad" codewords, the probability that all "diagonal" codewords are "bad" is upper bounded by

$$\frac{2^{t_\Sigma}\epsilon}{\prod_{i=1}^n 2^{t_i}} \cdot \frac{2^{t_\Sigma}\epsilon - 1}{\prod_{i=1}^n (2^{t_i} - 1)} \cdot \frac{2^{t_\Sigma}\epsilon - 2}{\prod_{i=1}^n (2^{t_i} - 2)} \cdots \frac{2^{t_\Sigma}\epsilon - (r^2 - 1)}{\prod_{i=1}^n (2^{t_i} - (r^2 - 1))}$$
$$\leq \left(\prod_{i=1}^n \frac{2^{t_i}}{2^{t_i} - (r^2 - 1)}\right)^{r^2} \epsilon^{r^2},$$

which is further upper bounded by $(2\epsilon)^{r^2}$ for r sufficiently large (since $2^{t_i}/(2^{t_i} - (r^2 - 1)) \to 1$ as $r \to \infty$).

Next, since every subcodebook has the same distribution, the expected number of subcodebooks for which all of their constituent codewords are "bad" is upper bounded by

$$\frac{2^{t_\Sigma}}{r^{2n}}(2\epsilon)^{r^2},$$

which is the product of the number of all subcodebooks and the probability bound of $(2\epsilon)^{r^2}$ we computed above. Since this bound tends to zero as $r \to \infty$, there exists at least one permutation tuple $(\pi_1, \ldots, \pi_n)$ such that every subcodebook has at least one codeword that is "good." Hence, we can define the new encoder $\phi'(x'_1, \ldots, x'_n)$ that maps each message tuple $(x'_1, \ldots, x'_n) \in [2^{t_1}/r^2] \times \cdots \times [2^{t_n}/r^2]$ to a "good" codeword in the subcodebook $\overline{\mathcal{C}}(x'_1, \ldots, x'_n)$. This completes the proof of Lemma 1.2.

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2

Mathematical Preliminaries

2.1 Directed Graphs

Throughout, unless specified otherwise, a graph $G = (V, E)$ shall mean a directed, finite, and simple graph, where $V = V(G)$ is the set of vertices (nodes) and $E = E(G) \subseteq V \times V$ is the set of directed edges. An edge (i, j) will be sometimes denoted in a more visual form $i \to j$. A graph $G = (V, E)$ is said to be *unidirectional* if $(i, j) \in E$ implies $(j, i) \notin E$. Similarly, G is said to be *bidirectional* if $(i, j) \in E$ implies $(j, i) \in E$. Given G, its associated undirected graph $U = U(G)$ is defined by $V(U) = V(G)$ and $E(U) = \{\{i, j\} \colon (i, j) \in E(G)\}$. Two vertices i and j of an undirected graph U are said to be *adjacent* if $\{i, j\} \in E(U)$. A bidirectional graph G will be sometimes identified with its associated undirected graph $U(G)$. The *transpose* G^T of a graph G is defined by $V(G^T) = V(G)$ and $(i, j) \in E(G^T)$ iff $(j, i) \in E(G)$. Thus, a graph G is bidirectional iff $G = G^T$ and is unidirectional iff $E(G) \cap E(G^T) = \emptyset$. The *complement* $\overline{G}$ of a graph G is defined by $V(\overline{G}) = V(G)$ and for $i \neq j$, $(i, j) \in E(\overline{G})$ iff $(i, j) \notin E(G)$. A graph G' is a *subgraph* of G if $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G) \cap (V(G') \times V(G'))$. A subgraph G' is said to be *induced* by $J \subseteq V(G)$ if $V(G') = J$ and $E(G') = E(G) \cap (J \times J)$. The subgraph G' induced by J is denoted as $G|_J$. A

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path in a graph G is a sequence of vertices $i_1 \to i_2 \to \cdots \to i_k$ such that $(i_j, i_{j+1}) \in E(G)$, $j \in [k-1]$. A *cycle* is a *closed* path, namely, $i_1 \to i_2 \to \cdots \to i_k = i_1$. A graph is said to be *acyclic* if it does not contain a cycle.

A *tournament* is a unidirectional graph in which every pair of distinct vertices is connected by a single (directed) edge.

Lemma 2.1 (Stearns [143], Erdös and Moser [67]). Every tournament on n vertices includes an acyclic induced subgraph on $1 + \lfloor \log_2 n \rfloor$ vertices.

An *independent set* of a graph G is a set of vertices with no edge among them. The *independence number* $\alpha(G)$ is the size of the largest independent set of the graph G. A *clique* K of a graph G is a set of vertices such that there is a (directed) edge from every vertex in K to every other vertex in K. The *clique number* $\omega(G)$ is the size of the largest clique of the graph G. It is easy to see that K is a clique of G iff it is an independent set of $\overline{G}$ and thus that

$$\omega(G) = \alpha(\overline{G}). \tag{2.1}$$

For an undirected graph U, the notions of graph complement $\overline{U}$, independent set, independence number $\alpha(U)$, clique, and clique number $\omega(U)$ are similarly defined. It can be readily checked that (2.1) continues to hold.

2.2 Graph Coloring

A (vertex) coloring of an undirected (finite simple) graph U is a mapping that assigns a color to each vertex such that no two adjacent vertices share the same color. The *chromatic number* $\chi(U)$ is the minimum number of colors such that a coloring of the graph U exists. More generally, a b-fold coloring assigns a set of b colors to each vertex such that no two adjacent vertices share some color in common. The b-fold chromatic number $\chi^{(b)}(U)$ is the minimum number of colors such that a b-fold coloring exists. The *fractional chromatic number* of the graph is defined as

$$\chi_f(U) = \inf_b \frac{\chi^{(b)}(U)}{b} = \lim_{b \to \infty} \frac{\chi^{(b)}(U)}{b},$$

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where the limit exists by Fekete's lemma [69] since $\chi^{(b)}(U)$ is subadditive, i.e., $\chi^{(b_1+b_2)}(U) \leq \chi^{(b_1)}(U) + \chi^{(b_2)}(U)$. Consequently,

$$\chi_f(U) \leq \chi(U). \tag{2.2}$$

For any coloring, the vertices of the same color form an independent set. Let $\mathcal{I}$ be the collection of all independent sets in U. The chromatic number and the fractional chromatic number are characterized via the following optimization problem

$$\begin{aligned} \text{minimize} \quad & \sum_{J \in \mathcal{I}} \gamma_J \\ \text{subject to} \quad & \sum_{J \in \mathcal{I}: i \in J} \gamma_J \geq 1, \quad i \in V. \end{aligned}$$

When the optimization variables γ_J, $J \in \mathcal{I}$, take values in $\{0, 1\}$, then the (integral) solution to the integer programming is the chromatic number. If this constraint is relaxed and $\gamma_J \in [0, 1]$, then the (rational) solution to this linear programming is the fractional chromatic number [128]. As the vertices in a clique should be colored distinctly,

$$\omega(U) \leq \chi_f(U) \leq \chi(U) \tag{2.3}$$

for any undirected graph U. The fractional chromatic number can be also related to the independence number.

Lemma 2.2 (Scheinerman and Ullman [128])**.** For any undirected graph U,

$$\chi_f(U) \geq \frac{|V(U)|}{\alpha(U)}. \tag{2.4}$$

An *automorphism* of an undirected graph U is a bijective function $\pi : V(U) \to V(U)$ such that for any two vertices $i, j \in V(U)$, $\pi(i)$ and $\pi(j)$ are adjacent iff i and j are adjacent. An undirected graph U is said to be *vertex transitive* if for any two vertices i and j of U, there exists an automorphism $\pi : V(U) \to V(U)$ such that $\pi(i) = j$.

Remark 2.1. The inequality in (2.4) holds with equality if the graph U is vertex transitive [128].

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2.3 Perfect Graphs

An undirected graph $U = (V, E)$ is said to be *perfect* if for every induced subgraph $U|_J$, $J \subseteq V$, the clique number equals the chromatic number, i.e., $\omega(U|_J) = \chi(U|_J)$. Perfect graphs can be characterized as follows.

Proposition 2.1 (Chudnovsky, Robertson, Seymour, and Thomas [46])**.** An undirected graph U is perfect iff no induced subgraph of U is an odd cycle of length at least five (also known as odd *hole*) or the complement of one (also known as odd *antihole*).

Let U be an undirected graph on n vertices. For each clique K of U, the *incidence vector* $\mathbf{x}(K) = (x_1(K), \ldots, x_n(K))$ is defined by

$$x_i(K) = \begin{cases} 1 & \text{if } i \in K \\ 0 & \text{otherwise.} \end{cases}$$

Let $\mathcal{K}$ be the collection of all cliques of U (and $\emptyset$). The *clique polytope* of U is defined by the convex hull of the incidence vectors of cliques of U as

$$P^*(U) = \left\{ \sum_{K \in \mathcal{K}} \delta_K \, \mathbf{x}(K) : \delta_K \geq 0 \text{ and } \sum_{K \in \mathcal{K}} \delta_K = 1 \right\}. \tag{2.5}$$

Another (convex) polytope associated with U is defined as

$$P(U) = \left\{ \mathbf{x} \in \mathbb{R}^n_{\geq 0} \colon \sum_{i \in I} x_i \leq 1 \text{ for all independent sets } I \right\}. \tag{2.6}$$

Since the incidence vector $\mathbf{x} = (x_1, \ldots, x_n)$ of every clique satisfies $\sum_{i \in I} x_i \leq 1$ for all independent sets I, $P^*(U) \subseteq P(U)$ for every U. Lovász's perfect graph theorem states that equality holds iff U is perfect.

Lemma 2.3 (Lovász [102])**.** For any graph U the following statements are equivalent:

- U is perfect.
- $P^*(U) = P(U)$.
- $\overline{U}$ is perfect.

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2.4 Graph Products

Generally speaking, a graph product is a binary operation on two (undirected) graphs U_1 and U_2 that produces a graph with the vertex set $V(U_1) \times V(U_2)$ and the edge set constructed from the original edge sets according to certain rules. In the following, $i \sim j$ denotes that i and j are adjacent.

Given two undirected graphs U_1 and U_2, the *disjunctive (OR) product* $U = U_1 \vee U_2$ is defined [122, 128] as $V(U) = V(U_1) \times V(U_2)$ and $(i_1, i_2) \sim (j_1, j_2)$ iff

$$i_1 \sim j_1 \quad \text{or} \quad i_2 \sim j_2.$$

We use the notion $U^{\vee k}$ for the disjunctive product of k copies of U. The fractional chromatic number of the disjunctive product is multiplicative.

Lemma 2.4 (Scheinerman and Ullman [128, Cor. 3.4.2])**.**

$$\chi_f(U_1 \vee U_2) = \chi_f(U_1)\chi_f(U_2).$$

Note that the (integral) chromatic number satisfies the following relationship [128, Prop. 3.4.4]:

$$\chi(U_1 \vee U_2) \leq \chi(U_1)\chi(U_2). \tag{2.7}$$

The chromatic and fractional chromatic numbers of the power of a graph scale in the same exponential rate.

Lemma 2.5 (Scheinerman and Ullman [128, Cor. 3.4.3])**.**

$$\chi_f(U) = \lim_{k\to\infty} \sqrt[k]{\chi(U^{\vee k})} = \inf_k \sqrt[k]{\chi(U^{\vee k})}.$$

The *Cartesian product* $U = U_1 \wedge U_2$ is defined by $(i_1, i_2) \sim (j_1, j_2)$ iff

$$(i_1 = j_1 \text{ and } i_2 \sim j_2) \quad \text{or} \quad (i_1 \sim j_1 \text{ and } i_2 = j_2).$$

This product does not increase the chromatic number.

Lemma 2.6 (Sabidussi [126, Lemma 2.6])**.**

$$\chi(U_1 \wedge U_2) = \max\{\chi(U_1), \chi(U_2)\}.$$

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The *lexicographic product* $U = U_1 \circ U_2$ is defined [78] by $(i_1, i_2) \sim (j_1, j_2)$ iff

$$i_1 \sim j_1 \quad \text{or} \quad (i_1 = j_1 \text{ and } i_2 \sim j_2)\,.$$

Note that the lexicographic product of graphs is not *commutative*, i.e., $U_1 \circ U_2 \neq U_2 \circ U_1$. Nonetheless, its fractional chromatic number is still multiplicative.

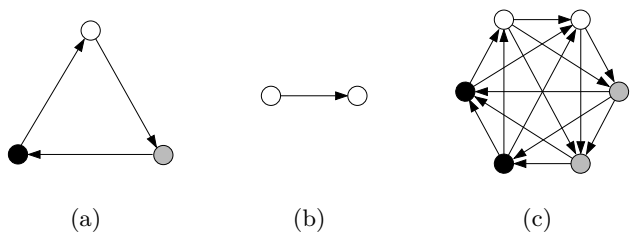
Lemma 2.7 (Scheinerman and Ullman [128, Cor. 3.4.5])**.**

$$\chi_f(U_1 \circ U_2) = \chi_f(U_1)\chi_f(U_2).$$

The lexicographic product can be also defined for directed graphs G_0 and G_1 as $((i_1, i_2), (j_1, j_2)) \in E(G_0 \circ G_1)$ iff

$$(i_1, j_1) \in E(G_0) \quad \text{or} \quad (i_1 = j_1 \text{ and } (i_2, j_2) \in E(G_1))\,.$$

In other words, each vertex in G_0 is replaced by a copy of G_1 and all vertices in one copy of G_1 are connected to vertices in another copy according to $E(G_0)$; see Figure 2.1.



(a) (b) (c)

Figure 2.1: (a) The 3-vertex graph G_0, (b) the 2-vertex graph G_1, and (c) the lexicographic product $G_0 \circ G_1$.

2.5 Information Measures

Let X be a random variable on a finite alphabet $\mathcal{X}$ drawn according to the probability mass function (pmf) $p(x) = \mathsf{P}\{X = x\}$, $x \in \mathcal{X}$. The *entropy* [137] of X (or more precisely, the entropy of $p(x)$) is

$$H(X) = \sum_{x \in \mathcal{X}} p(x) \log \frac{1}{p(x)} = \mathsf{E}\Big[\log \frac{1}{p(X)}\Big].$$

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By continuity, we use the convention $0 \log 0 = 0$ in the above summation. Let (X, Y) be a pair of random variables on finite alphabets. Their *joint entropy* is

$$H(X,Y) = \sum_{x,y} p(x,y) \log \frac{1}{p(x,y)} = \mathsf{E}\Big[\log \frac{1}{p(X,Y)}\Big],$$

which is the entropy of a single random variable $Z = (X, Y)$. By the chain rule, we have

$$H(X,Y) = H(X) + H(Y|X),$$

where $H(Y|X) = \mathsf{E}[\log(1/p(Y|X))]$ is the *conditional entropy* of Y given X. The conditional entropy satisfies

$$0 \le H(Y|X) \le H(Y).$$

The lower bound is tight (i.e., $H(Y|X) = 0$) iff Y is a function of X, and the upper bound is tight iff Y is independent of X.

The *mutual information* [137] between a pair of discrete random variables X and Y is

$$I(X;Y) = \mathsf{E}\left[\log \frac{p(X,Y)}{p(X)p(Y)}\right] = I(Y;X).$$

It can be also expressed in terms of the entropies as

$$I(X;Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X,Y).$$

The mutual information satisfies

$$0 \le I(X;Y) \le H(Y).$$

The lower bound is tight iff X and Y are independent, and the upper bound is tight iff Y is a function of X. As with entropy, the mutual information between X and (Y, Z) can be computed by treating (Y, Z) as a single random variable, and is expressed simply as $I(X; Y, Z)$. By the chain rule,

$$I(X;Y,Z) = I(X;Y) + I(X;Z|Y),$$

where

$$I(X;Z|Y) = \mathsf{E}\left[\log \frac{p(X,Z|Y)}{p(X|Y)p(Z|Y)}\right]$$

is the *conditional mutual information* between X and Z given Y.

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3

Multiletter Characterizations of the Capacity

In this section, we characterize the capacity region of a general index coding problem in two orthogonal approaches. First, we introduce the notion of confusion graph and characterize the capacity region via the chromatic number and the fractional chromatic number of the confusion graph of an index coding problem. Second, we present an information-theoretic characterization of the capacity region via Shannon's random coding technique.

3.1 Confusion Graphs

The notion of confusion graph for the index coding problem was originally introduced by Alon, Hassidim, Lubetzky, Stav, and Weinstein [3]. In the context of guessing games (cf. Section 11.2), an equivalent notion was introduced independently by Gadouleau and Riis [73]. The use of confusion graphs in information theory traces back to the work by Shannon [138]; see also [158, 5, 88].

Consider an index coding problem represented by the side information graph $G = (V, E)$ with $V = [n]$ and $E = \{(j, i) : j \in A_i, i \in [n]\}$. Let $\mathbf{t} = (t_1, \ldots, t_n)$ be a length-n positive integer tuple. We say that

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two binary message n-tuples $x^n, z^n \in \prod_{i=1}^{n}\{0,1\}^{t_i}$ are *confusable at position* $l \in [t_i]$ *of message* $i \in [n]$ if

$$x_{il} \neq z_{il} \quad \text{and} \quad x_j = z_j \text{ for all } j \in A_i.$$

Equivalently, x^n and z^n are *not* confusable at position l of message i if

$$x_{il} = z_{il} \quad \text{or} \quad x_j \neq z_j \text{ for some } j \in A_i.$$

As an example, consider the 3-message index coding problem with side information graph shown in Figure 3.1. Let $t_i = 1$, $i \in [3]$. Two message tuples $(x_1, x_2, x_3) = (0, 0, 0)$ and $(z_1, z_2, z_3) = (0, 1, 1)$ are confusable at position 1 of message 2, whereas two message tuples $(x_1, x_2, x_3) = (0, 0, 0)$ and $(z_1, z_2, z_3) = (1, 0, 1)$ are not confusable at position 1 of message 2 (and as a matter of fact, not confusable at any position).

The confusion relation between two tuples x^n and z^n are represented by the *confusion graph* $\Gamma_{\mathbf{t}}^{(il)}(G)$ *at position* l *of message* i, which is defined as an undirected graph with $\prod_{i=1}^{n} 2^{t_i}$ vertices such that every vertex corresponds to a binary tuple $x^n \in \prod_{i=1}^{n}\{0,1\}^{t_i}$ and two vertices are connected iff the corresponding binary tuples are confusable at position l of receiver i. Figure 3.2(a), (b), and (c) show the corresponding confusion graphs with $\mathbf{t} = (1, 1, 1)$ for the index coding problem in Figure 3.1. Note that each $\Gamma_{\mathbf{t}}^{(il)}(G)$ is a bipartite graph.

Combining the confusability over all messages and all positions, we say that x^n and z^n are *confusable* if they are confusable at some position l of some message i. The aggregate confusion relation among all binary tuples is represented by the *confusion graph*

$$E\left(\Gamma_{\mathbf{t}}(G)\right) = \bigcup_{i=1}^{n} \bigcup_{l=1}^{t_i} E(\Gamma_{\mathbf{t}}^{(il)}(G)). \tag{3.1}$$

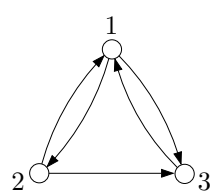


Figure 3.1: The index coding problem $(1|2, 3), (2|1), (3|1, 2)$.

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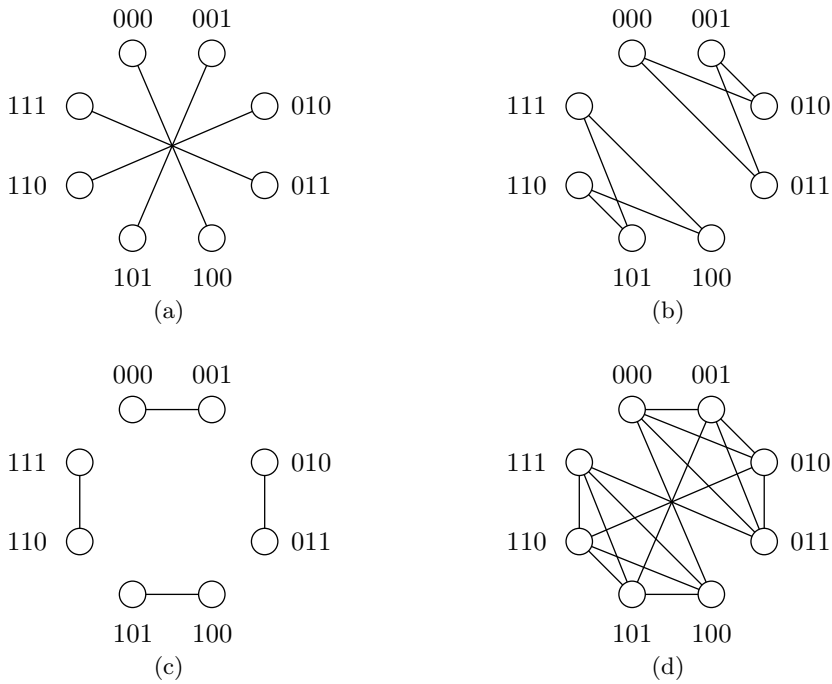


Figure 3.2: Confusion graphs for the directed graph G shown in Figure 3.1 corresponding to the integer tuple $\mathbf{t} = (t_1, t_2, t_3) = (1, 1, 1)$. (a) $\Gamma_{\mathbf{t}}^{(11)}(G)$. (b) $\Gamma_{\mathbf{t}}^{(21)}(G)$. (c) $\Gamma_{\mathbf{t}}^{(31)}(G)$. (d) $\Gamma_{\mathbf{t}}(G)$.

The confusion graph $\Gamma_{\mathbf{t}}(G)$ corresponding to $\mathbf{t} = (1, 1, 1)$ for the graph of Figure 3.1 is shown in Figure 3.2(d).

If the message tuples x^n and z^n are confusable, then some receiver, say, receiver i, cannot figure out whether its message is x_i or z_i based on the side information $x(A_i)$ or $z(A_i)$. In comparison, if the message tuples x^n and z^n are not confusable, then they can be distinguished (if distinct) from side information at each and every receiver. In Figure 3.2, $(x_1, x_2, x_3) = (0, 0, 0)$ and $(z_1, z_2, z_3) = (1, 0, 1)$ are not confusable. Receiver 1 can distinguish x_1 and z_1 based on x_3 and z_3. Receiver 2 has nothing to distinguish as $x_2 = z_2$. Receiver 3 can distinguish x_3 and z_3 based on x_1 and z_1.

If $\mathbf{t} = (t, \ldots, t)$, then $\Gamma_{\mathbf{t}}(G)$ is denoted simply by $\Gamma_t(G)$.

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3.2 Capacity via Confusion Graphs

Let

$$\beta_t(G) := \inf_{(t,r)\ \text{codes}} \frac{r}{t} \tag{3.2}$$

be the smallest number of transmissions per message bit that is achieved by the optimal index code for t-bit messages. As discussed in (1.2), the broadcast rate is the asymptotic limit

$$\beta(G) = \inf_t \beta_t(G) = \lim_{t\to\infty} \beta_t(G) \tag{3.3}$$

and for every t,

$$\beta(G) \le \beta_t(G). \tag{3.4}$$

Alon, Hassidim, Lubetzky, Stav, and Weinstein [3] used the notion of confusion graph to characterize $\beta_t(G)$ and its limit $\beta(G)$. Suppose that each message consists of t bits. Consider a vertex coloring of the confusion graph $\Gamma = \Gamma_t(G)$ with $\chi(\Gamma)$ colors. This partitions the vertices of Γ into $\chi(\Gamma)$ independent sets. By the definition of confusion graph, no two message tuples in each independent set are confusable. Among all message tuples that belong to an independent set, every receiver can figure out the desired message based on its side information alone. Therefore, assigning a distinct codeword to each independent set yields a valid index code. The total number of codewords of this index code is $\chi(\Gamma)$, which requires $r = \lceil \log \chi(\Gamma) \rceil$ bits to be broadcast. Conversely, consider any (t, r) index code that assigns (at most) 2^r distinct codewords to message tuples. By definition, all the message tuples mapped to a codeword form an independent set of the confusion graph $\Gamma = \Gamma_t(G)$. Moreover, every message tuple is mapped to some codeword so that these independent sets partition $V(\Gamma)$. Thus, $\chi(\Gamma) \le 2^r$, or equivalently, $r \ge \lceil \log \chi(\Gamma) \rceil$.

Proposition 3.1 ([3])**.** For any index coding problem with side information graph G,

$$\beta_t(G) = \frac{1}{t} \lceil \log \chi(\Gamma_t(G)) \rceil. \tag{3.5}$$

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Combined with (3.3), Proposition 3.1 characterizes the broadcast rate as

$$\beta(G) = \inf_t \frac{1}{t} \log \chi(\Gamma_t(G)) = \lim_{t\to\infty} \frac{1}{t} \log \chi(\Gamma_t(G)). \tag{3.6}$$

Here the equality between the infimum and the limit was justified in (1.2). Or more directly,

$$\chi(\Gamma_{t_1+t_2}) \le \chi(\Gamma_{t_1} \vee \Gamma_{t_2}) \tag{3.7}$$

$$\le \chi(\Gamma_{t_1})\chi(\Gamma_{t_2}), \tag{3.8}$$

where (3.7) follows since $E(\Gamma_{t_1+t_2}) \subseteq E(\Gamma_{t_1} \vee \Gamma_{t_2})$, and (3.8) follows by (2.7). Hence, $\log \chi(\Gamma_t)$ is subadditive and the second identity in (3.6) follows by Fekete's lemma [69]. Combined with (3.4), Proposition 3.1 also establishes an upper bound on the broadcast rate as

$$\beta(G) \le \frac{1}{t} \lceil \log \chi(\Gamma_t(G)) \rceil. \tag{3.9}$$

It turns out that the broadcast rate can also be characterized via the *fractional chromatic number* of the confusion graph as

$$\beta(G) = \lim_{t\to\infty} \frac{1}{t} \log \chi_f(\Gamma_t(G)). \tag{3.10}$$

Similar to (3.6), we can argue that the limit in (3.10) exists by Fekete's lemma. Recall that for a general graph Γ, $\chi_f(\Gamma) \le \chi(\Gamma)$ so that one direction of (3.10), i.e., "$\ge$" always holds. To prove the other direction, consider $\epsilon > 0$. For each t and the corresponding confusion graph $\Gamma_t(G)$, Lemma 2.5 implies that for some integer k sufficiently large,

$$\sqrt[k]{\chi(\Gamma_t^{\vee k}(G))} \le \chi_f(\Gamma_t(G)) + \epsilon. \tag{3.11}$$

It can be also checked that $E(\Gamma_{kt}(G)) \subseteq E(\Gamma_t^{\vee k}(G))$, which, when combined with (3.11), implies that $\sqrt[k]{\chi(\Gamma_{kt}(G))} \le \chi_f(\Gamma_t(G)) + \epsilon$. Hence,

$$\frac{1}{kt} \lceil \log \chi(\Gamma_{kt}(G)) \rceil \le \frac{1}{t} \big(\log \big(\chi_f(\Gamma_t(G)) + \epsilon \big) + 1 \big).$$

Therefore, by (3.9),

$$\beta(G) \le \lim_{t\to\infty} \frac{1}{t} \log \big(\chi_f(\Gamma_t(G)) + \epsilon \big).$$

Taking $\epsilon \to 0$ completes the proof of the other direction of (3.10).

The discussion thus far is summarized in the following.

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Proposition 3.2. For the index coding problem with side information graph G, the broadcast rate is

$$\beta(G) = \lim_{t\to\infty} \frac{1}{t} \log \chi(\Gamma_t(G)) \tag{3.12}$$

$$= \lim_{t\to\infty} \frac{1}{t} \log \chi_f(\Gamma_t(G)). \tag{3.13}$$

Another graph-theoretic quantity intimately related to integral and fractional chromatic numbers $\chi(\Gamma)$ and $\chi_f(\Gamma)$ is the clique number $\omega(\Gamma)$ (cf. (2.3)). For a general graph, there exist strict gaps between χ and χ_f, and between χ_f and ω. With the first gap asymptotically vanishing for confusion graphs in Proposition 3.2, the following question on the second gap arises naturally.

Open Problem 3.1. Show that

$$\beta(G) = \lim_{t\to\infty} \frac{1}{t} \log \omega(\Gamma_t(G))$$

or provide a counterexample.

Computing $\beta_t(G)$ in (3.2), using either Proposition 3.1 or any other means, is a computationally challenging problem; see [93] for the difficulty of even approximating it. Regardless of its computational complexity, $\beta_t(G)$, in light of (3.9) and Proposition 3.2, provides a sequence of computable bounds that converges to the broadcast rate from above. With no known convergent sequence of bounds from below, however, it is not known [55, Sec. III] if the broadcast rate is *computable* in general.

Open Problem 3.2. Does there exist a finite, terminating algorithm that computes the broadcast rate of an arbitrary n-message index coding problem within any desired precision?

A related question is as follows.

Open Problem 3.3. For a general index coding problem G, is there an integer $t = t(G)$ such that $\beta(G) = \beta_t(G)$?

Proposition 3.2 can be readily generalized to the following graph-theoretic characterization of the capacity region of the index coding problem.

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Theorem 3.1. The capacity region $\mathscr{C}$ of the index coding problem G is the closure of all rate tuples $(R_1, \ldots, R_n)$ such that

$$R_i \leq \frac{t_i}{\log \chi(\Gamma_{\mathbf{t}}(G))}, \quad i \in [n], \tag{3.14}$$

for some $\mathbf{t} = (t_1, \ldots, t_n)$.

The proof of this result follows the same argument that established (3.9).

Remark 3.1. By adapting (3.11), the capacity region can also be characterized in terms of the fractional chromatic number of the confusion graph, i.e., $\chi(\Gamma_{\mathbf{t}}(G))$ can be replaced with $\chi_f(\Gamma_{\mathbf{t}}(G))$ in (3.14) [12].

3.3 Capacity via Random Coding

In this section, we use Shannon's random coding idea [137] to establish an information-theoretic characterization of the capacity region of the index coding problem. Generating an asymptotically good code randomly from a random ensemble was the key step in establishing the channel coding theorem for point-to-point channels (see, for example, [50] for the proof). Indeed, for a memoryless channel $p(v|u)$, a random codebook generated according to an input distribution $p(u)$ can achieve any rate less than $I(U;V)$ with vanishing probability of decoding error. The same coding technique can be applied to the index coding problem in a straightforward manner.

Theorem 3.2. The capacity region of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G is the closure of

$$\bigcup_{r=1}^{\infty} \mathscr{C}_r(G),$$

where $\mathscr{C}_r(G)$ is the set of all rate tuples $(R_1, \ldots, R_n)$ satisfying

$$R_i \leq \frac{1}{r} I(U_i; V|U(A_i)), \quad i \in [n], \tag{3.15}$$

for some pmf $p(u_1) \cdots p(u_n)$ and some function $f : \mathcal{U}_1 \times \cdots \times \mathcal{U}_n \to \mathcal{V}$ that maps the n-tuple $(U_1, \ldots, U_n)$ to V such that the cardinalities of $U_1, \ldots, U_n$, and V are upper bounded by 2^r.

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The random variables $U_1, \ldots, U_n$, and V in Theorem 3.2 do not appear in the index coding problem definition, and consequently are referred to as *auxiliary* random variables. We first prove the achievability of rate tuples in $\mathscr{C}_r$ for each $r = 1, 2, \ldots$ based on random coding with these auxiliary variables as inputs and outputs of point-to-point communication channels. Fix a pmf $p(u_1) \cdots p(u_n)$ and a function $v = f(u_1, \ldots, u_n)$ under the prescribed cardinality constraints. For each $i \in [n]$, randomly and independently generate 2^{krR_i} sequences $u_i^k(x_i)$, $x_i \in [2^{krR_i}]$, each according to $\prod_{j=1}^k p_{U_i}(u_{ij})$. These codewords constitute the codebook, which is shared among all communicating parties. To communicate the message tuple $(x_1, \ldots, x_n)$, we transmit $y = v^k(u_1^k(x_1), \ldots, u_n^k(x_n)) \in [2^{kr}]$, where $v_j = f(u_{1j}(x_1), \ldots, u_{nj}(x_n))$, $j = 1, \ldots, k$.

Receiver $i \in [n]$ finds the most likely x_i based on the received sequence v^k and side information sequences $u_j^k(x_j)$, $j \in A_i$. Since the message x_i is encoded by a random codebook with input distribution $p(u_i)$ and given the input U_i, the output $(U(A_i), V)$ is distributed according to the conditional distribution $p(u(A_i), v|u_i) = p(u(A_i))p(v|u_i, u(A_i))$, by the same argument as in the proof of Shannon's channel coding theorem, decoding is successful with high probability (w.h.p.) if

$$rR_i < I(U_i; U(A_i), V) = I(U_i; V|U(A_i)),$$

where the last identity follows since U_i and $U(A_i)$ are independent. The same conclusion can be also drawn by using the suboptimal joint typicality decoding (see, for example, [61, Sec. 3.1]).

To prove the converse, namely, to show that any achievable rate tuple $(R_1, \ldots, R_n)$ must lie in the closure of some $\mathscr{C}_r$, we note that any valid $(t_1, \ldots, t_n, r)$ index code satisfies

$$H(X_i|Y, X(A_i)) = 0, \quad i \in [n].$$

Hence,

$$rR_i \le t_i = H(X_i) = I(X_i; Y|X(A_i)), \quad i \in [n].$$

By identifying $U_i = X_i$, $i \in [n]$, and $V = Y$, the cardinalities of which are all upper bounded by 2^r, we can conclude that

$$R_i \le \frac{1}{r} I(U_i; V|U(A_i)), \quad i \in [n],$$

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for some $p(u_1)\cdots p(u_n)$ and $v = f(u_1, \ldots, u_n)$ such that the cardinalities are bounded by 2^r. This completes the proof of Theorem 3.2.

Remark 3.2. The rate bounds in (3.15) can be replaced by

$$R_i \leq \frac{I(U_i; V|U(A_i))}{H(V)}, \quad i \in [n].$$

The proof of the converse follows trivially from that of Theorem 3.2 since $H(V) \leq r$. For the proof of achievability, we can compress the V^k sequence *losslessly* in $kH(V)$ bits with vanishing probability of error, and conclude from Lemma 1.2 that this encoding error does not affect the capacity region.

The characterization of the capacity region in Theorem 3.2, in light of the preceding converse proof, is essentially automatic. As in the graph-theoretic characterization in Theorem 3.1, the union $\cup_r \mathscr{C}_r$ consists of an increasing and convergent sequence of inner bounds on the capacity region, and therefore is not computable. Such expressions are often referred to as *multiletter characterizations* in network information theory.

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4

Structural Properties of Index Coding Capacity

In this section, we discuss certain structures in the index coding problem that allow the problem to be decomposed into subproblems and its capacity to be characterized via the capacities of those subproblems.

4.1 Generalized Lexicographic Products

Consider the index coding problems in Figure 4.1, in which the side information graphs can be decomposed into two parts with no, one-way, complete one-way, or complete two-way interaction. As we will show in the next section, the broadcast rate for the first three cases is the sum of the broadcast rates of the subproblems, while the broadcast rate for the last case is the maximum of the broadcast rates of the subproblems. All these cases can be unified by a new notion of graph product that allows a natural decomposition of some index coding problems into subproblems. Suppose that G_0 is a graph with vertex set $V(G_0) = [m]$ and that $G_1, \ldots, G_m$ are arbitrary graphs on distinct vertices. The *generalized lexicographic product* [130, 74] $G = G_0 \circ (G_1, \ldots, G_m)$ is

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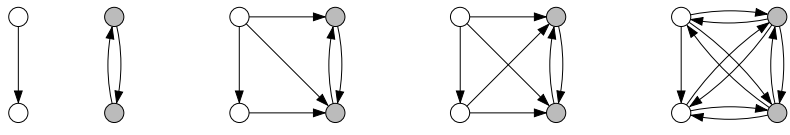


Figure 4.1: Side information graphs with (a) no interaction, (b) one-way interaction, (c) complete one-way interaction, and (d) complete two-way interaction among the two parts (white and gray).

defined by $V(G) = \cup_{i=1}^{m} V(G_i)$ and $E(G)$ consisting of (i, j) such that

$$i, j \in V(G_k) \text{ for some } k \text{ and } (i, j) \in E(G_k)$$

or

$$i \in V(G_k), j \in V(G_l) \text{ for some } k \neq l \text{ and } (k, l) \in E(G_0).$$

In other words, vertex $i \in V(G_0)$ is replaced by a copy of G_i and every vertex in the copy of G_k is connected to every vertex in the copy of G_l according to $E(G_0)$ (see Figure 4.2). This notion of generalized lexicographic product extends that of lexicographic product (see Section 2.4). By relabeling the vertices, $G_0 \circ G_1 = G_0 \circ (G_1^{(1)}, \ldots, G_1^{(m)})$, where $G_1^{(1)}, \ldots, G_1^{(m)}$ are copies of G_1 over disjoint vertex sets.

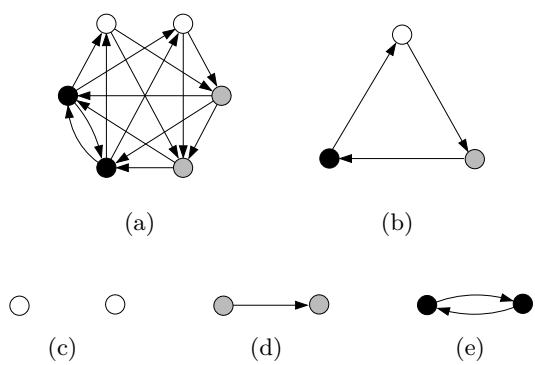


Figure 4.2: (a) A 6-vertex graph that is the generalized lexicographic product $G_0 \circ (G_1, G_2, G_3)$, (b) the 3-vertex graph G_0, (c) the 2-vertex graph G_1, (d) the 2-vertex graph G_2, and (e) the 2-vertex graph G_3.

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4.2 Special Cases

Suppose that G_1 and G_2 are two vertex-induced subgraphs of G such that $V(G_1)$ and $V(G_2)$ partition $V(G) = [n]$ and G has no edge between $V(G_1)$ and $V(G_2)$ (see Figure 4.1(a)). Then G can be viewed as $G_0 \circ (G_1, G_2)$, where G_0 is the two-vertex graph in Figure 4.3(a). Consider a message tuple $x^n = (\mathbf{x}_1, \mathbf{x}_2)$, where $\mathbf{x}_1$ corresponds to the index coding problem G_1 and $\mathbf{x}_2$ corresponds to the index coding problem G_2. Similarly let $z^n = (\mathbf{z}_1, \mathbf{z}_2)$, where $\mathbf{z}_1$ and $\mathbf{z}_2$ correspond to G_1 and G_2, respectively. By the definition of confusability, x^n and z^n are confusable iff they are confusable at some receiver $i \in V(G_1)$ or confusable at some receiver $i \in V(G_2)$. Since there is no edge between G_1 and G_2, these "local" confusability conditions are equivalent to the confusability of $\mathbf{x}_1$ and $\mathbf{z}_1$ for the subproblem G_1 and the confusability of $\mathbf{x}_2$ and $\mathbf{z}_2$ for the subproblem G_2, respectively. Thus, x^n and z^n are confusable for G iff $\mathbf{x}_1$ and $\mathbf{z}_1$ are confusable for G_1 or $\mathbf{x}_2$ and $\mathbf{z}_2$ are confusable for G_2. By the definitions of confusion graph and disjunctive product, $\Gamma_{\mathbf{t}}(G) = \Gamma_{\mathbf{t}}(G_1) \vee \Gamma_{\mathbf{t}}(G_2)$. Now by Lemma 2.4,

$$\log \chi_f(\Gamma_t(G)) = \log \chi_f(\Gamma_t(G_1)) + \log \chi_f(\Gamma_t(G_2)),$$

which, along with Proposition 3.2, implies that

$$\begin{aligned} \beta(G) &= \lim_{t\to\infty} \frac{1}{t} \log \chi_f(\Gamma_t(G)) \\ &= \lim_{t\to\infty} \frac{1}{t} \log \chi_f(\Gamma_t(G_1)) + \lim_{t\to\infty} \frac{1}{t} \log \chi_f(\Gamma_t(G_2)) \\ &= \beta(G_1) + \beta(G_2). \end{aligned} \tag{4.1}$$

In words, the broadcast rate is additive in those of subproblems with no interaction, which is not surprising.

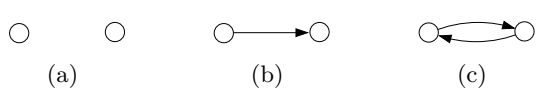


Figure 4.3: (a) A 2-vertex graph with no edge, (b) a 2-vertex graph with one edge, (c) a 2-vertex graph with two edges.

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Example 4.1. The side information graph G shown in Figure 4.1(a), is vertex-partitioned by 2-vertex subgraphs G_1 and G_2 with one edge and two edges, respectively. The subgraphs G_1 and G_2 can be further partitioned by two 1-vertex subgraphs with (complete) one-way and complete two-way interactions, respectively. As we will see shortly, $\beta(G_1) = 1 + 1 = 2$ (see (4.2)) and $\beta(G_2) = \max\{1, 1\} = 1$ (see (4.7)). Therefore, by (4.1) we have $\beta(G) = \beta(G_1) + \beta(G_2) = 3$.

Example 4.2. If $G = (V, E)$ with $E = \emptyset$, then by using (4.1) inductively, we have $\beta(G) = |V|$.

Next, consider a graph G vertex-partitioned by subgraphs G_1 and G_2 such that there exists an edge from every vertex in G_1 to every vertex in G_2 and no edge from G_2 to G_1 (see Figure 4.1(c)) . Then G can be viewed as $G = G_0 \circ (G_1, G_2)$, where G_0 is the 2-vertex graph in Figure 4.3(b). Since every vertex in G_2 has every vertex (message) in G_1 as side information and no vertex in G_1 has any vertex in G_2 as side information, $x^n = (\mathbf{x}_1, \mathbf{x}_2)$ and $z^n = (\mathbf{z}_1, \mathbf{z}_2)$ are confusable for G iff $\mathbf{x}_1$ and $\mathbf{z}_1$ are confusable for G_1, or $\mathbf{x}_1 = \mathbf{z}_1$ and $\mathbf{x}_2$ and $\mathbf{z}_2$ are confusable for G_2. By the definitions of confusion graph and lexicographic product, $\Gamma_{\mathbf{t}}(G) = \Gamma_{\mathbf{t}_1}(G_1) \circ \Gamma_{\mathbf{t}_2}(G_2)$. Thus, by Lemma 2.7 and Proposition 3.2,

$$\log \chi_f(\Gamma_{\mathbf{t}}(G)) = \log \chi_f(\Gamma_{\mathbf{t}_1}(G_1)) + \log \chi_f(\Gamma_{\mathbf{t}_2}(G_2))$$

and

$$\beta(G) = \lim_{t\to\infty} \frac{1}{t} \log \chi_f(\Gamma_t(G)) = \beta(G_1) + \beta(G_2). \tag{4.2}$$

Example 4.3. For the side information graph G shown in Figure 4.1(c) we have $\beta(G) = \beta(G_1) + \beta(G_2) = 3$.

As a generalization of both (4.1) and (4.2), consider a graph G vertex-partitioned by subgraphs G_1 and G_2 for which there exists no edge from G_2 to G_1, while there may be some edges from G_1 to G_2 (see Figure 4.1(b)). Then, $\beta(G)$ can be sandwiched as $\beta(G'') \le \beta(G) \le \beta(G')$, where G' is the graph with no edge between $V(G_1)$ and $V(G_2)$, whereas G'' is the graph for which there is an edge from every vertex in G_1 to every vertex in G_2 but there is no edge from G_2 to G_1. Thus, by (4.1) and (4.2),

$$\beta(G) = \beta(G_1) + \beta(G_2). \tag{4.3}$$

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Example 4.4. For the side information graph G shown in Figure 4.1(b) we have $\beta(G) = \beta(G_1) + \beta(G_2) = 3$.

Now suppose that the graph G can be vertex-partitioned by m subgraphs $G_1, \ldots, G_m$ such that if $i < j$ there exists no edge from G_j to G_i, while there may be some edges from G_i to G_j. Therefore, G can be partitioned into two graphs $(G_1, \ldots, G_{m-1})$ and G_m with no edges from the latter to the former. Hence, by (4.3),

$$\beta(G) = \beta((G_1, \ldots, G_{m-1})) + \beta(G_m).$$

Since $(G_1, \ldots, G_{m-1})$ can also be partitioned by two subgraphs with one-way interaction, by repeating the same argument inductively we have the following.

Proposition 4.1 (One-way interaction)**.** Let G be a side information graph that can be vertex-partitioned by m subgraphs $G_1, \ldots, G_m$ such that if $i < j$ there exists no edge from G_j to G_i, while there may be some edges from G_i to G_j. Then

$$\beta(G) = \beta(G_1) + \cdots + \beta(G_m). \tag{4.4}$$

In particular, if G_0 is an acyclic directed graph with $V(G_0) = [m]$,

$$\beta(G_0 \circ (G_1, \ldots, G_m)) = \beta(G_1) + \cdots + \beta(G_m). \tag{4.5}$$

Remark 4.1. Let G be a graph as described in Proposition 4.1. Equation (4.4) can be generalized to characterize the capacity region $\mathscr{C}$ of the index coding problem G as

$$\mathscr{C} = \left\{ (\rho_1 \mathbf{R}_1, \ldots, \rho_m \mathbf{R}_m) \colon \mathbf{R}_i \in \mathscr{C}_i, i \in [m], \sum_{i=1}^{m} \rho_i \leq 1 \right\}. \tag{4.6}$$

In other words, the capacity region of G is achieved by time division between the optimal coding schemes for subproblems $G_1, \ldots, G_m$.

Example 4.5. For the index coding problem with side information graph G in Figure 4.4 we have $\beta(G) = 4$. In general, if G is acyclic, then $\mathscr{C}(G) = \{(R_1, \ldots, R_n) \colon \sum_{i=1}^{n} R_i \leq 1\}$ and $\beta(G) = |V(G)|$.

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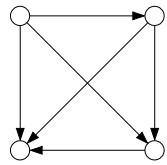


Figure 4.4: A 4-vertex acyclic graph.

Next, consider a graph G that can be vertex-partitioned by subgraphs G_1 and G_2 such that there are edges from every vertex in G_1 to every vertex in G_2 and vice versa. Then G can be viewed as $G = G_0 \circ (G_1, G_2)$, where G_0 is the 2-vertex graph in Figure 4.3(c). Since every vertex in G_1 has every message in G_2 as side information and every vertex in G_2 has every message in G_1 as side information, $x^n = (\mathbf{x}_1, \mathbf{x}_2)$ and $z^n = (\mathbf{z}_1, \mathbf{z}_2)$ are confusable for G iff $\mathbf{x}_1 = \mathbf{z}_1$ and $\mathbf{x}_2$ and $\mathbf{z}_2$ are confusable for G_2, or $\mathbf{x}_2 = \mathbf{z}_2$ and $\mathbf{x}_1$ and $\mathbf{z}_1$ are confusable for G_1. By the definitions of confusion graph and Cartesian product, $\Gamma_{\mathbf{t}}(G) = \Gamma_{\mathbf{t}_1}(G_1) \wedge \Gamma_{\mathbf{t}_2}(G_2)$. Thus, by Lemma 2.6 and (3.6),

$$\chi(\Gamma_{\mathbf{t}}(G)) = \max\{\chi(\Gamma_{\mathbf{t}_1}(G_1)), \chi(\Gamma_{\mathbf{t}_2}(G_2))\}$$

and

$$\beta(G) = \lim_{t\to\infty} \frac{1}{t} \log \chi(\Gamma_t(G)) = \max\{\beta(G_1), \beta(G_2)\}. \tag{4.7}$$

Example 4.6. For the side information graph G shown in Figure 4.1(d) we have $\beta(G) = \max\{\beta(G_1), \beta(G_2)\} = 2$.

Now suppose that G_0 is the complete graph with m vertices, i.e., $E(G_0) = \{(i, j) : i \neq j \in [m]\}$. Then, the graph $G_0 \circ (G_1, \ldots, G_m)$ can be partitioned into two graphs $G_0|_{[m-1]} \circ (G_1, \ldots, G_{m-1})$ and G_m with complete two-way interaction among the two parts. Hence, by (4.7),

$$\beta(G_0 \circ (G_1, \ldots, G_m)) = \max\{\beta(G_0|_{[m-1]} \circ (G_1, \ldots, G_{m-1})), \beta(G_m)\}.$$

Since $G_0|_{[j]}$, $j \in [m-1]$, is also complete, by repeating the same argument we have the following.

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Proposition 4.2 (Complete two-way interaction)**.** Let G_0 be a complete graph with m vertices. Then

$$\beta(G_0 \circ (G_1, \ldots, G_m)) = \max\{\beta(G_1), \ldots, \beta(G_m)\}. \tag{4.8}$$

Remark 4.2. Let G_0 be a complete graph with m vertices. Equation (4.8) can be generalized to characterize the capacity region $\mathscr{C}$ of the index coding problem $G_0 \circ (G_1, \ldots, G_m)$ as

$$\mathscr{C} = \{(\mathbf{R}_1, \ldots, \mathbf{R}_m) \colon \mathbf{R}_i \in \mathscr{C}_i, i \in [m]\}. \tag{4.9}$$

In other words, the capacity region of $G_0 \circ (G_1, \ldots, G_m)$ is achieved by simultaneously using the optimal coding schemes for $G_1, \ldots, G_m$.

4.3 Capacity of a Generalized Lexicographic Product

We now consider the generalized lexicographic product $G = G_0 \circ (G_1, \ldots, G_m)$ with an arbitrary directed graph G_0 with m vertices.

Theorem 4.1 (Generalized lexicographic product [15])**.** Let G_0 be the side information graph of an index coding problem with m messages and capacity region $\mathscr{C}_0$. Let $G_1, \ldots, G_m$ be the side information graphs of m index coding problems with capacity regions $\mathscr{C}_1, \ldots, \mathscr{C}_m$, respectively. Then the capacity region of the index coding problem with side information graph $G = G_0 \circ (G_1, \ldots, G_m)$ is

$$\begin{aligned} \mathscr{C}(G) &= \mathscr{C}_0 \circ (\mathscr{C}_1, \ldots, \mathscr{C}_m) \\ &:= \{(\rho_1 \mathbf{R}_1, \ldots, \rho_m \mathbf{R}_m) \colon \rho \in \mathscr{C}_0, \mathbf{R}_i \in \mathscr{C}_i, i \in [m]\}. \end{aligned} \tag{4.10}$$

Remark 4.3. Since $\mathscr{C}_0, \mathscr{C}_1, \ldots \mathscr{C}_m$ are compact, so is the RHS of (4.10).

Remark 4.4. If $\mathscr{C}_0, \mathscr{C}_1, \ldots, \mathscr{C}_m$ are polytopes of the form

$$\begin{aligned} \mathscr{C}_0 &= \{\rho : T_0 \rho \leq \mathbf{1}\}, \\ \mathscr{C}_i &= \{\mathbf{R}_i : T_i \mathbf{R}_i \leq \mathbf{1}\}, \quad i \in [m], \end{aligned}$$

where $\mathbf{1} = (1, \ldots, 1)$ and the inequalities hold elementwise, then $\mathscr{C}$ is also a polytope in $\mathbf{R} = (\mathbf{R}_1, \ldots, \mathbf{R}_m)$ characterized by Fourier–Motzkin elimination of m variables $\rho = (\rho_1, \ldots, \rho_m)$ from the linear inequalities

$$\begin{aligned} T_0 \rho &\leq \mathbf{1}, \\ T_i \mathbf{R}_i &\leq \rho_i \mathbf{1}, \quad i \in [m]. \end{aligned}$$

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Remark 4.5. Theorem 4.1 can be specialized to the broadcast rate of $G = G_0 \circ (G_1, \ldots, G_m)$. If $\boldsymbol{\beta} = (\beta(G_1), \ldots, \beta(G_m))$, then

$$\beta(G) = \frac{1}{C_0(\boldsymbol{\beta})} \leq \max_{i \in [m]} \beta(G_0)\beta(G_i),$$

where $C_0(\boldsymbol{\beta})$ is the $\boldsymbol{\beta}$-directed capacity of the index coding problem G_0 (cf. Section 1.2). This implies Propositions 4.1 and 4.2 for acyclic and complete G_0, respectively.

Setting $G_1, \ldots, G_m$ in Theorem 4.1 to be identical copies of each other, we see that the broadcast rate is multiplicative under the lexicographic product of index coding side information graphs.

Corollary 4.1. For any two directed graphs G_0 and G_1,

$$\beta(G_0 \circ G_1) = \beta(G_0)\beta(G_1).$$

Example 4.7. The graph in Figure 2.1(c) can be viewed as the lexicographic product $G = G_0 \circ G_1$ of two smaller graphs G_0 and G_1 in Figures 2.1(a) and 2.1(b), respectively, with $\beta(G_0) = 2$ and $\beta(G_1) = 2$. Hence, by Theorem 4.1, $\beta(G) = \beta(G_0)\beta(G_1) = 4$.

Achievability of (4.10) follows by constructing a code for problem G by concatenating the index codes for problems $G_1, \ldots, G_m$ as the "inner" codes and the index code for problem G_0 as the "outer" code [15] as illustrated in Figure 4.5. The proof of the converse, which is presented in Appendix 4.A, is based on the information theoretic characterization of the capacity region in Theorem 3.2 and the following observation.

Lemma 4.1 ([15])**.** For any n-message index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G, let

$$(\phi(x^n), \psi_1(y, x(A_1)), \ldots, \psi_n(y, x(A_n)))$$

be a $(t_1, \ldots, t_n, r)$ index code under a *relaxed* decoding condition

$$\psi_i(\phi(x^n), x(A_i)) = x_i, \quad i \in J,$$

for some subset $J \subseteq [n]$ of the messages. Then,

$$\frac{(t_i : i \in J)}{r} \in \mathscr{C}(G|_J).$$

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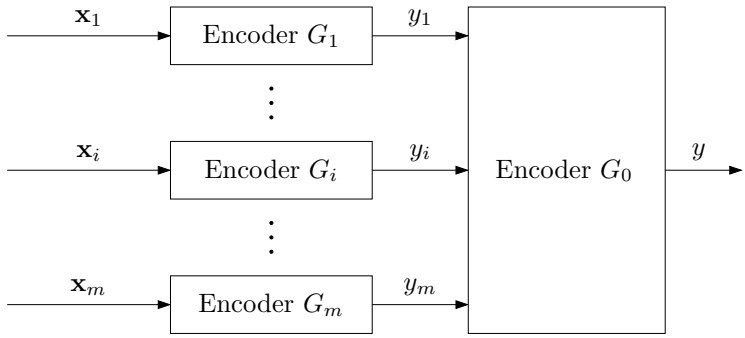


Figure 4.5: Construction of an index code for index coding problem $G_0 \circ (G_1, \ldots, G_m)$ by concatenating the index codes for problems $G_1, \ldots, G_m$ as the inner codes and the index code for problem G_0 as the outer code.

The following corollary, which is a generalization of Remarks 4.1 and 4.2, extends application of Theorem 4.1 beyond side information graphs in the form of a generalized lexicographic product.

Corollary 4.2. For $i = 0, 1, \ldots, m$, let G_i' and G_i'' be side information graphs of index coding problems with capacity regions $\mathscr{C}_i'$ and $\mathscr{C}_i''$, respectively, such that $V(G_i') = V(G_i'')$ and $E(G_i') \subseteq E(G_i'')$. Suppose that $|V(G_0')| = |V(G_0'')| = m$ and let

$$G' = G_0' \circ (G_1', \ldots, G_m')$$

and

$$G'' = G_0'' \circ (G_1'', \ldots, G_m'').$$

Then the capacity region of any index coding problem G such that

$$V(G) = V(G') = V(G'')$$

and

$$E(G') \subseteq E(G) \subseteq E(G'')$$

is bounded as

$$\mathscr{C}_0' \circ (\mathscr{C}_1', \ldots, \mathscr{C}_m') = \mathscr{C}(G') \subseteq \mathscr{C}(G) \subseteq \mathscr{C}(G'') = \mathscr{C}_0'' \circ (\mathscr{C}_1'', \ldots, \mathscr{C}_m'').$$

In particular, if $\mathscr{C}_i' = \mathscr{C}_i'' = \mathscr{C}_i$, $i = 0, 1, \ldots, m$, then

$$\mathscr{C}(G) = \mathscr{C}(G') = \mathscr{C}(G'') = \mathscr{C}_0 \circ (\mathscr{C}_1, \ldots, \mathscr{C}_m).$$

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Example 4.8. Consider the side information graph depicted in Figure 4.6(a), which cannot be viewed as the generalized lexicographic product of smaller graphs. Let G' and G'' be the graphs depicted in Figures 4.6(b) and (c), respectively. Since the graph G satisfies $V(G) = V(G') = V(G'')$ and $E(G') \subseteq E(G) \subseteq E(G'')$, its capacity region is sandwiched between the capacity regions $\mathscr{C}(G')$ and $\mathscr{C}(G'')$. Now the graphs G' and G'' are generalized lexicographic products of smaller graphs as $G' = G'_0 \circ (G'_1, G'_2, G'_3)$ and $G'' = G''_0 \circ (G''_1, G''_2, G''_3)$, where G'_0, G''_0, $G'_1 = G''_1$, G'_2, G''_2, and $G'_3 = G''_3$ are depicted in Figure 4.7. Note that for each $i = 0, 1, 2, 3$, $V(G'_i) = V(G''_i)$ and $E(G'_i) \subseteq E(G''_i)$. Furtheremore, the capacity regions of problems G'_i and G''_i can be shown

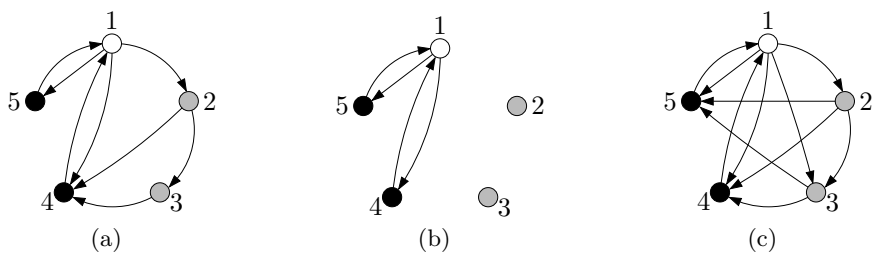


Figure 4.6: (a) The 5-vertex graph G is sandwiched between (b) $G' = G'_0 \circ (G'_1, G'_2, G'_3)$ and (c) $G'' = G''_0 \circ (G''_1, G''_2, G''_3)$.

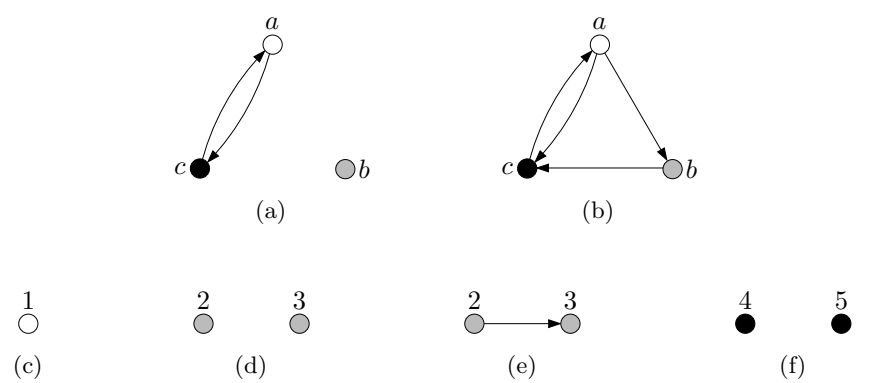


Figure 4.7: (a) The 3-vertex graph G'_0, (b) the 3-vertex graph G''_0, (c) the 1-vertex graph $G'_1 = G''_1$, (d) the 2-vertex graph G'_2, (e) the 2-vertex graph G''_2, and (f) the 2-vertex graph $G'_3 = G''_3$.

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to be identical as

$$\begin{aligned}
\mathscr{C}_0 &= \mathscr{C}_0' = \mathscr{C}_0'' = \{(\rho_a, \rho_b, \rho_c) : \rho_a + \rho_b \leq 1, \rho_b + \rho_c \leq 1\},\\
\mathscr{C}_1 &= \mathscr{C}_1' = \mathscr{C}_1'' = \{R_1 : R_1 \leq 1\},\\
\mathscr{C}_2 &= \mathscr{C}_2' = \mathscr{C}_2'' = \{(R_2, R_3) : R_2 + R_3 \leq 1\},\\
\mathscr{C}_3 &= \mathscr{C}_3' = \mathscr{C}_3'' = \{(R_4, R_5) : R_4 + R_5 \leq 1\}.
\end{aligned}$$

Hence, by Corollary 4.2, the capacity region $\mathscr{C}(G)$ of index coding problem G is equal to $\mathscr{C}(G') = \mathscr{C}(G'')$, which is the set of all rate tuples $(R_1, R_2, R_3, R_4, R_5)$ such that

$$\begin{aligned}
R_1 &\leq \rho_a,\\
R_2 + R_3 &\leq \rho_b,\\
R_4 + R_5 &\leq \rho_c
\end{aligned}$$

for some (ρ_a, ρ_b, ρ_c) such that $\rho_a + \rho_b \leq 1$ and $\rho_b + \rho_c \leq 1$. By Remark 4.4, this region simplifies to the set of $(R_1, \ldots, R_5)$ such that

$$\begin{aligned}
R_1 + R_2 + R_3 &\leq 1,\\
R_2 + R_3 + R_4 + R_5 &\leq 1.
\end{aligned}$$

So far, we considered the union and product operations of side information graphs that act on the capacity region additively and multiplicatively. This allows the capacity region of a larger index coding problem in terms of those of smaller problems. There are other simple operations that may act on the capacity region in some canonical manners. We recall from Section 2.1 the definitions of graph transpose G^T, namely,

$$(i, j) \in E(G^T) \text{ iff } (j, i) \in E(G),$$

and graph complement $\overline{G}$, namely,

$$(i, j) \in E(\overline{G}) \text{ iff } (i, j) \notin E(G).$$

Open Problem 4.1. Show that

$$\beta(G^T) = \beta(G)$$

or provide a counterexample. Repeat the same question for $\mathscr{C}(G^T)$.

Open Problem 4.2. Is $\mathscr{C}(\overline{G})$ characterizable in terms of $\mathscr{C}(G)$?

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4.A Proof of the Converse for Theorem 4.1

Let $G = G_0 \circ (G_1, \ldots, G_m)$ and $V(G) = V(G_1) \cup \cdots \cup V(G_m)$. Let A_j denote the side information set of receiver $j \in V(G)$ for the index coding problem G, and let A'_i denote the side information set of receiver $i \in [m]$ for the index coding problem G_0. In this notation, if $j \in V(G_i)$ for some $i \in [m]$, then by the definition of generalized lexicographic product, the side information set of receiver j can be decomposed as

$$A_j = (A_j \cap V(G_i)) \cup \Big(\bigcup_{l \in A'_i} V(G_l) \Big),$$

where the first term denotes the side information from within the subgraph G_i and the second term denotes the side information from other subgraphs. As earlier, we write $\mathbf{x}_i$ for $(x_j : j \in V(G_i))$ and $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_m)$. We also write $\mathbf{x}(A'_i)$ for

$$(\mathbf{x}_l : l \in A'_i) = (x_j : j \in \cup_{l \in A'_i} V(G_l)).$$

For any $(\mathbf{t}_1, \ldots, \mathbf{t}_m, r)$ index code for $G = G_0 \circ (G_1, \ldots, G_m)$, we argue that the corresponding rate tuple can be factored as

$$\frac{\mathbf{t}_i}{r} = \frac{s_i}{r} \frac{\mathbf{t}_i}{s_i}, \quad i \in [m],$$

for some $(s_1, \ldots, s_m)$, so that

$$\left(\frac{s_1}{r}, \ldots, \frac{s_m}{r} \right) \in \mathscr{C}_0, \tag{4.11a}$$

and for any $\epsilon > 0$,

$$\frac{(1-\epsilon)\mathbf{t}_i}{s_i} \in \mathscr{C}_i, \quad i \in [m]. \tag{4.11b}$$

Consequently,

$$(1-\epsilon) \left(\frac{\mathbf{t}_1}{r}, \ldots, \frac{\mathbf{t}_m}{r} \right) \in \mathscr{C}_0 \circ (\mathscr{C}_1, \ldots, \mathscr{C}_m).$$

Since $\epsilon > 0$ is arbitrary, this would establish the desired proof of the converse.

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We now verify (4.11) for an appropriate choice of $(s_1, \ldots, s_m)$. Let $Y = \phi(\mathbf{X}_1, \ldots, \mathbf{X}_m) \in \{0,1\}^r$ be the encoder output of the given index code for independent and uniformly distributed messages, which induces the joint distribution of the form

$$p(\mathbf{x}_1, \ldots, \mathbf{x}_m, y) = p(\mathbf{x}_1) \cdots p(\mathbf{x}_m) p(y|\mathbf{x}_1, \ldots, \mathbf{x}_m) \tag{4.12}$$

such that Y is a function of $(\mathbf{X}_1, \ldots, \mathbf{X}_m)$ and X_j is a function of $(Y, X(A_j))$ for every $j \in V(G)$. Now let

$$s_i = I(\mathbf{X}_i; Y | \mathbf{X}(A_i')), \quad i \in [m],$$

where the mutual information is evaluated under the joint distribution in (4.12). Then, by Theorem 3.2 (with $U_i = \mathbf{X}_i$ and $V = Y$), we have (4.11a).

For (4.11b), we can adapt random coding for rate–distortion theory [139] and joint typicality encoding [61, Sec. 3.6] over k copies of $(\mathbf{X}_1, \ldots, \mathbf{X}_m, Y)$ to establish the following.

Lemma 4.2 ([15]). Let $\epsilon > 0$ and $s_i = I(\mathbf{X}_i; Y|\mathbf{X}(A_i'))$. Then there exist mappings

$$\phi_i'(\mathbf{x}^k) \in \{0,1\}^{ks_i/(1-\epsilon)}, \quad i \in [m],$$

and

$$\psi_j'(y_i', x^k(A_j)) \in \{0,1\}^{kt_j}, \quad j \in V(G_i),$$

such that

$$\psi_j'(\phi_i'(\mathbf{x}^k), x^k(A_j)) = x_j^k, \quad i \in [m],\ j \in V(G_i),$$

for k sufficiently large.

In words, the mappings $\phi_i'(\mathbf{x}^k)$ and $\psi_j'(y_i', x^k(A_j))$, $j \in V(G_i)$, form a $(k\mathbf{t}_1, \ldots, k\mathbf{t}_m, ks_i/(1-\epsilon))$ index code for G under the relaxed decoding condition that only $\mathbf{x}_i^k = (x_j^k : j \in V(G_i))$ is required to be recovered correctly. Hence, by applying Lemma 4.1 for each $i \in [m]$, we can conclude that (4.11b) holds. This completes the proof of Theorem 4.1.

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5

Performance Bounds

We discuss several lower bounds on the broadcast rate and outer bounds on the capacity region. These bounds stipulate the limits, with varying degrees of tightness, on the performance of the optimal index code.

5.1 MAIS Bound

The simplest outer bound on the capacity region is a direct corollary of the structural properties established in the previous section. First Lemma 4.1 implies that for any achievable rate tuple $\mathbf{R} = (R_1, \ldots, R_n)$ of an n-message index coding problem G and any subgraph G' induced by $V' \subseteq V(G)$, the subtuple $(R_i : i \in V')$ is included in the capacity region of the index coding problem G'. This fact, together with Remark 4.1 and Example 4.5, implies that for any *acyclic* vertex induced subgraph G' of the index coding problem G, we have $\mathscr{C}(G) \subseteq \{\mathbf{R}\colon \sum_{i \in V(G')} R_i \leq 1\}$. Considering all acyclic subgraphs G' of G, we establish the following outer bound on the capacity region in general and lower bound on the broadcast rate in particular.

Theorem 5.1 (Maximal acyclic induced subgraph (MAIS) bound [23])**.** The capacity region $\mathscr{C}(G)$ of the index coding problem $(i|A_i)$, $i \in [n]$,

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with side information graph G is included in the rate region $\mathscr{R}_{\mathrm{MAIS}}(G)$ that consists of all nonnegative rate tuples $(R_1, \ldots, R_n)$ satisfying

$$\sum_{i \in J} R_i \leq 1 \tag{5.1}$$

for all $J \subseteq [n]$ such that $G|_J$ is acyclic. In particular, the broadcast rate is lower bounded as

$$\beta_{\mathrm{MAIS}}(G) := \max_{J \subseteq V(G): G|_J \text{ is acyclic}} |J| \leq \beta(G). \tag{5.2}$$

Remark 5.1. The inequality in (5.1) is active iff J is *maximal*, namely, there exists no superset $J' \supsetneq J$ such that $G|_{J'}$ is acyclic.

Remark 5.2. Since every independent set is acyclic, the MAIS bound in (5.2) can be relaxed as $\alpha(G) \leq \beta(G)$.

Application of the MAIS bound is illustrated by the following.

Example 5.1. For the index coding problem depicted in Figure 5.1, the subgraphs $G|_{\{1\}}$ and $G|_{\{2,3\}}$ are acyclic. Hence, $\beta(G) \geq \beta_{\mathrm{MAIS}}(G) = 2$ and $\mathscr{C}(G) \subseteq \mathscr{R}_{\mathrm{MAIS}}(G)$, where $\mathscr{R}_{\mathrm{MAIS}}(G)$ consists of all rate tuples (R_1, R_2, R_3) satisfying

$$\begin{aligned} R_1 &\leq 1, \\ R_2 + R_3 &\leq 1. \end{aligned}$$

Similar to the broadcast rate β, the MAIS bound β_{MAIS} is multiplicative under the lexicographic product of side information graphs. The proof is relegated to Appendix 5.A.

Proposition 5.1. For any two side information graphs G_0 and G_1, $\beta_{\mathrm{MAIS}}(G_0 \circ G_1) = \beta_{\mathrm{MAIS}}(G_0)\beta_{\mathrm{MAIS}}(G_1)$.

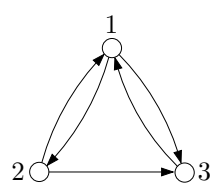


Figure 5.1: The index coding problem $(1|2,3),(2|1),(3|1,2)$.

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The MAIS bound is not tight in general (see Example 5.2 in the next section). Using Corollary 4.1 and Proposition 5.1, we can show that the gap between the MAIS bound and the broadcast rate of the index coding problem can be magnified to a multiplicative factor that grows polynomially in the number of messages of the problem.

Proposition 5.2. Let G_0 be the side information graph of an index coding problem with n_0 messages for which $\beta(G_0)/\beta_{\text{MAIS}}(G_0) = \rho > 1$, and let $G = G_0^{\circ k}$ with $n = n_0^k$ vertices. Then

$$\frac{\beta(G)}{\beta_{\text{MAIS}}(G)} = \frac{(\beta(G_0))^k}{(\beta_{\text{MAIS}}(G_0))^k} = \rho^k = n^{\log_{n_0}(\rho)}.$$

5.2 Polymatroidal Bound

Let $(R_1, \ldots, R_n)$ be an achievable rate tuple for the index coding problem $(i|A_i)$, $i \in [n]$. Then, there exists a $(t_1, \ldots, t_n, r)$ index code with encoding function ϕ and decoding functions ψ_i, $i \in [n]$, such that

$$R_i \leq \frac{t_i}{r}.$$

Let X_i be the uniform random variable over $\{0,1\}^{t_i}$ that represents message i. Then,

$$H(X_i) = t_i, \quad i \in [n].$$

Moreover, by the independence of the random variables $X_1, \ldots, X_n$,

$$H(X_1, \ldots, X_n) = \sum_{i=1}^{n} H(X_i).$$

Let $Y = \phi(X_1, \ldots, X_n)$ be the random variable over $\{0,1\}^r$ that represents the encoder output. Then,

$$H(Y) \leq r$$

and

$$H(Y|X_1, \ldots, X_n) = 0.$$

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Since receiver i can recover its desired message based on the received codeword and its side information, we also have the following decodability conditions

$$H(X_i|Y, X(A_i), X(C_i)) = H(X_i|Y, X(A_i)) = 0, \quad i \in [n], \tag{5.3}$$

for every $C_i \subseteq B_i = [n] \setminus (A_i \cup \{i\})$. Now consider

$$\begin{aligned} R_i &\leq \frac{t_i}{r} \\ &= \frac{1}{r} H(X_i) \\ &= \frac{1}{r} [H(X_i) - H(X_i|Y, X(A_i))] \\ &= \frac{1}{r} [H(X_i|X(A_i)) - H(X_i|Y, X(A_i))] \\ &= \frac{1}{r} I(X_i; Y|X(A_i)) \\ &= \frac{1}{r} [H(Y|X(A_i)) - H(Y|X_i, X(A_i))] \\ &\leq \frac{1}{H(Y)} [H(Y|X(A_i)) - H(Y|X_i, X(A_i))]. \end{aligned} \tag{5.4}$$

Define a set function $f : 2^{[n]} \to [0, 1]$ as

$$f(J) = \frac{H(Y|X(\bar{J}))}{H(Y)}, \quad J \subseteq [n], \tag{5.5}$$

where $\bar{J} = [n] \setminus J$. Then (5.4) can be rewritten as

$$R_i \leq f(B_i \cup \{i\}) - f(B_i), \quad i \in [n].$$

In the following, we explore a few properties of the set function f defined in (5.5). First,

$$f(\emptyset) = \frac{H(Y|X([n]))}{H(Y)} = 0 \tag{5.6}$$

and

$$f([n]) = \frac{H(Y)}{H(Y)} = 1. \tag{5.7}$$

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Second, since conditioning reduces entropy, for any $k \notin J \subseteq [n]$,

$$\begin{aligned} f(J) &= H(Y|X(\bar{J})) \\ &\leq H(Y|X(\bar{J} \setminus \{k\})) \\ &= f(J \cup \{k\}). \end{aligned} \tag{5.8}$$

Third, for any $J, K \subseteq [n]$,

$$\begin{aligned} &H(Y)(f(J \cup K) - f(J)) \\ &= H(Y|X(\bar{J} \cap \overline{K})) - H(Y|X(\bar{J})) \\ &= I(Y; X(\bar{J} \setminus \overline{K})|X(\bar{J} \cap \overline{K})) \\ &= H(X(\bar{J} \setminus \overline{K})|X(\bar{J} \cap \overline{K})) - H(X(\bar{J} \setminus \overline{K})|Y, X(\bar{J} \cap \overline{K})) \\ &= H(X(\bar{J} \setminus \overline{K})|X(\overline{K})) - H(X(\bar{J} \setminus \overline{K})|Y, X(\bar{J} \cap \overline{K})) \\ &\leq H(X(\bar{J} \setminus \overline{K})|X(\overline{K})) - H(X(\bar{J} \setminus \overline{K})|Y, X(\overline{K})) \\ &= I(Y; X(\bar{J} \setminus \overline{K})|X(\overline{K})) \\ &= H(Y|X(\overline{K})) - H(Y|X(\bar{J} \cup \overline{K})) \\ &= H(Y)(f(K) - f(J \cap K)), \end{aligned}$$

or equivalently,

$$f(J \cup K) + f(J \cap K) \leq f(J) + f(K), \tag{5.9}$$

which shows that f is also *submodular* [129, 72]. The submodularity can be equivalently expressed as

$$f(J) + f(J \cup \{i\} \cup \{k\}) \leq f(J \cup \{i\}) + f(J \cup \{k\}), \tag{5.10}$$

for every $J \subseteq [n]$ and distinct $i, k \notin J$ [129, Theorem 44.1]. Fourth, by the decodability conditions (5.3) and the independence of $X_1, \ldots, X_n$,

$$\begin{aligned} &H(Y)(f(B_i \cup \{i\}) - f(B_i)) \\ &= H(Y|X(A_i)) - H(Y|X_i, X(A_i)) \\ &= I(X_i; Y|X(A_i)) \\ &= I(X_i; Y|X(A_i), X(B_i \setminus C_i)) \\ &= H(Y|X(A_i), X(B_i \setminus C_i)) - H(Y|X_i, X(A_i), X(B_i \setminus C_i)) \\ &= H(Y)(f(C_i \cup \{i\}) - f(C_i)) \end{aligned} \tag{5.11}$$

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for every $C_i \subseteq B_i$. By the submodularity in (5.9), we have

$$f(B_i \cup \{i\}) - f(B_i) \leq f(C_i \cup \{i\}) - f(C_i) \leq f(\{i\}), \quad C_i \subseteq B_i.$$

Hence, the condition in (5.11) can be expressed succinctly as

$$f(B_i \cup \{i\}) - f(B_i) = f(\{i\}). \tag{5.12}$$

In summary, if the rate tuple $(R_1, \ldots, R_n)$ is achievable, then $R_i \leq f(B_i \cup \{i\}) - f(B_i)$, $i \in [n]$, for some submodular nondecreasing set function f satisfying (5.6)–(5.12). This yields an outer bound on the capacity region (a lower bound on the broadcast rate) referred to as the polymatroidal bound. The name follows the polymatroidal axioms used in its expression, but the corresponding rate region is not a polymatroid in general. This bound has been introduced in [165], [56], and [29] in the context of distributed source coding, network coding, and index coding, respectively.

Theorem 5.2 (Polymatroidal bound)**.** The capacity region $\mathscr{C}(G)$ of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G is included in $\mathscr{R}_{\mathrm{PM}}(G)$ that consists of all rate tuples $(R_1, \ldots, R_n)$ satisfying

$$R_i \leq f(B_i \cup \{i\}) - f(B_i), \quad i \in [n], \tag{5.13a}$$

for some set function $f : 2^{[n]} \to [0, 1]$ such that

$$f(\emptyset) = 0, \tag{5.13b}$$

$$f([n]) = 1, \tag{5.13c}$$

$$f(J) \leq f(J \cup \{k\}), \qquad k \notin J \subseteq [n], \tag{5.13d}$$

$$f(J) + f(J \cup \{i\} \cup \{k\}) \leq f(J \cup \{i\}) + f(J \cup \{k\}), \quad i, k \notin J \subseteq [n], \tag{5.13e}$$

$$f(\{i\}) = f(B_i \cup \{i\}) - f(B_i), \quad i \in [n]. \tag{5.13f}$$

Remark 5.3. Given an index coding problem, the polymatroidal bound $\mathscr{R}_{\mathrm{PM}}$ is computed by eliminating the f variables from (5.13a)–(5.13f) through Fourier–Motzkin elimination [61, Appendix D]. A looser bound can be computed in lower complexity by solving multiple linear programs corresponding to different weighted sums of $R_1, \ldots, R_n$, for example, all linear combinations with 0–1 coefficients.

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Corollary 5.1. For any index coding problem G, the broadcast rate is lower bounded as

$$\beta_{\mathrm{PM}}(G) := \min_{f} \max_{i\in[n]} \frac{1}{f(B_i \cup \{i\}) - f(B_i)} \leq \beta(G),$$

where the minimum is over all set functions f satisfying (5.13b)–(5.13f).

Remark 5.4. The quantity $1/\beta_{\mathrm{PM}}(G)$ is the solution to the following linear program with $2^n + 1$ variables:

$$\begin{aligned} \text{maximize} \quad & R \\ \text{subject to} \quad & R \leq f(B_i \cup \{i\}) - f(B_i), \quad i \in [n], \\ & f \text{ satisfies (5.13b)–(5.13f).} \end{aligned}$$

The last inequality (5.13f) in the polymatroidal bound is rather technical, but it is not known if it is inactive and can be removed to result in the same bound.

Open Problem 5.1. Find the minimal set of inequalities from (5.13) to characterize the polymatroidal bound in general.

The polymatroidal bound is tighter than the MAIS bound.

Proposition 5.3. For any index coding problem G,

$$\mathscr{R}_{\mathrm{PM}}(G) \subseteq \mathscr{R}_{\mathrm{MAIS}}(G)$$

and consequently $\beta_{\mathrm{MAIS}}(G) \leq \beta_{\mathrm{PM}}(G)$.

The proof is presented in Appendix 5.B. The inclusion in Proposition 5.3 can be strict as illustrated by the following.

Example 5.2. For the index coding problem depicted in Figure 5.2, we have $\beta_{\mathrm{MAIS}} = 2$ and the MAIS outer bound is characterized by

$$\begin{aligned} &R_1 + R_3 \leq 1, \quad R_1 + R_4 \leq 1, \quad R_2 + R_4 \leq 1, \\ &R_2 + R_5 \leq 1, \quad R_3 + R_5 \leq 1. \end{aligned} \tag{5.14}$$

However, it can be shown that the polymatroidal bound yields the tighter bound $\beta_{\mathrm{PM}} = 5/2$ on the broadcast rate and the polymatroidal outer bound $\mathscr{R}_{\mathrm{PM}}$ is characterized by

$$R_1 + R_2 + R_3 + R_4 + R_5 \leq 2 \tag{5.15}$$

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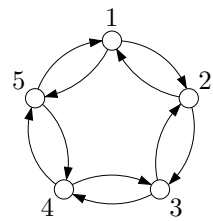


Figure 5.2: The index coding problem $(1|2,5),(2|1,3),(3|2,4),(4|3,5),(5|1,4)$.

in addition to the five inequalities in (5.14). In Section 6.3, we will see that for this example, the polymatroidal bound is tight, i.e., $\beta = 5/2$ and the capacity region $\mathscr{C}$ is characterized by (5.14) and (5.15).

Example 5.3. For the index coding problem depicted in Figure 5.3, the MAIS outer bound is characterized by

$$\begin{aligned} &R_1 + R_3 \leq 1, \qquad R_1 + R_4 \leq 1, \qquad R_2 + R_4 \leq 1, \\ &R_2 + R_5 \leq 1, \qquad R_3 + R_4 \leq 1, \qquad R_3 + R_5 \leq 1, \end{aligned}$$

while the polymatroidal outer bound has the additional inequality

$$R_1 + R_2 + R_3 + R_4 + R_5 \leq 2.$$

Again the capacity region can be shown to coincide with the polymatroidal outer bound.

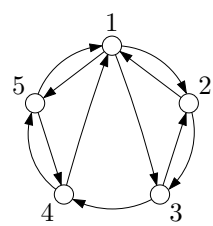


Figure 5.3: The index coding problem $(1|2,4,5),(2|1,3),(3|1,2),(4|3,5),(5|1,4)$.

Blasiak, Kleinberg, and Lubetzky [29] showed that the polymatroidal bound satisfies the following structural property under the lexicographic product (cf. Corollary 4.1 and Proposition 5.1).

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Proposition 5.4. For any pair of side information graphs G_0 and G_1,

$$\beta_{\mathrm{PM}}(G_0 \circ G_1) \geq \beta_{\mathrm{PM}}(G_0)\beta_{\mathrm{PM}}(G_1).$$

Open Problem 5.2. Does the inequality in Proposition 5.4 hold with equality?

Example 5.2 shows that there is a gap between the polymatroidal bound and the MAIS bound. Using Propositions 5.1 and 5.4, this gap can be magnified to a multiplicative factor that grows polynomially in the number of messages of the problem, which also implies Proposition 5.2 indirectly.

Proposition 5.5 ([29]). Let G_0 be the side information graph of an n_0-message index coding problem for which $\beta_{\mathrm{PM}}(G_0)/\beta_{\mathrm{MAIS}}(G_0) = \rho > 1$ and let $G = G_0^{\circ k}$ with $n = n_0^k$ vertices. Then

$$\frac{\beta_{\mathrm{PM}}(G)}{\beta_{\mathrm{MAIS}}(G)} \geq \frac{(\beta_{\mathrm{PM}}(G_0))^k}{(\beta_{\mathrm{MAIS}}(G_0))^k} = \rho^k = n^{\log_{n_0}(\rho)}.$$

5.3 Information Inequalities and Index Coding

For any joint distribution for the collection of n random variables $X_1, \ldots, X_n$, $(2^n - 1)$ entropies $H(X(J))$, $\emptyset \neq J \subseteq [n]$, satisfy

$$\begin{aligned} &H(X(J \cup K \cup L)) - H(X(K \cup L)) \\ &\quad \leq H(X(J \cup K)) - H(X(K)), \quad J, K, L \subseteq [n], \end{aligned} \tag{5.16a}$$

or equivalently,

$$I(X(J); X(L)|X(K)) \geq 0. \tag{5.16b}$$

Any convex combination of these linear inequalities is a valid linear inequality on $H(X(J))$, $J \subseteq [n]$, and is referred to as a *Shannon-type* inequality. Surprisingly, there are some linear inequalities on the entropies that cannot be written as such convex combinations. These inequalities are referred to as *non-Shannon-type* inequalities, including the following example discovered by Zhang and Yeung [170]:

$$\begin{aligned} &3H(X_1, X_3) + 3H(X_1, X_4) + 3H(X_3, X_4) + H(X_2, X_3) + H(X_2, X_4) \\ &\quad \geq 2H(X_3) + 2H(X_4) + H(X_1, X_2) + H(X_1) \\ &\qquad + H(X_2, X_3, X_4) + 4H(X_1, X_3, X_4). \end{aligned} \tag{5.17}$$

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Matúš [110] showed that even for only four random variables, there are infinitely many independent non-Shannon-type information inequalities.

Let $(R_1, \ldots, R_n)$ be an achievable rate tuple for the index coding problem $(i|A_i)$, $i \in [n]$. Then, there exists a $(t_1, \ldots, t_n, r)$ index code with encoding function ϕ and decoding functions ψ_i, $i \in [n]$, such that

$$R_i \leq \frac{t_i}{r}. \tag{5.18}$$

As discussed in the previous section, a given $(t_1, \ldots, t_n, r)$ index code induces $n+1$ random variables X_i, $i \in [n]$, and $X_0 = Y = \phi(X_1, \ldots, X_n)$. One can form an outer bound on the capacity region (a lower bound on the broadcast rate) of the index coding problem by considering all Shannon-type inequalities that hold for $(2^{n+1} - 1)$ entropies of these $n + 1$ random variables.

Define a set function $h : 2^{\{0\}\cup[n]} \to \mathbb{R}_{\geq 0}$ as

$$h(J) = H(X(J)), \quad J \subseteq \{0\} \cup [n]. \tag{5.19}$$

Noting that $H(X_i) = t_i, i \in [n]$, and $H(X_0) \leq r$, we can rewrite (5.18) as

$$R_i \leq \frac{h(\{i\})}{h(\{0\})}, \quad i \in [n].$$

The independence of the random variables $X_1, \ldots, X_n$ can be expressed as

$$h([n]) = H(X_1, \ldots, X_n) = \sum_{i=1}^{n} H(X_i) = \sum_{i=1}^{n} h(\{i\}). \tag{5.20}$$

Since $X_0 = \phi(X_1, \ldots, X_n)$, we have $H(X_0|X_1, \ldots, X_n) = 0$, or equivalently,

$$h(\{0\} \cup [n]) = h([n]). \tag{5.21}$$

The decodability condition at each receiver $i \in [n]$ can be expressed as $H(X_i|Y, X(A_i)) = 0$, or equivalently,

$$h(\{0\} \cup \{i\} \cup A_i) = h(\{0\} \cup A_i), \quad i \in [n]. \tag{5.22}$$

The conditions in (5.20)–(5.22) embody the problem formulation of index coding. In addition, any entropy function satisfies (5.16), namely,

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it is nonnegative (which is already specified in the range of the h function), monotonic, i.e.,

$$h(J) \le h(J \cup \{k\}), \quad k \notin J \subseteq \{0\} \cup [n], \tag{5.23}$$

and submodular [129, Theorem 44.1], i.e.,

$$h(J) + h(J \cup \{i\} \cup \{k\}) \le h(J \cup \{i\}) + h(J \cup \{k\})$$

for every distinct $i, k \notin J \subseteq \{0\} \cup [n]$. Summarizing these conditions yields the following outer bound on the capacity region and lower bound on the broadcast rate based on Shannon-type inequalities.

Theorem 5.3. The capacity region $\mathscr{C}(G)$ of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G is included in $\mathscr{R}_{\mathrm{Sh}}(G)$ that consists of all rate tuples $(R_1, \ldots, R_n)$ satisfying

$$R_i \le \frac{h(\{i\})}{h(\{0\})}, \quad i \in [n], \tag{5.24a}$$

for some set function $h : 2^{\{0\}\cup[n]} \to \mathbb{R}_{\ge 0}$ such that

$$h(\emptyset) = 0, \tag{5.24b}$$

$$h([n]) = \sum_{i=1}^{n} h(\{i\}), \tag{5.24c}$$

$$h(\{0\} \cup [n]) = h([n]), \tag{5.24d}$$

$$h(\{0\} \cup \{i\} \cup A_i) = h(\{0\} \cup A_i), \quad i \in [n], \tag{5.24e}$$

$$h(J) \le h(J \cup \{k\}), \quad k \notin J \subseteq \{0\} \cup [n], \tag{5.24f}$$

$$h(J) + h(J \cup \{i\} \cup \{k\}) \le h(J \cup \{i\}) + h(J \cup \{k\}), \quad i, k \notin J \subseteq \{0\} \cup [n]. \tag{5.24g}$$

Remark 5.5. Under (5.24g), (5.24c) is equivalent to the more general condition

$$h(J) = \sum_{i \in J} h(\{i\}), \quad J \subseteq [n]. \tag{5.24h}$$

Similarly, under (5.24g), (5.24e) is equivalent to

$$h(\{0\} \cup \{i\} \cup C_i) = h(\{0\} \cup C_i), \quad A_i \subseteq C_i \subseteq [n], \tag{5.24i}$$

which implies that

$$h(\{0\} \cup [n]) = h(\{0\} \cup ([n] \setminus \{i\})), \quad i \in [n]. \tag{5.24j}$$

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Remark 5.6. As in Remark 5.3, the outer bound $\mathscr{R}_{\mathrm{Sh}}$ can be computed by Fourier–Motzkin elimination or linear programs.

Remark 5.7. For any index coding problem G, the broadcast rate is lower bounded as

$$\beta_{\mathrm{Sh}}(G) := \min_{h} \max_{i\in[n]} \frac{h(\{0\})}{h(\{i\})} \leq \beta(G),$$

where the minimum is over all set functions h satisfying (5.24b)–(5.24g). Its reciprocal $1/\beta_{\mathrm{Sh}}(G)$ is the solution to the corresponding linear program involving $2^{n+1}+1$ variables (cf. Remark 5.4).

In establishing the polymatroidal outer bound $\mathscr{R}_{\mathrm{PM}}$ in Theorem 5.2, we did not explicitly use all Shannon-type inequalities. Nonetheless, $\mathscr{R}_{\mathrm{PM}}$ is as tight as the apparently stronger bound $\mathscr{R}_{\mathrm{Sh}}$, as will be verified in Appendix 5.C.

Proposition 5.6. $\mathscr{R}_{\mathrm{Sh}} = \mathscr{R}_{\mathrm{PM}}$.

The polymatroidal bound (which, by Proposition 5.6, effectively uses all Shannon-type inequalities) is not tight in general and can be improved by considering non-Shannon-type inequalities, as illustrated by the following examples.

Example 5.4. Sun and Jafar [145] showed that for the index coding problem

$$\begin{aligned}
&(1|2,3,6,7,8,9,10,11), && (2|1,3,4,6,7,8,9,10,11),\\
&(3|1,2,4,5,6,7,8,9,10,11), && (4|1,2,3,5,6,7,8,9,10,11),\\
&(5|1,3,4,6,7,8,9,10,11), && (6|1,4,5,7,8,9,10,11),\\
&(7|2,4,5,6,8,9,10,11), && (8|1,3,5,6,7,9,10,11),\\
&(9|1,2,5,6,7,8,10,11), && (10|1,2,4,6,7,8,9,11),\\
&(11|1,2,3,5,7,8,9,10), &&
\end{aligned}$$

$\beta_{\mathrm{PM}} = 2.5$, whereas using the Zhang–Yeung non-Shannon-type information inequality in (5.17) yields a tighter outer bound of $28/11 \simeq 2.5455$.

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Example 5.5. Baber et al. [18] showed that for the index coding problem

$$\begin{array}{lll}(1|2,3,4,5,6,7), & (2|1,3,7,9,10), & (3|1,2,4,8,9),\\ (4|1,3,5,8,10), & (5|1,4,6,9,10), & (6|1,5,7,8,9),\\ (7|1,2,6,8,10), & (8|3,4,6,7,9,10), & (9|2,3,5,6,8),\\ (10|2,4,5,7,8). & & \end{array}$$

which has an undirected side information graph, the polymatroidal bound is $\beta_{\mathrm{PM}} = 56/17 \simeq 3.2941$, whereas using the Zhang–Yeung inequality (5.17) yields the lower bound of $598/181 \simeq 3.3038$ and using the 214 non-Shannon-type information inequalities established by Dougherty, Freiling, and Zeger [57] yields an even tighter lower bound of $29523/8929 \simeq 3.3064$ on the broadcast rate. Note that in [18], this example is discussed in the context of guessing games [125]. We converted it here to the corresponding index coding problem; see Section 11 for the exact relationship between index coding and guessing games.

Any tuple of $(2^{n+1} - 1)$ entropies $H(X(J))$, $\emptyset \neq J \subseteq \{0\} \cup [n]$ satisfies all Shannon and non-Shannon inequalities. These inequalities form the set of *entropic vectors* Γ^*_{n+1} [164, Chapter 13]. As a point in $\mathbb{R}^{2^{n+1}-1}_{\geq 0}$, each entropic vector can be *realized* by some tuple of $(n+1)$ random variables, potentially beyond $(X_0, X_1, \ldots, X_n)$ induced by all index codes. The closure $\overline{\Gamma}^*_{n+1}$ of the set of entropic vectors is a (closed) convex cone.

The tightest outer bound on the capacity region of a general index coding problem can be established by augmenting the monotonicity and submodularity conditions (5.24f) and (5.24g) in Theorem 5.3 by the all-encompassing condition that the h function is entropic. Every outer bound we have encountered thus far (including Examples 5.4 and 5.5) is some relaxation of this outer bound. As a matter of fact, this outer bound coincides with the capacity region, as shown in our third (cf. Theorems 3.1 and 3.2) noncomputable characterization of the index coding capacity region via entropic vectors.

Theorem 5.4. The capacity region $\mathscr{C}(G)$ of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G consists of all rate tuples

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$(R_1, \ldots, R_n)$ satisfying

$$R_i \leq \frac{h(\{i\})}{h(\{0\})}, \quad i \in [n], \tag{5.25a}$$

for some set function $h : 2^{\{0\}\cup[n]} \to \mathbb{R}_{\geq 0}$ such that

$$h(\emptyset) = 0, \tag{5.25b}$$

$$h([n]) = \sum_{i=1}^{n} h(\{i\}), \tag{5.25c}$$

$$h(\{0\} \cup [n]) = h([n]), \tag{5.25d}$$

$$h(\{0\} \cup \{i\} \cup A_i) = h(\{0\} \cup A_i), \quad i \in [n], \tag{5.25e}$$

$$(h(\{0\}), h(\{1\}), \ldots, h(\{0\} \cup [n])) \in \overline{\Gamma}^*_{n+1}. \tag{5.25f}$$

Indeed, by (5.24e), the rate bounds in Theorem 5.3 can be rewritten as

$$R_i \leq \frac{h(\{i\}) + h(\{0\} \cup A_i) - h(\{0\} \cup A_i \cup \{i\})}{h(\{0\})}, \quad i \in [n],$$

the numerator of which can be expressed as $I(U_i; V|U(A_i))$ and the denominator of which can be expressed as $H(V)$ for some tuple of $(n+1)$ random variables $U_1, \ldots, U_n$, and V such that $U_1, \ldots, U_n$ are independent (cf. (5.24c) and (5.25c)), $H(V|U_1, \ldots, U_n) = 0$ (cf. (5.24d) and (5.25d)), and $H(U_i|V, U(A_i)) = 0$, $i \in [n]$ (cf. (5.24e) and (5.25e)). As shown by Theorem 3.2 and Remark 3.2, any rate tuple satisfying these conditions is achievable — even without explicitly specifying the decodability condition in (5.24e) and (5.25e)!

In light of Theorem 5.4, the set of entropic vectors $\overline{\Gamma}^*_{n+1}$ plays a rather fundamental role in characterizing the capacity region, as illustrated by Chan and Grant [41] for the general multisource network coding. Therefore, many fundamental questions on the capacity region of a general index coding problem can be translated to those on the entropic vectors.

Open Problem 5.3. Is there an n-message index coding problem for which the characterization of the capacity region in Theorem 5.4 involves infinitely many non-Shannon-type inequalities?

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If the answer to this question is positive, then the corresponding index coding capacity region is not a polyhedron. Nonetheless, this capacity region could be computable in the usual sense.

Open Problem 5.4. Is there a computable procedure to enumerate all extremal inequalities in $\overline{\Gamma}^*_{n+1}$?

If the answer to this question is positive, then it also solves the computability question of the broadcast rate in Open Problem 3.2. By adding more and more such extremal inequalities and computing the corresponding *lower* bound on the broadcast rate, we can approach the broadcast rate from below.

Shannon-type inequalities form an important first step in that procedure. It remains open to determine how loose the corresponding polymatroidal bound can be.

Open Problem 5.5. Is there a sequence of n-message index coding problems G_n for which $\beta(G_n)/\beta_{\mathrm{PM}}(G_n) \geq n^{\epsilon}$ for some $\epsilon > 0$?

If the inequality in Proposition 5.4 holds with equality (see Open Problem 5.2), then as in Proposition 5.2, the gap between the broadcast rate and the polymatroidal bound (or equivalently, the gap between Shannon-type and non-Shannon-type inequalities) can be magnified to a polynomial multiplicative factor.

5.A Proof of Proposition 5.1

Let $G = G_0 \circ G_1$. We establish the desired identity by proving inequalities in both directions.

Proof of "$\geq$". Suppose that V_0 and V_1 induce maximum acyclic subgraphs of G_0 and G_1, respectively, i.e., $\beta_{\mathrm{MAIS}}(G_i) = |V_i|$, $i = 0, 1$. By the definition of lexicographic product, $V_0 \times V_1$ induces an acyclic subgraph of G, which implies that $\beta_{\mathrm{MAIS}}(G) \geq \beta_{\mathrm{MAIS}}(G_0)\beta_{\mathrm{MAIS}}(G_1)$.

Proof of "$\leq$". Let $V^* \subseteq V(G_0) \times V(G_1)$ be a maximum set such that $G|_{V^*}$ is acyclic, i.e., $\beta_{\mathrm{MAIS}}(G) = |V^*|$. Assume without loss of generality that $V(G_0) = [m]$. Then, V^* can be partitioned as $V^* = V_1 \cup \cdots \cup V_m$, where V_i induces an acyclic subgraph of the i-th copy of G_1 in the

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lexicographic product G, and thus

$$|V_i| \le \beta_{\mathrm{MAIS}}(G_1), \quad i \in [m].$$

Let $J = \{i \colon V_i \neq \emptyset\}$. Since $G|_{V^*}$ is acyclic, J induces an acyclic subgraph of G_0 and thus

$$|J| \le \beta_{\mathrm{MAIS}}(G_0).$$

Hence, we have

$$\begin{aligned}\beta_{\mathrm{MAIS}}(G) &= \sum_{i \in J} |V_i| \\ &\le \sum_{i \in J} \beta_{\mathrm{MAIS}}(G_1) \\ &\le \beta_{\mathrm{MAIS}}(G_0)\beta_{\mathrm{MAIS}}(G_1).\end{aligned}$$

This completes the proof of the proposition.

5.B Proof of Proposition 5.3

We show that every inequality of the MAIS bound can be derived using the inequalities of the polymatroidal bound, i.e., (5.13a)–(5.13f). First note that by the submodularity of the set function f in (5.13e),

$$f(\{i\} \cup K) - f(K) \le f(\{i\} \cup J) - f(J), \quad J \subseteq K \subseteq [n], \tag{5.26}$$

for any $i \notin K$. Suppose that $J \subseteq V(G)$ induces an acyclic subgraph of G. Assume without loss of generality that $J = [m]$ and that $j \in B_i$ for every $i < j \in [m]$. Let $J_i = \{i+1, \dots, m\}$. Then, by the assumption, $J_i \subseteq B_i$. Now we have

$$\begin{aligned}\sum_{i \in J} R_i &\le \sum_{i=1}^{m} \big(f(\{i\} \cup B_i) - f(B_i)\big) && (5.27)\\ &\le \sum_{i=1}^{m} \big(f(\{i\} \cup J_i) - f(J_i)\big) && (5.28)\\ &= f(\{1\} \cup J_1) - f(J_m) && (5.29)\\ &\le 1, && (5.30)\end{aligned}$$

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where (5.27) follows by (5.13a) and (5.28) follows by (5.26). Furthermore, (5.29) holds by telescoping since $J_i = \{i+1\} \cup J_{i+1}$. Finally, (5.30) follows by (5.13b), since $J_m = \emptyset$. This completes the proof of the proposition.

5.C Proof of Proposition 5.6

We first verify that $\mathscr{R}_{\mathrm{Sh}} \subseteq \mathscr{R}_{\mathrm{PM}}$. Suppose that $(R_1, \ldots, R_n) \in \mathscr{R}_{\mathrm{Sh}}$. Then there exists a set function $h(J)$, $J \subseteq \{0\} \cup [n]$, satisfying (5.24b)–(5.24g) such that

$$R_i \leq \frac{h(\{i\})}{h(\{0\})}, \quad i \in [n]. \tag{5.31}$$

Define a set function $f : 2^{[n]} \to [0,1]$ as

$$f(J) = \frac{h(\bar{J} \cup \{0\}) - h(\bar{J})}{h(\{0\})}, \tag{5.32}$$

where $\bar{J} = [n] \setminus J$ for $J \subseteq [n]$. Then (5.31) can be rewritten as

$$\begin{aligned} R_i &\leq \frac{h(\{i\})}{h(\{0\})} \\ &= \frac{h(A_i \cup \{i\}) - h(A_i)}{h(\{0\})} && (5.33) \\ &= \frac{h(A_i \cup \{0\}) - h(A_i)}{h(\{0\})} - \frac{h(A_i \cup \{i\} \cup \{0\}) - h(A_i \cup \{i\})}{h(\{0\})} && (5.34) \\ &= f(\{i\} \cup B_i) - f(B_i), && (5.35) \end{aligned}$$

where (5.33) follows by (5.24h), (5.34) follows by (5.24e), and (5.35) follows by the definition of f in (5.32). We now show that the set function f defined in (5.32) satisfies (5.13b)–(5.13f) one by one. First, by (5.24d),

$$f(\emptyset) = \frac{h([n] \cup \{0\}) - h([n])}{h(\{0\})} = 0$$

and by (5.24b),

$$f([n]) = \frac{h(\{0\}) - h(\emptyset)}{h(\{0\})} = 1.$$

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Next, for $k \notin J \subseteq [n]$ and $K = J \cup \{k\}$,

$$\begin{aligned} f(J) &= \frac{h(\bar{J} \cup \{0\}) - h(\bar{J})}{h(\{0\})} \\ &= \frac{h(\overline{K} \cup \{k\} \cup \{0\}) - h(\overline{K}) - h(\{k\})}{h(\{0\})} && (5.36) \\ &\leq \frac{h(\overline{K} \cup \{0\}) + h(\{k\}) - h(\overline{K}) - h(\{k\})}{h(\{0\})} && (5.37) \\ &= \frac{h(\overline{K} \cup \{0\}) - h(\overline{K})}{h(\{0\})} \\ &= f(K), \end{aligned}$$

where (5.36) follows by (5.24h) since $\bar{J} = \overline{K} \cup \{k\}$, and (5.37) follows by (5.24g). Next, for $i, k \notin J \subseteq [n]$ and $K = \bar{J} \setminus \{i, k\} \subseteq [n]$,

$$\begin{aligned} &h(\{0\})(f(J \cup \{i\} \cup \{k\}) + f(J)) \\ &= h(K \cup \{0\}) - h(K) + h(\bar{J} \cup \{0\}) - h(\bar{J}) \\ &= h(K \cup \{0\}) - h(K) \\ &\qquad + h(K \cup \{i\} \cup \{k\} \cup \{0\}) - h(K) - h(\{i\}) - h(\{k\}) && (5.38) \\ &\leq h(K \cup \{k\} \cup \{0\}) - h(K) - h(\{k\}) \\ &\qquad + h(K \cup \{i\} \cup \{0\}) - h(K) - h(\{i\}) && (5.39) \\ &= h(K \cup \{k\} \cup \{0\}) - h(K \cup \{k\}) \\ &\qquad + h(K \cup \{i\} \cup \{0\}) - h(K \cup \{i\}) && (5.40) \\ &= h(\{0\})(f(J \cup \{i\}) + f(J \cup \{k\})), \end{aligned}$$

where (5.38) and (5.40) follow by (5.24h), and (5.39) follows by (5.24g). Finally, for $i \in [n]$,

$$\begin{aligned} &h(\{0\})(f(B_i \cup \{i\}) - f(B_i)) \\ &= h(A_i \cup \{0\}) - h(A_i) - h(A_i \cup \{i\} \cup \{0\}) + h(A_i \cup \{i\}) \\ &= h(A_i \cup \{i\}) - h(A_i) && (5.41) \\ &= h(\{i\}) && (5.42) \\ &= h([n]) - h([n] \setminus \{i\}) && (5.43) \\ &= h(([n] \setminus \{i\}) \cup \{0\}) - h([n] \setminus \{i\}) && (5.44) \\ &= h(\{0\}) f(\{i\}), \end{aligned}$$

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where (5.41) follows by (5.24e), (5.42) and (5.43) follow by (5.24h), and (5.44) follows by (5.24j).

We now move on to show $\mathscr{R}_{\mathrm{PM}} \subseteq \mathscr{R}_{\mathrm{Sh}}$. Suppose that $(R_1, \ldots, R_n) \in \mathscr{R}_{\mathrm{PM}}$. Then there exists a set function $f(J)$, $J \subseteq [n]$, satisfying (5.13b)–(5.13f) such that

$$R_i \leq f(B_i \cup \{i\}) - f(B_i) = f(\{i\}). \tag{5.45}$$

Define a set function $h : 2^{\{0\}\cup[n]} \to \mathbb{R}_{\geq 0}$ as

$$h(\emptyset) = 0, \tag{5.46a}$$

$$h(\{i\}) = \frac{f(\{i\})}{\sum_{i=1}^{n} f(\{i\})}, \quad i \in [n], \tag{5.46b}$$

$$h(J) = \sum_{i \in J} h(\{i\}), \quad J \subseteq [n], \tag{5.46c}$$

$$h(J \cup \{0\}) = h(J) + \frac{f(\bar{J})}{\sum_{i=1}^{n} f(\{i\})}, \quad J \subseteq [n]. \tag{5.46d}$$

By definition,

$$h(\{0\}) = \frac{1}{\sum_{i=1}^{n} f(\{i\})}$$

and (5.45) can be rewritten as

$$R_i \leq \frac{h(\{i\})}{h(\{0\})}, \quad i \in [n].$$

We now show that the set function h defined in (5.46a)–(5.46d) satisfies (5.24b)–(5.24g). The first three conditions follow by the definition of h in (5.46a), (5.46c), and (5.46d), respectively. Next, for $i \in [n]$,

$$\begin{aligned}
h(\{i\} \cup A_i \cup \{0\}) &= h(\{i\} \cup A_i) + \frac{f(B_i)}{\sum_{k=1}^{n} f(\{k\})} \\
&= h(\{i\} \cup A_i) + \frac{f(B_i \cup \{i\}) - f(\{i\})}{\sum_{k=1}^{n} f(\{k\})} && (5.47) \\
&= h(A_i) + h(\{i\}) + \frac{f(B_i \cup \{i\}) - f(\{i\})}{\sum_{k=1}^{n} f(\{k\})} && (5.48) \\
&= h(A_i) + \frac{f(B_i \cup \{i\})}{\sum_{k=1}^{n} f(\{k\})} && (5.49) \\
&= h(A_i \cup \{0\}),
\end{aligned}$$

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where (5.47) follows by (5.13f), (5.48) follows by (5.46c), and (5.49) follows by (5.46b).

Next, for (5.24f), we consider the following cases:

- If $k = 0$ and $J \subseteq [n]$, then

$$\begin{aligned} h(J) &\leq h(J) + \frac{f(\bar{J})}{\sum_{i=1}^{n} f(\{i\})} \\ &= h(J \cup \{0\}) \\ &= h(J \cup \{k\}). \end{aligned}$$

- If $0 \neq k \notin J \subseteq [n]$, then

$$\begin{aligned} h(J) &= \sum_{j \in J} h(\{j\}) \\ &\leq \sum_{j \in J \cup \{k\}} h(\{j\}) \\ &= h(J \cup \{k\}). \end{aligned}$$

- Otherwise, $k \neq 0 \in J = K \cup \{0\}$ for some $K \subseteq [n]$ and

$$\begin{aligned} h(J) &= h(K \cup \{0\}) \\ &= h(K) + \frac{f(\overline{K})}{\sum_{i=1}^{n} f(\{i\})} \\ &= h(K \cup \{k\}) - h(\{k\}) + \frac{f(\overline{K})}{\sum_{i=1}^{n} f(\{i\})} \qquad (5.50) \\ &= h(K \cup \{k\}) + \frac{f(\overline{K}) - f(\{k\})}{\sum_{i=1}^{n} f(\{i\})} \\ &\leq h(K \cup \{k\}) + \frac{f(\overline{K} \setminus \{k\})}{\sum_{i=1}^{n} f(\{i\})} \qquad (5.51) \\ &= h(K \cup \{k\} \cup \{0\}) \\ &= h(J \cup \{k\}), \end{aligned}$$

  where (5.50) follows by (5.24h), and (5.51) follows by the submodularity of f in (5.13e).

Hence, property (5.24f) holds.

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Finally, in order to show that the set function h defined in (5.46) satisfies (5.24g), we consider the following cases:

- If $0 \notin J$ and $i, k \neq 0$, then (5.24g) holds with equality by (5.46c).
- If $0 \notin J$, $i = 0$, and $k \neq 0$, then

$$\begin{aligned}
&\big(h(J) + h(J \cup \{i\} \cup \{k\})\big)\Big(\sum_{j=1}^{n} f(\{j\})\Big) \\
&= \sum_{j \in J} f(\{j\}) + \sum_{j \in J} f(\{j\}) + f(\{k\}) + f(\bar{J} \setminus \{k\}) \\
&\leq \sum_{j \in J} f(\{j\}) + \sum_{j \in J} f(\{j\}) + f(\{k\}) + f(\bar{J}) \qquad (5.52) \\
&= \big(h(J \cup \{i\}) + h(J \cup \{k\})\Big(\sum_{j=1}^{n} f(\{j\})\Big),
\end{aligned}$$

  where (5.52) follows by the monotonicity of f in (5.13d).

- Otherwise ($0 \in J$ and $i, k \neq 0$), $J = K \cup \{0\}$ for some $K \subseteq [n]$ and

$$\begin{aligned}
&\big(h(J) + h(J \cup \{i\} \cup \{k\})\big)\Big(\sum_{j=1}^{n} f(\{j\})\Big) \\
&= \sum_{j \in K} f(\{j\}) + f(\overline{K}) + \sum_{j \in K \cup \{i,k\}} f(\{j\}) + f(\overline{K} \setminus \{i,k\}) \\
&\leq \sum_{j \in K \cup \{i\}} f(\{j\}) + f(\overline{K} \setminus \{i\}) + \sum_{j \in K \cup \{k\}} f(\{j\}) + f(\overline{K} \setminus \{k\}) \qquad (5.53) \\
&= \big(h(J \cup \{i\}) + h(J \cup \{k\})\big)\Big(\sum_{j=1}^{n} f(\{j\})\Big),
\end{aligned}$$

  where (5.53) follows by (5.13e).

This completes the proof of the proposition.

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6

Coding Schemes

In this section, we review some of the most interesting index coding schemes, summarized in Figure 6.1. Each coding scheme corresponds to an upper bound on the broadcast rate. Some schemes can handle asymmetric rate tuples and yield inner bounds on the capacity region.

6.1 MDS Codes

Consider the index coding problem with 3 messages represented by the side information graph shown in Figure 6.2, in which every receiver has one piece of side information. Consider a systematic $(5, 3)$ code $(x_1, x_2, x_3, p_1, p_2)$ over the finite field $\mathbb{F}_4 = \{0, 1, \alpha, \alpha + 1\}$, where $p_1 = x_1 + x_2 + x_3$ and $p_2 = x_1 + \alpha x_2 + (\alpha + 1)x_3$. This code has the maximum distance separable (MDS) property, which guarantees that any three out of the five code symbols are sufficient to recover all three message symbols. We now employ this code for the 3-message index coding problem and transmit the parity symbols p_1 and p_2. Then, every receiver has three code symbols (including its side information) and by the MDS property, can successfully recover its desired message (along with two other messages).

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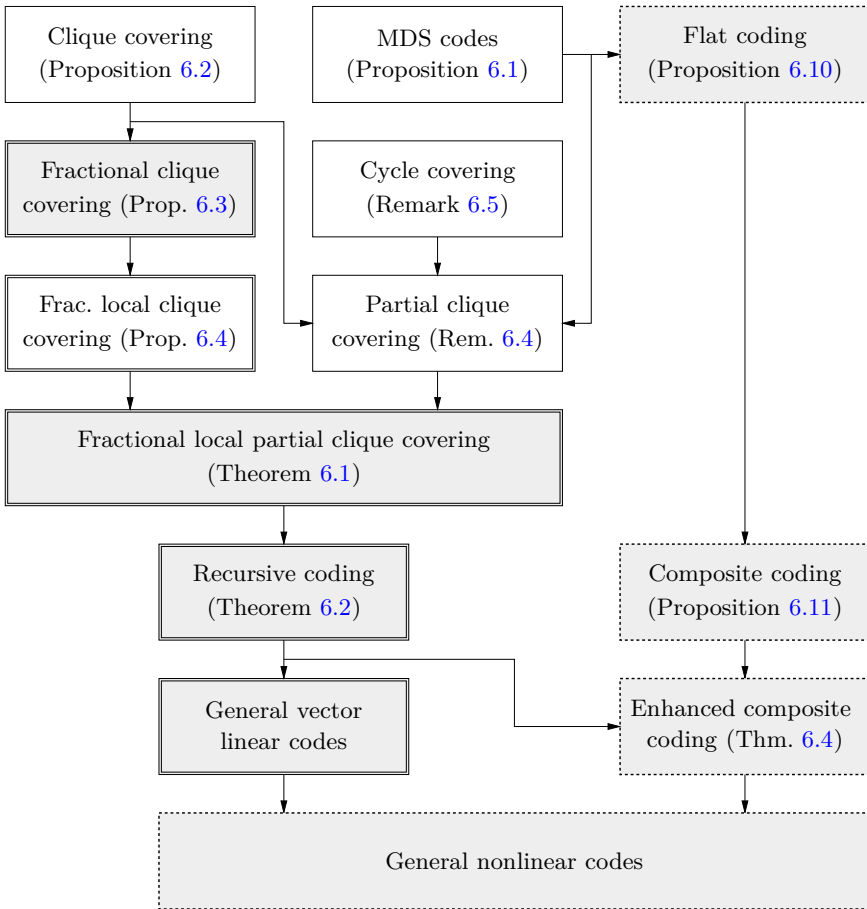


Figure 6.1: Summary of the coding schemes. Here a directed edge from scheme A to scheme B indicates that scheme A can be viewed as a special case of scheme B, that is, the upper bound on the broadcast rate achieved by scheme A is also achievable by scheme B. The schemes in solid-lined boxes are linear (single line for scalar and double line for vector linear coding). The schemes in dashed-lined boxes are nonlinear. For the schemes in shaded boxes, corresponding inner bounds on the capacity region are presented in addition to upper bounds on the broadcast rate. Interference alignment (Section 6.8) is not included here since it is not applicable concretely to all index coding problems as a general-purpose coding scheme. Multiletter nonlinear coding schemes based on coloring of confusion graphs and on generic random coding can be found in Section 3.

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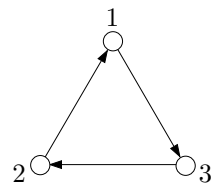


Figure 6.2: The index coding problem $(1|2), (2|3), (3|1)$ with $\beta = 2$, which is achievable by a $(5, 3)$ MDS code.

For a general index coding problem G, let

$$\text{minindeg}(G) := \min_{i\in[n]} |\{j\colon (j,i) \in E(G)\}| = \min_{i\in[n]} |A_i|$$

be the minimum number of side information messages over all receivers. Consider a systematic $(n+\kappa, n)$ MDS code with n message symbols and κ parity symbols. Then, given any n out of the $n+\kappa$ code symbols, one can recover all the messages. Such an MDS code exists over a sufficiently large alphabet. As every receiver has at least $\text{minindeg}(G)$ messages as side information, if we employ the $(n+\kappa, n)$ MDS code and transmit $\kappa = \kappa(G) := n - \text{minindeg}(G)$ parity symbols, then every receiver will have n code symbols and thus can recover all the message symbols that it does not have, including its desired message. This establishes the following upper bound on the broadcast rate by Birk and Kol [26].

Proposition 6.1 (MDS bound [26])**.** For the n-message index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G,

$$\beta(G) \le \beta_{\text{MDS}}(G) := \kappa(G) = n - \text{minindeg}(G) = n - \min_{i\in[n]} |A_i|.$$

Remark 6.1. If the graph G is a *clique*, then $\text{minindeg}(G) = n-1$ and transmitting the parity symbol $x_1 + \cdots + x_n$ of an $(n+1, n)$ MDS code over $\mathbb{F}_2$ achieves $\beta(G) = 1$.

6.2 Clique Covering

Consider the 3-message index coding problem G shown in Figure 6.3. Since receiver 3 has no side information, $\text{minindeg}(G) = 0$. Therefore, using an MDS code, we need to transmit three symbols, which is no

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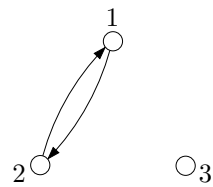


Figure 6.3: The index coding problem $(1|2), (2|1), (3|-)$ with $\beta = 2$, which is achievable by the clique covering scheme.

better than uncoded transmission of the messages. We can instead partition the vertices into two cliques, namely, $\{1, 2\}$ and $\{3\}$, and transmit $x_1 + x_2$ and x_3. Then, receiver 3 receives its desired message x_3 directly. Since receiver 1 has message x_2 as side information, it can successfully recover its desired message x_1. Similarly, receiver 2 can recover its desired message x_2. This scheme requires two transmissions, which matches the MAIS bound in Theorem 5.1 and is thus optimal.

Generalizing this idea, we partition the vertices of the side information graph G by cliques and transmit the sum (parity) of all the messages in each clique. Since every receiver has all the other messages inside its clique as side information, all the messages can be successfully recovered by their corresponding receivers. This coding scheme, which can be viewed as *time division* over a clique partition (one parity symbol per clique), achieves the following upper bound on the broadcast rate by Birk and Kol [26].

Proposition 6.2 (Clique covering bound [26])**.** The broadcast rate of the n-message index coding problem G is upper bounded by the minimum number of cliques that partition G (or equivalently, the chromatic number of the undirected complement of G), which is the solution $\beta_{\mathrm{C}}(G)$ to the integer program

$$\text{minimize} \quad \sum_{J \in \mathcal{K}} \gamma_J \tag{6.1a}$$

$$\text{subject to} \quad \sum_{J \in \mathcal{K}: i \in J} \gamma_J \geq 1, \quad i \in [n], \tag{6.1b}$$

$$\gamma_J \in \{0, 1\}, \quad J \in \mathcal{K}, \tag{6.1c}$$

where $\mathcal{K}$ is the collection of all cliques in G.

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Remark 6.2. For the index coding problem depicted in Figure 6.2, using an MDS code outperforms the clique covering scheme that requires three transmissions. Thus, the clique covering scheme and the MDS code for index coding are not comparable.

6.3 Fractional Clique Covering

We revisit the index coding problem in Example 5.2, the side information graph of which is reproduced in Figure 6.4. We can partition the graph into three cliques, say, $\{1,2\}, \{3,4\}$, and $\{5\}$, and thus, the clique covering scheme takes three transmissions. Instead, for messages represented in two bits as $x_i = (x_{i1}, x_{i2}) \in \mathbb{F}_2^2$, consider the vector linear coding scheme that transmits $x_{11} + x_{22}$, $x_{21} + x_{32}$, $x_{31} + x_{42}$, $x_{41} + x_{52}$, and $x_{51} + x_{12}$. Then every receiver can successfully recover its two bits. This scheme achieves the upper bound of $5/2$ on the broadcast rate, which matches the polymatroidal bound in Theorem 5.2.

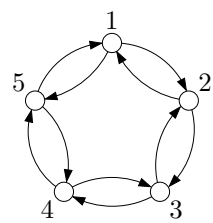


Figure 6.4: The index coding problem $(1|2,5), (2|1,3), (3|2,4), (4|3,5), (5|1,4)$ with $\beta = 5/2$, which is achievable by the fractional clique covering scheme.

More generally, Blasiak, Kleinberg, and Lubetzky [30] extended the clique covering scheme and its achievable rate in (6.1) by *time sharing* over all cliques so that the combined rate of each message over all parity symbols it participates in is at least one. The resulting rate β_{F} corresponds to the solution to the *linear program* obtained by relaxing the integer constraints in (6.1c).

Proposition 6.3 (Fractional clique covering bound [30])**.** The broadcast rate of the n-message index coding problem G is upper bounded by the minimum number of cliques that fractionally partition G (or equiv-

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alently, $\chi_f(\overline{G})$), which is the solution $\beta_{\mathrm{F}}(G)$ to the linear program

$$\text{minimize} \quad \sum_{J \in \mathcal{K}} \gamma_J \tag{6.2a}$$

$$\text{subject to} \quad \sum_{J \in \mathcal{K}: i \in J} \gamma_J \geq 1, \quad i \in [n], \tag{6.2b}$$

$$\gamma_J \in [0,1], \quad J \in \mathcal{K}. \tag{6.2c}$$

For the index coding problem in Figure 6.4, using the vector linear code is equivalent to setting

$$\gamma(J) = \begin{cases} 1/2 & J = \{1,2\},\{2,3\},\{3,4\},\{4,5\},\{5,1\}, \\ 0 & \text{otherwise}, \end{cases}$$

which satisfies the constraints of (6.2) and achieves $\beta = 5/2$.

Remark 6.3. Fractional clique covering can be readily extended to asymmetric rates, which establishes the inner bound on the capacity region that consists of all rate tuples $(R_1, \ldots, R_n)$ satisfying

$$R_i \leq \sum_{J \in \mathcal{K}: i \in J} \delta_J, \quad i \in [n], \tag{6.3a}$$

for some $(\delta_J \in [0,1], J \in \mathcal{K})$ such that

$$\sum_{J \in \mathcal{K}} \delta_J \leq 1. \tag{6.3b}$$

For the index coding problem in Figure 6.4, by eliminating the δ_J variables, the inner bound in (6.3) simplifies to the polymatroidal outer bound on the capacity region in (5.14) and (5.15).

6.4 Local Clique Covering

Consider the 6-message index coding problem G depicted in Figure 6.5. The graph G can be partitioned into three cliques $\{1,2\},\{3,4\}$, and $\{5,6\}$, and hence, by the clique covering scheme, it suffices to transmit

$$p_1 = x_1 + x_2, \quad p_2 = x_3 + x_4, \quad \text{and} \quad p_3 = x_5 + x_6.$$

However, since only two of the three parity symbols p_1, p_2, and p_3 are

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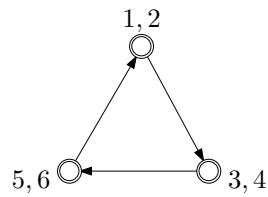


Figure 6.5: The index coding problem $(1|2,5,6)$, $(2|1,5,6)$, $(3|1,2,4)$, $(4|1,2,3)$, $(5|3,4,6)$, $(6|3,4,5)$ with $\beta = 2$, which is achievable by the local clique covering scheme. Here each double circle represents a 2-vertex clique.

missing at each receiver, we can reduce the number of parity transmissions by using a two-erasure correcting MDS code with two hyper-parity symbols, say, $p_1 + p_2 = (x_1 + x_2) + (x_3 + x_4)$ and $p_1 + p_3 = (x_1 + x_2) + (x_5 + x_6)$. Then, each receiver can recover all parity symbols p_1, p_2, p_3, and thus its desired message. This upper bound of 2 on the broadcast rate matches the MAIS bound and thus is tight.

More generally, consider a partition of the side information graph G into cliques. Suppose that each receiver has all the messages of at least one clique as side information. Then each receiver can recover some parity symbols from its side information and the number of transmissions for the parity symbols can be reduced to exclude those parity symbols available locally at each receiver. Based on this idea, Shanmugam, Dimakis, and Langberg [133] extended the clique covering scheme to *local clique covering* by applying an MDS code to parity symbols for cliques. Combining this scheme with fractional covering yields the following.

Proposition 6.4 (Fractional local clique covering bound [133])**.** The broadcast rate of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G is upper bounded by the solution $\beta_{\mathrm{FL}}(G)$ to the linear program

$$\text{minimize} \quad \max_{i\in[n]} \sum_{J\in\mathcal{K}:\, J\not\subseteq A_i} \gamma_J \tag{6.4a}$$

$$\text{subject to} \quad \sum_{J\in\mathcal{K}:\, i\in J} \gamma_J \geq 1, \quad i \in [n], \tag{6.4b}$$

$$\gamma_J \in [0,1], \quad J \in \mathcal{K}. \tag{6.4c}$$

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This coding scheme achieves the *fractional local chromatic number* [66, 89] of $\overline{G}$. The improvement over fractional clique covering is captured by the summation of γ_J over cliques $J \not\subseteq A_i$ in (6.4a) rather than all cliques in (6.2a). Hence, compared to the regular time sharing of clique covering, local clique covering can be viewed as *local time sharing*, in which the total transmission time is split only among those parity symbols not available locally at each receiver.

6.5 Partial Clique Covering

Birk and Kol [26] extended clique covering in Section 6.2 by partitioning G with induced subgraphs (or *partial cliques*). By using an MDS code, each subgraph $G|_J$ then requires $\kappa(G|_J) = |J| - \text{minindeg}(G|_J)$ transmissions. If $G|_J$ is a clique, then $\kappa(G|_J) = 1$ as in clique covering.

As an example, consider the 5-message index coding problem G in Figure 6.6. It can be shown that fractional local clique covering requires $\beta_{\text{FL}}(G) = 4$ transmissions. This upper bound on the broadcast rate can be improved by time sharing over *partial cliques*. Since

$$\kappa(G|_{\{1,2\}}) = \kappa(G|_{\{1,5\}}) = \kappa(G|_{\{3\}}) = 1,$$
$$\kappa(G|_{\{1,3,4\}}) = \kappa(G|_{\{2,4,5\}}) = 2,$$

setting

$$\gamma_J = \begin{cases} 1/2, & J = \{1,2\},\{1,5\},\{1,3,4\},\{2,4,5\},\{3\}, \\ 0, & \text{otherwise} \end{cases}$$

achieves the upper bound of $7/2$ on the broadcast rate.

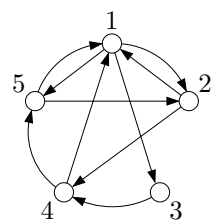


Figure 6.6: The index coding problem $(1|2,4,5),(2|1,5),(3|1),(4|2,3),(5|1,4)$ with $\beta_{\text{FLP}} = 7/2 < \beta_{\text{FL}} = 4$.

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In general, local time sharing over MDS codes of arbitrary subgraphs (partial cliques) yields the following.

Theorem 6.1 (Fractional local partial clique covering bound [9])**.** The broadcast rate of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G is upper bounded by the solution $\beta_{\mathrm{FLP}}(G)$ to the linear program

$$\text{minimize} \quad \max_{i\in[n]} \sum_{J\subseteq[n]:J\not\subseteq A_i} \gamma_J \cdot \kappa(G|_J) \tag{6.5a}$$

$$\text{subject to} \quad \sum_{J\subseteq[n]:i\in J} \gamma_J \geq 1, \quad i \in [n], \tag{6.5b}$$

$$\gamma_J \in [0,1], \quad J \subseteq [n]. \tag{6.5c}$$

Remark 6.4. Time division (instead of time sharing or fractional partitioning of G) over MDS codes of partial cliques yields the *partial clique covering* bound, which is a special case of the fractional local partial clique covering that achieves the rate characterized by the corresponding integer program with $\gamma_J \in \{0,1\}$ in place of (6.5c).

Remark 6.5. When the vertices in $J \subseteq V(G)$ form a cycle, a simple linear code over all but one edges in the cycle can achieve the rate of $\kappa(G|_J) = |J|-1$. As an example, for the subgraph $G|_{\{2,4,5\}}$ in Figure 6.6, it suffices to transmit $x_2 + x_4$ and $x_4 + x_5$ in $\mathbb{F}_2$, from which $x_2 + x_5$ can be computed. By restricting J to be cycles (or single-vertex sets) in (6.5) and saving one symbol per cycle, we recover the *cycle covering* scheme [44, 117].

Remark 6.6. Fractional local partial clique covering can be readily extended to the corresponding inner bound $\mathscr{R}_{\mathrm{FLP}}(G)$ on the capacity region (cf. Remark 6.3) that consists of all rate tuples $(R_1, \ldots, R_n)$ satisfying

$$R_i \leq \sum_{J\subseteq[n]:i\in J} \frac{\delta_J}{\kappa(G|_J)}, \quad i \in [n], \tag{6.6a}$$

for some $\delta_J \in [0,1]$, $J \subseteq [n]$, such that

$$\sum_{J\subseteq[n]:J\not\subseteq A_i} \delta_J \leq 1, \quad i \in [n]. \tag{6.6b}$$

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6.6 Recursive Codes

We can extend the fractional local partial clique covering scheme in Section 6.5 by performing local time sharing over subproblem solutions *recursively.*

Theorem 6.2 (Recursive coding inner bound [9])**.** The capacity region of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G includes the rate region $\mathscr{R}_{\mathrm{R}}(G)$ that is recursively defined as the set of all nonnegative rate tuples $(R_1, \dots, R_n)$ satisfying

$$R_i \leq \sum_{J \subsetneq [n] : i \in J} \delta_J \, T_{iJ}, \qquad i \in [n], \tag{6.7a}$$

for some $\delta_J \geq 0$, $J \subsetneq [n]$, and $T_{iJ} \geq 0$, $i \in J \subsetneq [n]$, such that

$$\sum_{J \subsetneq [n] : J \not\subseteq A_i} \delta_J \leq 1, \qquad i \in [n], \tag{6.7b}$$

$$(T_{iJ} \colon i \in J) \in \mathscr{R}_{\mathrm{R}}(G|_J), \qquad J \subsetneq [n], \tag{6.7c}$$

where $\mathscr{R}_{\mathrm{R}}(G|_J)$ is the rate region for subproblem $G|_J$ with recursion termination condition $\mathscr{R}_{\mathrm{R}}(G|_{\{i\}}) = [0, 1]$. In particular,

$$\beta_{\mathrm{R}}(G) = \min_{R : (R, \dots, R) \in \mathscr{R}_{\mathrm{R}}(G)} \frac{1}{R}. \tag{6.8}$$

Given all the fascinating coding schemes we have discussed so far in this section and given the simplicity of recursive coding, it is perhaps anticlimactic to note that recursive codes, which can be viewed as vector linear index codes over some finite field, outperform fractional local partial clique covering in general (cf. Remark 6.6).

Proposition 6.5 ([9])**.** For any index coding problem G,

$$\mathscr{R}_{\mathrm{FLP}}(G) \subseteq \mathscr{R}_{\mathrm{R}}(G).$$

The proof is relegated to Appendix 6.A.

Remark 6.7. Recursive coding can be limited to symmetric rates at every step of recursion. This symmetric recursive coding scheme achieves

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the upper bound $\beta_{\mathrm{SR}}(G)$ on the broadcast rate, which is defined recursively as the solution to the linear program

$$\text{minimize} \quad \max_{i\in[n]} \sum_{J\subsetneq[n]:J\not\subseteq A_i} \gamma_J \cdot \beta_{\mathrm{SR}}(G|_J) \tag{6.9a}$$

$$\text{subject to} \quad \sum_{J\subsetneq[n]:i\in J} \gamma_J \geq 1, \qquad i \in [n], \tag{6.9b}$$

$$\gamma_J \in [0,1], \qquad J \subsetneq [n], \tag{6.9c}$$

where $\beta_{\mathrm{SR}}(G|_J)$ is the solution for subproblem $G|_J$ and $\beta_{\mathrm{SR}}(G|_{\{i\}}) = 1$, $i \in [n]$.

The structure of the linear program (6.9) is very similar to that for fractional local partial clique covering in (6.5), except that local time sharing over different subproblems is over symmetric recursive codes of rates $\beta_{\mathrm{SR}}(G|_J)$ instead of MDS codes of rates $\kappa(G|_J)$. By induction, $\beta_{\mathrm{SR}}(G|_J) \leq \kappa(G|_J)$ for every $J \subseteq [n]$ (including $J = [n]$) and hence

$$\beta_{\mathrm{SR}}(G) \leq \beta_{\mathrm{FLP}}(G).$$

This inequality is strict in general [9], as illustrated by the 5-message index coding problem in Figure 6.7.

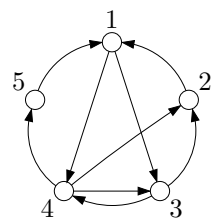


Figure 6.7: The index coding problem $(1|2,5),(2|3,4),(3|1,4),(4|1,3),(5|4)$ with $\beta_{\mathrm{SR}} = 3 < \beta_{\mathrm{FLP}} = 7/2$. Here the bounds are computed by solving the respective linear programs.

At the same time, recursion over asymmetric rates (Theorem 6.2) is richer than recursion over symmetric rates (Remark 6.7), and we have

$$\beta_{\mathrm{R}}(G) \leq \beta_{\mathrm{SR}}(G),$$

the formal proof of which can be found in [9, Proposition 1]. This inequality is strict in general, as illustrated by the 7-message index coding problem in Figure 6.8.

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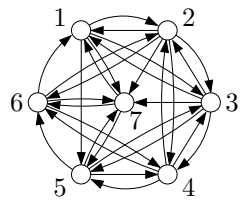


Figure 6.8: The index coding problem $(1|2,3,6,7)$, $(2|1,3,4,6)$, $(3|1,2,4,5)$, $(4|2,3,5,6)$, $(5|1,3,4,7)$, $(6|2,4,5,7)$, $(7|1,2,3,5,6)$ with $\beta_{\mathrm{PM}} = \beta = \beta_{\mathrm{R}} = 5/2 < \beta_{\mathrm{SR}} = 8/3$.

Remark 6.8. If the capacity region $\mathscr{C}(G|_J)$ is known for some or all subproblems $J \subsetneq [n]$, then the inner bound in Theorem 6.2 can be tightened by replacing $\mathscr{R}_{\mathrm{R}}(G|_J)$ with $\mathscr{C}(G|_J)$ in (6.7). The corresponding inner bound $\mathscr{R}_{\mathrm{LTS}}$ on the capacity region (along with the associated upper bound β_{LTS} on the broadcast rate) represents the rate region achievable by local time sharing over optimal codes for subproblems. As illustrated by the 5-message index coding problem in Figure 6.9, however, local time sharing over subproblems is suboptimal in general.

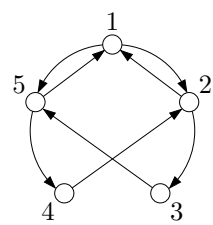


Figure 6.9: The index coding problem $(1|2,5), (2|1,4), (3|2), (4|5), (5|1,3)$ with $\beta = 3 < \beta_{\mathrm{SR}} = \beta_{\mathrm{R}} = \beta_{\mathrm{LTS}} = 7/2$. Here β is achieved by a scalar linear code $(x_1 + x_2 + x_5, x_2 + x_3, x_4 + x_5)$ over $\mathbb{F}_2$.

6.7 General Linear Codes

So far we have discussed scalar or vector linear codes based on decomposition of the side information graph. A general linear code, however, can utilize the global structure of the given index coding problem. As a simple illustration, we revisit the 3-message index coding problem $(1|2), (2|3), (3|1)$ in Figure 6.2. Consider any 3-by-3 matrix M such

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that

$$M_{ii} = 1 \quad \text{and} \quad M_{ij} = 0 \text{ if } j \notin A_i. \tag{6.10}$$

We can easily check that the matrices

$$M_1 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \quad \text{and} \quad M_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

satisfy (6.10).

Consider the three linear combinations of the messages generated by multiplying any such matrix M by the message vector $[x_1\ x_2\ x_3]^T$ from right. For example, M_1 generates x_1+x_2, x_2+x_3, and x_1+x_3, and M_2 generates x_1, x_2, and x_1+x_3. If we transmit these three generated (parity) symbols, receiver i can use the linear combination from row i to recover x_i since this linear combination contains only x_i and $x(A_i)$. Now note that M_1 is of rank 2 and thus that any two of the three linear combinations can determine the third; for example $(x_1 + x_3) = (x_1 + x_2) + (x_2 + x_3)$. Thus, it suffices to transmit the first two parity symbols.

More generally, if M is of rank r, then we can transmit linear combinations of the messages from r independent rows of M. Therefore, any matrix M satisfying (6.10) defines a scalar linear code and $\text{rank}(M)$ is an upper bound on the broadcast rate. Conversely, each encoder of a scalar linear code can be expressed as $y = [y_1\ \cdots\ y_r]^T = E[x_1\ \cdots\ x_n]^T$ and each decoder $i \in [n]$ can be expressed as a linear combination of y and a subset of side information $x(A_i)$. Hence, the row span of E must include all the rows of some matrix M of the form (6.10), or equivalently, $r \geq \text{rank}(M)$. By optimizing over all such matrices M, we establish the following upper bound on the linear broadcast rate $\lambda_{\mathbb{F}_2}(G)$ and thus on the broadcast rate $\beta(G)$.

Proposition 6.6 (Binary minrank bound [23])**.** For the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G,

$$\lambda_{\mathbb{F}_2}(G) \leq \text{minrank}_{\mathbb{F}_2}(G) := \min\{\text{rank}(M) \colon M \in \mathbb{F}_2^{n\times n} \text{ satisfies (6.10)}\}.$$

Remark 6.9. Peeters [124] showed that computing $\text{minrank}_{\mathbb{F}_2}(G)$ is NP-hard in $|V(G)|$. El Rouayheb, Chaudhry, and Sprintson [63] showed that this computation problem is in fact NP-complete.

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Remark 6.10. Haviv and Langberg [79] showed that the minrank of a directed Erdös–Rényi random graph $G(n,p)$ is lower bounded by $\Omega(\sqrt{n})$ with high probability (w.h.p.). This bound was improved recently by Golovnev, Regev, and Weinstein [75] as $\text{minrank}_{\mathbb{F}_2}(G(n,p)) = \Theta(n/\log n)$ w.h.p.

A scalar linear code can be similarly constructed in an arbitrary finite field $\mathbb{F}$. Consider any matrix $M \in \mathbb{F}^{n\times n}$ that satisfies

$$M_{ii} \neq 0 \quad \text{and} \quad M_{ij} = 0 \text{ if } j \notin A_i. \tag{6.11}$$

As in the binary case, any such matrix M induces a scalar linear code in $\mathbb{F}$, and conversely any scalar linear code can be represented by one. There is no loss of generality in assuming that $M_{ii} = 1$. Optimizing over all such matrices M, we have the following upper bound on the linear broadcast rate $\lambda_{\mathbb{F}}(G)$ and on the broadcast rate $\beta(G)$.

Proposition 6.7 (General minrank bound [104])**.** For the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G,

$$\lambda_{\mathbb{F}}(G) \leq \text{minrank}_{\mathbb{F}}(G) := \min\{\text{rank}(M)\colon M \in \mathbb{F}^{n\times n} \text{ satisfies (6.11)}\}.$$

The $\text{minrank}_{\mathbb{F}}(G)$ of an undirected graph G was introduced by Haemers [76, 77] as an upper bound of the Shannon capacity $\Theta(G)$ of G [138]. For any $\mathbb{F}$ and undirected (bidirectional) graph G,

$$\alpha(G) \leq \Theta(G) \leq \text{minrank}_{\mathbb{F}}(G) \leq \chi(\overline{G}).$$

In particular, for the 5-message index coding problem in Figure 5.2, since $\Theta(G) = \sqrt{5}$ as shown in the celebrated work of Lovász [103] and $\chi(\overline{G}) = 3$, $\text{minrank}_{\mathbb{F}}(G) = 3$ for every $\mathbb{F}$ (since it must be an integer). Therefore, a scalar linear code is in general insufficient and a vector linear code is needed.

As a further extension of a scalar linear code matrix M satisfying (6.11), Maleki, Cadambe, and Jafar [107] constructed a general vector linear code over $\mathbb{F}$ by considering a block matrix $\mathbb{F}^{nt\times nt}$ consisting of n^2 submatrices $M_{ij} \in \mathbb{F}^{t\times t}$, $i,j \in [n]$, such that

$$\det(M_{ii}) \neq 0 \quad \text{and} \quad M_{ij} = 0 \text{ if } j \neq A_i. \tag{6.12}$$

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Once again there is no loss of generality in assuming that $M_{ii} = I$, the identity matrix. For each $t = 1, 2, \ldots$, the linear broadcast rate is upper bounded as

$$\begin{aligned}\lambda_{\mathbb{F}}(G) &\leq \lambda_{\mathbb{F},t}(G) \\ &:= \frac{1}{t}\min\{\text{rank}(M)\colon M \in \mathbb{F}^{nt\times nt} \text{ satisfies (6.12)}\}. \end{aligned} \tag{6.13}$$

By subadditivity, we have the following multiletter characterization of the linear broadcast rate.

Theorem 6.3 (Linear broadcast rate)**.** For the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G, the linear broadcast rate over $\mathbb{F}$ is

$$\lambda_{\mathbb{F}}(G) = \lim_{t\to\infty} \lambda_{\mathbb{F},t}(G).$$

As for the broadcast rate, the upper bound in (6.13) converges to the *linear* broadcast rate from above. With no known convergent sequence of bounds from below, however, the computability of the linear broadcast rate is not known in general (cf. Open Problem 3.2).

Open Problem 6.1. Does there exist a finite, terminating algorithm that computes the linear broadcast rate of an arbitrary n-message index coding problem within any desired precision?

All linear coding schemes from scalar (integral) MDS codes to vector (fractional) recursive codes discussed in Sections 6.1 through 6.6 are essentially independent of the code alphabet $\mathbb{F}$. As long as MDS coding, or more generally, local time sharing, can be performed, the achievable rates of the coding schemes are invariant in $\mathbb{F}$. For general linear codes, however, the minrank or linear broadcast rate depends heavily on the code alphabet.

Most notably, Lubetzky and Stav [104] showed that there exists a sequence of n-message index coding problem G_n such that

$$\text{minrank}_{\mathbb{F}}(G_n) \leq n^{\epsilon} \quad \text{and} \quad \text{minrank}_{\mathbb{F}'}(G_n) \geq n^{1-\epsilon}$$

for $\epsilon \ll 1$ and two finite fields $\mathbb{F}$ and $\mathbb{F}'$ of distinct characteristics. This example, inter alia, shows that the performance of optimal scalar linear codes depends on the choice of the finite field in a strong sense.

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The linear broadcast rate $\lambda_{\mathbb{F}}$ itself (achieved in general by a sequence of *vector* linear codes) depends on $\mathbb{F}$ as well. This dependence relies on a deep connection between the linear representation of matroids and network coding, which was originally explored in the seminal work by Dougherty, Freiling, and Zeger [58]; see also [41]. This connection has been adapted to index coding by El Rouayheb, Sprintson, Georghiades [62] and subsequently by Blasiak, Kleinberg, and Lubetzky [29], who showed that there exists an index coding problem for which the linear broadcast rate $\lambda_{\mathbb{F}}$ is a function of $\mathbb{F}$.

The alphabet dependence of linear index codes has been strengthened and amplified to establish the insufficiency of linear codes compared to general nonlinear codes. For scalar linear codes, Lubetzky and Stav [104] constructed a sequence of n-message index coding problems G_n such that $\mathrm{minrank}_{\mathbb{F}}(G_n) \geq \sqrt{n}$ for any $\mathbb{F}$, yet $\beta(G_n) \leq n^{\epsilon}$ for $\epsilon \ll 1$, which separates the minrank from the broadcast rate in a rather strong sense. For vector linear codes, the strongest separation result by Blasiak, Kleinberg, and Lubetzky [29] considers *multiple groupcast* (multiple receivers per message) and shows that there exists an n_0-message problem G_0 such that its lexicographic product $G_n = G_0^{\circ k}$ with $n = n_0^k$ messages satisfies

$$\frac{\lambda_{\mathbb{F}}(G_n)}{\beta(G_n)} \geq \rho^k = n^{\log_{n_0}(\rho)}$$

for some $\rho > 1$. This result can be translated to the regular (multiple-unicast) index coding problem using a technique by Maleki, Cadambe, and Jafar [107] that associates an arbitrary groupcast problem with some multiple-unicast problem. Hence, linear codes are insufficient in general and the optimal rates of scalar and vector linear codes are both polynomially separated from the broadcast rate. It is not known how much vector coding improves upon scalar coding.

Open Problem 6.2. Is there a sequence of n-message index coding problems G_n such that

$$\frac{\mathrm{minrank}_{\mathbb{F}}(G_n)}{\lambda_{\mathbb{F}}(G_n)} \geq n^{\epsilon}$$

for some $\epsilon > 0$?

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6.8 Linear Codes Based on Interference Alignment

Motivated by interference alignment coding schemes for wireless interference channels [105, 35], Maleki, Cadambe, and Jafar [107], [83, Sec. IV] proposed a class of vector linear codes that can be constructed in a systematic manner. In this section, we discuss some examples to demonstrate how the interference alignment ideas can be used in designing such codes, as opposed to the intractable combinatorial optimization problem of minimizing the rank over all nt-by-nt matrices in Theorem 6.3. The general idea is to align undesired messages (interference) as much as possible while keeping the desired message separate from the interference.

To present the first interference alignment idea, which constructs a scalar linear code, we introduce the following notions: For distinct $i, j \in [n]$, we denote the *alignment relation* $i \Leftrightarrow j$ iff $i \notin A_k$ and $j \notin A_k$ for some $k \neq i, j$. The set of messages is partitioned into *alignment subsets* such that if $i \Leftrightarrow j$, then i and j belong to the same alignment subset. In particular, if $i \Leftrightarrow j$ and $j \Leftrightarrow k$, then i, j, and k belong to the same alignment subset.

As an example, consider the index coding problem depicted in Figure 6.10. In this example, the only alignment relation is $2 \Leftrightarrow 3$ (since $2, 3 \notin A_1$) and G can be partitioned into alignment subsets $\{1\}$, $\{2, 3\}$, and $\{4\}$. The alignment subset $\{2, 3\}$ is the only one with more than one member and we have $2 \in A_3$ and $3 \in A_2$. Consider three vectors $\mathbf{v}_1 = [1\ 0]^T$, $\mathbf{v}_2 = [0\ 1]^T$, and $\mathbf{v}_3 = [1\ 1]^T$ on $\mathbb{F}_2$, with the property that every two of them are linearly independent. Transmitting x_1 along $\mathbf{v}_1$, x_2 and x_3 along $\mathbf{v}_2$, and x_4 along $\mathbf{v}_3$, we have

$$y = x_1\mathbf{v}_1 + (x_2 + x_3)\mathbf{v}_2 + x_4\mathbf{v}_3.$$

Receiver 1 has x_4 as side information; so it can cancel the term $x_4\mathbf{v}_3$ from the received sum. Since vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent, receiver 1 can successfully recover x_1 (as well as unnecessary $x_2 + x_3$). Receiver 2 has x_1 and x_3 as side information and can cancel the terms $x_1\mathbf{v}_1$ and $x_3\mathbf{v}_2$. Since vectors $\mathbf{v}_2$ and $\mathbf{v}_3$ are linearly independent, receiver 2 can successfully recover x_2. Similarly, we can argue that receivers 3 and 4 can also recover their desired message successfully.

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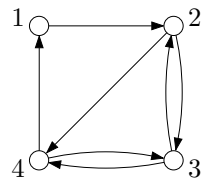


Figure 6.10: The index coding problem $(1|4), (2|1,3), (3|2,4), (4|2,3)$ with $\beta = 2$, which is achievable by the scalar interference alignment scheme.

In general, it suffices to transmit two linear combinations of the messages if every pair in the same alignment subset has each other as side information.

Proposition 6.8 ([83, 107])**.** Suppose that G can be partitioned into alignment subsets such that for every pair (i, j) in an alignment subset, $i \in A_j$ and $j \in A_i$. Then $\beta(G) \leq 2$.

To prove this, partition the set of messages $V(G) = [n]$ into alignment subsets $V_1, \ldots, V_m$. Define the mapping $f : [n] \to [m]$ so that $i \in V_{f(i)}$, $i \in [n]$. Choose the field size q sufficiently large so that there exist m vectors $\mathbf{v}_1, \ldots, \mathbf{v}_m$ in the two dimensional vector space over $\mathbb{F}_q$ with every pair linearly independent. We transmit the messages with indices in V_j along the vector $\mathbf{v}_j$, $j \in [m]$. Receiver i then receives

$$\begin{aligned}\sum_{j\in[m]} x_j \mathbf{v}_{f(j)} &= x_i \mathbf{v}_{f(i)} + \sum_{j\in A_i} x_j \mathbf{v}_{f(j)} + \sum_{j\notin\{i\}\cup A_i} x_j \mathbf{v}_{f(j)} \\ &= x_i \mathbf{v}_{f(i)} + \sum_{j\in A_i} x_j \mathbf{v}_{f(j)} + \left(\sum_{j\notin\{i\}\cup A_i} x_j\right) \mathbf{v}_l, \qquad (6.14)\end{aligned}$$

where (6.14) follows since all $j \notin \{i\} \cup A_i$ belong to the same alignment set V_l by definition. After canceling the second term that depends solely on side information messages, receiver i obtains a linear combination of the two vectors $\mathbf{v}_{f(i)}$ and $\mathbf{v}_l$. By the assumption that every pair $(j, k) \in V_l$ has each other as side information, the two partitions $V_{f(i)}$ and V_l are distinct, and thus $\mathbf{v}_{f(i)}$ and $\mathbf{v}_l$ are linearly independent. Hence, x_i can be successfully recovered at receiver $i \in [n]$ and the upper bound of 2 on the broadcast rate is achievable.

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Next, we use interference alignment to design vector linear index codes for a class of index coding problems with symmetric neighboring side information. Let $0 \leq d \leq u \leq u+d \leq n-2$ or $(u,d) = (n-1,0)$ (the latter of which corresponds to a complete G), and consider the index coding problem with

$$A_i = \{i \ominus d, \ldots, i \ominus 1, i \oplus 1, \ldots, i \oplus u\}, \quad i \in [n], \tag{6.15}$$

where $\oplus$ and $\ominus$ denote addition and subtraction modulo n (over $[n]$), respectively. As an example, consider the index coding problem depicted in Figure 6.11 with $n = 6$, $u = 2$, and $d = 1$. Consider 6 vectors $\mathbf{v}_1, \ldots, \mathbf{v}_6$ such that any 5 of them are linearly independent. For example, $\mathbf{v}_1 = [1\ 0\ 0\ 0\ 0]^T$, $\mathbf{v}_2 = [0\ 1\ 0\ 0\ 0]^T$, $\mathbf{v}_3 = [0\ 0\ 1\ 0\ 0]^T$, $\mathbf{v}_4 = [0\ 0\ 0\ 1\ 0]^T$, $\mathbf{v}_5 = [0\ 0\ 0\ 0\ 1]^T$, and $\mathbf{v}_6 = [1\ 1\ 1\ 1\ 1]^T$ over $\mathbb{F}_2$. Decompose message x_i into 2 independent messages x_{i0} and x_{i1} and transmit x_{i0} along $\mathbf{v}_i$ and x_{i1} along $\mathbf{v}_{i\oplus 1}$. Thus, receiver i receives

$$\begin{aligned}&(x_{10}+x_{61})\mathbf{v}_1 + (x_{20}+x_{11})\mathbf{v}_2 + (x_{30}+x_{21})\mathbf{v}_3\\ &\quad + (x_{40}+x_{31})\mathbf{v}_4 + (x_{50}+x_{41})\mathbf{v}_5 + (x_{60}+x_{51})\mathbf{v}_6.\end{aligned}$$

By symmetry, it suffices to show that receiver 1 can successfully recover its desired message $x_1 = (x_{10}, x_{11})$. Note that receiver 1 has side information of $x_{20}, x_{21}, x_{30}, x_{31}, x_{60}, x_{61}$. Canceling the effect of these messages, receiver 1 obtains

$$x_{10}\mathbf{v}_1 + x_{11}\mathbf{v}_2 + (x_{40}+x_{31})\mathbf{v}_4 + (x_{50}+x_{41})\mathbf{v}_5 + x_{51}\mathbf{v}_6,$$

which is a linear combination of 5 independent vectors and thus receiver 1 can successfully recover its desired messages x_{10} and x_{11}. Hence,

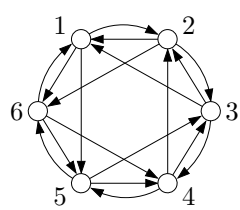


Figure 6.11: The index coding problem $(1|2,3,6)$,$(2|1,3,4)$,$(3|2,4,5)$,$(4|3,5,6)$, $(5|1,4,6)$, $(6|1,2,5)$ with symmetric neighbors ($u = 2$ and $d = 1$), for which $\beta_{\mathrm{PM}} = \beta = 5/2$ is achievable by the vector interference alignment scheme.

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this scheme achieves the upper bound of 5/2 on the broadcast rate, which coincides with the polymatroidal lower bound.

In general, if $(u, d) = (n-1, 0)$, then, by the clique covering scheme, $\beta = 1$ is achievable. If $u + d = n - 2$, then each message forms its own alignment subset and, by Proposition 6.8, the upper bound of 2 on β is achievable. Otherwise ($u+d \leq n-3$), let $r = n-(u-d)$ and $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be r-dimensional vectors over $\mathbb{F}_q$, where q is sufficiently large that any r of $\mathbf{v}_i$, $i \in [n]$, are linearly independent. Decompose message x_i into $d+1$ parts $x_{i0}, \ldots, x_{id}$ and transmit x_{ij} along $\mathbf{v}_{i\oplus j}$. Receiver i receives

$$\sum_{i=1}^{n} \sum_{j=0}^{d} x_{ij} \mathbf{v}_{i\oplus j}.$$

By symmetry, it suffices to show that receiver 1 can successfully recover x_1. Receiver 1 can cancel the effect of its side information x_{ij} for all $i \in \{n-d+1, \ldots, n, 2, 3, \ldots, u+1\}$ and obtain

$$\sum_{j=0}^{d} x_{1j} \mathbf{v}_{1\oplus j} + \sum_{i=u+2}^{n-d} \sum_{j=0}^{d} x_{ij} \mathbf{v}_{i\oplus j}.$$

This expression is the linear combination of $r = n - (u - d)$ vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{d+1}, \mathbf{v}_{u+2}, \ldots, \mathbf{v}_n$ that are linearly independent. Therefore, receiver 1 can recover its desired messages x_{1j}, $j = 0, \ldots, d$. This proves the following.

Proposition 6.9 ([83, 107]). For the index coding problem G with symmetric neighboring side information sets in (6.15), the broadcast rate is upper bounded as

$$\beta(G) \leq \frac{n - (u - d)}{d + 1}.$$

As illustrated so far, interference alignment provides a systematic and intuitive framework to construct scalar or vector linear codes for some index coding problems with certain structures (see [83, 107] for more examples). For a general index coding problem, however, it is unclear how the interference alignment perspective can be applicable beyond the obvious interpretation of the minrank characterization of the linear broadcast rate in Theorem 6.3, in which interfering messages are suppressed by the zero entries of the coding matrix M.

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6.9 Flat Coding

In this section and the next, we use Shannon's random coding idea [137] to prove the existence of index codes of certain rates. Earlier in Section 3.3, we applied a simple form of random coding to achieve the entire capacity region. This development was not quite satisfactory, however, since the capacity region was characterized in a noncomputable multiletter expression. Here we take the conceptually simple random coding approach to the next level and apply it in a more concrete form as in the original proof of the network coding theorem [2]. This development leads to rather straightforward derivation of a computable inner bound on the capacity region, at the cost of lacking the actual code construction as opposed to the coding schemes we have discussed in the preceding sections.

We first illustrate a simple random coding scheme, referred to as *flat coding*, through an example. Consider the index coding problem $(1|2), (2|1,3), (3|1)$ with side information graph depicted in Figure 6.12. We use the following random coding argument to find conditions under which a nonnegative rate triple (R_1, R_2, R_3) is achievable *with vanishing error probability* (see Appendix 1.A).

Fix an integer r and let $t_i = \lceil rR_i \rceil, i \in [3]$. For each message triple $(x_1, x_2, x_3) \in \Pi_{i=1}^{3}[2^{t_i}]$, generate a codeword $y(x_1, x_2, x_3)$ drawn uniformly at random from $[2^r]$. These randomly generated codewords constitute a *codebook*, which is shared among all communicating parties. Note that codebook generation is "flat" over all messages; hence this scheme is called flat coding. To communicate message triple (x_1, x_2, x_3), we transmit $y = y(x_1, x_2, x_3)$.

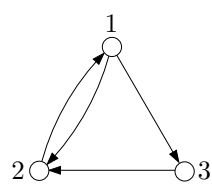


Figure 6.12: The index coding problem $(1|2), (2|1,3), (3|1)$, the capacity region of which is achieved by the flat coding scheme.

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Based on y and its side information x_2, receiver 1 finds a message pair $(\hat{x}_1, \hat{x}_3)$ such that $y(\hat{x}_1, x_2, \hat{x}_3) = y$. Assuming that the true message triple is (x_1, x_2, x_3), the probability of decoding error at receiver 1, that is, the probability that there exists another $(\tilde{x}_1, \tilde{x}_3) \neq (x_1, x_3)$ such that $y(\tilde{x}_1, x_2, \tilde{x}_3) = y$, is upper bounded as

$$\begin{aligned} P_1 &= \mathsf{P}\{y(\tilde{x}_1, x_2, \tilde{x}_3) = y \text{ for some } (\tilde{x}_1, \tilde{x}_3) \neq (x_1, x_3)\} \\ &\leq \sum_{(\tilde{x}_1, \tilde{x}_3) \neq (x_1, x_3)} \mathsf{P}\{y(\tilde{x}_1, x_2, \tilde{x}_3) = y\} \qquad (6.16) \\ &\leq \frac{2^{t_1} 2^{t_3} - 1}{2^r} \qquad (6.17) \\ &\leq \frac{2^{rR_1+1} 2^{rR_3+1}}{2^r}, \end{aligned}$$

where (6.16) follows by the union of the events bound and (6.17) holds since the number of wrong triples $(\tilde{x}_1, x_2, \tilde{x}_3)$ is $2^{t_1} 2^{t_3} - 1$ and codewords assigned to two different message triples are the same with probability $1/2^r$ from the random codebook construction. Thus, P_1 tends to zero as $r \to \infty$ if $R_1 + R_3 < 1$.

Based on y and its side information (x_1, x_3), receiver 2 finds a message $\hat{x}_2$ such that $y(x_1, \hat{x}_2, x_3) = y$, which is unique w.h.p. if $R_2 < 1$. Following similar steps, decoding at receiver 3 is successful w.h.p. if $R_2 + R_3 < 1$. Thus, if the specified rate constraints are satisfied simultaneously, the probability of error averaged over messages *and* codebooks tends to zero as $r \to \infty$, and there must exist a sequence of $(\lceil rR_1 \rceil, \ldots, \lceil rR_n \rceil, r)$ index codes such that the probability of error P_e, averaged over the messages, tends to zero as $r \to \infty$. By Lemma 1.2, this error probability can be made to be exactly zero without sacrificing the rates. In summary, this flat coding scheme asymptotically achieves the upper bound of 2 on the broadcast rate and the inner bound on the capacity region consisting of all nonnegative rate triples (R_1, R_2, R_3) such that

$$\begin{aligned} R_1 + R_3 &\leq 1, \\ R_2 + R_3 &\leq 1. \end{aligned}$$

Since the subgraphs $G|_{\{1,3\}}$ and $G|_{\{2,3\}}$ are acyclic, this inner bound matches the MAIS outer bound, characterizing the capacity region.

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More generally, consider an index coding problem with n messages of rates $R_1, \ldots, R_n$. Fix an integer r and let $t_i = \lceil rR_i \rceil, i \in [n]$. For each $(x_1, \ldots, x_n) \in [2^{t_1}] \times \cdots \times [2^{t_n}]$, generate a codeword $y(x_1, \ldots, x_n)$ drawn uniformly at random from $[2^r]$ and share the codebook among all communicating parties. To communicate message tuple $(x_1, \ldots, x_n)$, we transmit $y = y(x_1, \ldots, x_n)$.

Let $B_i = [n] \setminus (\{i\} \cup A_i)$ denote the set of interfering messages at receiver $i \in [n]$. Receiver i finds a message tuple $(\hat{x}_i, \hat{x}(B_i))$ such that $y(\hat{x}_i, x(A_i), \hat{x}(B_i)) = y$. Then the probability of decoding error at receiver i, which is the probability that there is another tuple $(\tilde{x}_i, \tilde{x}(B_i)) \neq (x_i, x(B_i))$ such that $y(\tilde{x}_i, x(A_i), \tilde{x}(B_i)) = y$, can be upper bounded as

$$\begin{aligned} P_i &= \mathsf{P}\{y(\tilde{x}_i, x(A_i), \tilde{x}(B_i)) = y \text{ for some } (\tilde{x}_i, \tilde{x}(B_i)) \neq (x_i, x(B_i))\} \\ &\leq \frac{(2^{t_i} \cdot \prod_{j \in B_i} 2^{t_j}) - 1}{2^r} \\ &\leq \frac{2^{rR_i+1} \cdot \prod_{j \in B_i} 2^{rR_j+1}}{2^r}. \end{aligned}$$

Therefore, P_i tends to zero as $r \to \infty$ if $R_i + \sum_{j \in B_i} R_j = \sum_{j \notin A_i} R_j < 1$. By taking similar steps as before, we can conclude the following.

Proposition 6.10 (Flat coding inner bound)**.** The capacity region of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G includes the rate region $\mathscr{R}_{\text{flat}}(G)$ that consists of all nonnegative rate tuples $(R_1, \ldots, R_n)$ satisfying

$$\sum_{j \notin A_i} R_j \leq 1, \quad i \in [n].$$

In particular,

$$\beta(G) \leq \beta_{\text{flat}}(G) := \kappa(G) = n - \min_{i \in [n]} |A_i|.$$

Remark 6.11. The flat coding upper bound $\beta_{\text{flat}}(G)$ on the broadcast rate is identical to the bound $\beta_{\text{MDS}}(G)$ achieved by the MDS codes (see Section 6.1). The flat coding scheme can be thus interpreted as a random construction of an *approximate* MDS code, which achieves $\beta_{\text{MDS}}(G)$ asymptotically.

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6.10 Composite Coding

We now discuss a more powerful random coding scheme, referred to as *composite coding*, that is built upon flat coding. For a concrete example, we revisit the index coding problem with side information graph depicted in Figure 6.10 on page 245. We develop a composite coding scheme that uses message rates $R_1, \ldots, R_4$ as well as two *composite rates* $S_{\{1,4\}}$ and $S_{\{2,3,4\}}$.

Fix an integer r and let $t_i = \lceil rR_i \rceil$, $i \in [4]$, and $s_J = \lceil rS_J \rceil$, $J = \{1,4\}$ and $\{2,3,4\}$. We use a *concatenated* codebook structure. As the inner code, we map (x_1, x_4) to a *composite* index $w_{\{1,4\}} = w_{\{1,4\}}(x_1, x_4) \in [2^{s_{\{1,4\}}}]$ and (x_2, x_3, x_4) to a composite index $w_{\{2,3,4\}} = w_{\{2,3,4\}}(x_2, x_3, x_4) \in [2^{s_{\{2,3,4\}}}]$. As the outer code, we map the composite indices $w_{\{1,4\}}$ and $w_{\{2,3,4\}}$ to a codeword $y = y(w_{\{1,4\}}, w_{\{2,3,4\}}) \in [2^r]$ and transmit it. As in flat coding, both inner and outer codes are generated uniformly at random from respective codeword sets and shared among all communicating parties.

Decoding is performed backwards. First, each receiver i decodes y to recover the composite indices $w_{\{1,4\}}$ as $\hat{w}_{\{1,4\}}$ and $w_{\{2,3,4\}}$ as $\hat{w}_{\{2,3,4\}}$, which is successful w.h.p. if

$$S_{\{1,4\}} + S_{\{2,3,4\}} < 1.$$

Second, receiver i recovers x_i and some other messages simultaneously using $x(A_i)$ and $(\hat{w}_{\{1,4\}}, \hat{w}_{\{2,3,4\}})$, assuming that the latter is correct. More specifically, receiver 1 recovers x_1 from $\hat{w}_{\{1,4\}}$ and side information x_4, which is successful w.h.p. if

$$R_1 < S_{\{1,4\}}.$$

Similarly, receiver 3 recovers x_3 from $\hat{w}_{\{2,3,4\}}$ and (x_2, x_4), which is successful w.h.p. if

$$R_3 < S_{\{2,3,4\}},$$

and receiver 4 recovers x_4 from $\hat{w}_{\{2,3,4\}}$ and (x_2, x_3), which is successful w.h.p. if

$$R_4 < S_{\{2,3,4\}}.$$

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Receiver 2 recovers x_2 and x_4 from $(\hat{w}_{\{1,4\}}, \hat{w}_{\{2,3,4\}})$ and (x_1, x_3). More specifically, it finds a message pair $(\hat{x}_2, \hat{x}_4)$ such that

$$w_{\{1,4\}}(x_1, \hat{x}_4) = \hat{w}_{\{1,4\}} \tag{6.18a}$$

and

$$w_{\{2,3,4\}}(\hat{x}_2, x_3, \hat{x}_4) = \hat{w}_{\{2,3,4\}}. \tag{6.18b}$$

To analyze the probability of error, we partition the event that some wrong pair $(\tilde{x}_2, \tilde{x}_4) \neq (x_2, x_4)$ satisfies (6.18) into three cases: (1) $\tilde{x}_2 = x_2, \tilde{x}_4 \neq x_4$, (2) $\tilde{x}_2 \neq x_2, \tilde{x}_4 = x_4$, and (3) $\tilde{x}_2 \neq x_2, \tilde{x}_4 \neq x_4$. Among these three cases, the first does not incur a decoding error. Thus, the probability of decoding error at receiver 2 can be decomposed as

$$\begin{aligned} P_2 = {} & \mathsf{P}\{\text{(6.18) holds for some } \tilde{x}_2 \neq x_2, \tilde{x}_4 = x_4\} \\ & + \mathsf{P}\{\text{(6.18) holds for some } \tilde{x}_2 \neq x_2, \tilde{x}_4 \neq x_4\}. \end{aligned} \tag{6.19}$$

The first term in (6.19) can be upper bounded by

$$\begin{aligned} \sum_{\tilde{x}_2 \neq x_2} \mathsf{P}\{w_{\{2,3,4\}}(\tilde{x}_2, x_3, x_4) = \hat{w}_{\{2,3,4\}}\} &\leq \frac{2^{t_2} - 1}{2^{s_{\{2,3,4\}}}} \\ &\leq \frac{2^{rR_2+1}}{2^{rS_{\{2,3,4\}}}}, \end{aligned}$$

which tends to zero as $r \to \infty$ if $R_2 < S_{\{2,3,4\}}$. The second term in (6.19) can be similarly upper bounded by

$$\begin{aligned} \sum_{\tilde{x}_2 \neq x_2, \tilde{x}_4 \neq x_4} \mathsf{P}\{w_{\{1,4\}}(x_1, \tilde{x}_4) = \hat{w}_{\{1,4\}} \text{ and } w_{\{2,3,4\}}(\tilde{x}_2, x_3, \tilde{x}_4) = \hat{w}_{\{2,3,4\}}\} & \\ \leq \frac{(2^{t_2} - 1)(2^{t_4} - 1)}{2^{s_{\{1,4\}}} 2^{s_{\{2,3,4\}}}} & \\ \leq \frac{2^{rR_2+1} 2^{rR_4+1}}{2^{rS_{\{1,4\}}} 2^{rS_{\{2,3,4\}}}}, & \end{aligned}$$

which tends to zero as $r \to \infty$ if $R_2 + R_4 < S_{\{1,4\}} + S_{\{2,3,4\}}$. In summary, decoding at receiver 2 is successful w.h.p. if

$$\begin{aligned} R_2 &< S_{\{2,3,4\}}, \\ R_2 + R_4 &< S_{\{1,4\}} + S_{\{2,3,4\}}. \end{aligned}$$

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Combining all the rate conditions, this composite coding scheme can achieve any nonnegative rate quadruple (R_1, R_2, R_3, R_4) satisfying

$$\begin{aligned} R_1 &< S_{\{1,4\}}, \\ R_2 &< S_{\{2,3,4\}}, \\ R_3 &< S_{\{2,3,4\}}, \\ R_4 &< S_{\{2,3,4\}}, \\ R_2 + R_4 &< S_{\{1,4\}} + S_{\{2,3,4\}} \end{aligned}$$

for some $S_{\{1,4\}} > 0$ and $S_{\{2,3,4\}} > 0$ such that

$$S_{\{1,4\}} + S_{\{2,3,4\}} < 1.$$

After Fourier–Motzkin elimination of the composite rates $S_{\{1,4\}}$ and $S_{\{2,3,4\}}$, and taking the limit of achievable rates, the resulting composite coding inner bound on the capacity region is characterized by

$$\begin{aligned} R_1 + R_2 &\le 1, \\ R_1 + R_3 &\le 1, \\ R_1 + R_4 &\le 1, \\ R_2 + R_4 &\le 1. \end{aligned}$$

This matches the MAIS outer bound and thus is the capacity region. Note that the broadcast rate is 2, which is not achievable by MDS codes or flat coding.

We now generalize the coding scheme by introducing composite indices for each nonempty $J \subseteq [n]$ and optimizing over the decoding set at each receiver. The coding scheme involves n message rates $R_1, \ldots, R_n$ and $(2^n - 1)$ composite rates S_J, $J \subseteq [n]$, $J \neq \emptyset$. For simplicity of notation, we set $S_\emptyset = 0$ by convention. The overall coding scheme is illustrated in Figure 6.13.

Fix an integer r and let $t_i = \lceil rR_i \rceil$, $i \in [n]$, and $s_J = \lceil rS_J \rceil$, $J \subseteq [n]$. As the inner code, for each J and $x(J)$, generate a composite index $w_J(x(J))$ drawn uniformly at random from $[2^{s_J}]$. As the outer code, for each $(w_J, J \subseteq [n])$, generate a codeword $y(w_J, J \subseteq [n])$ drawn uniformly at random from $[2^r]$. The codebook consisting of thus generated $w_J(\cdot)$, $J \subseteq [n]$, and $y(\cdot)$ is shared among all communicating parties. To

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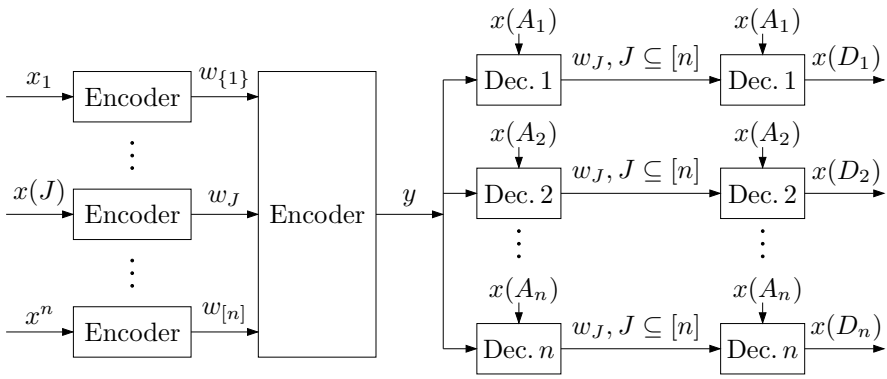


Figure 6.13: Composite coding scheme. Encoding is performed in two steps— first mapping $x(J)$ to w_J and then mapping $w_J, J \subseteq [n]$ to y. Decoding is also performed in two steps—first recovering $w_J, J \subseteq [n]$ and then recovering $x(D_i)$.

communicate $(x_1, \ldots, x_n)$, we first encode $w_J = w_J(x(J))$ for each J and then transmit $y = y(w_J, J \subseteq [n])$.

Decoding is performed in two steps. First, receiver i recovers the composite indices $w_J, J \subseteq [n]$, using y as well as its side information $x(A_i)$. More specifically, it finds a composite index tuple $(\hat{w}_J, J \nsubseteq A_i)$ such that $y((\hat{w}_J, J \nsubseteq A_i), (w_J(x(J)), J \subseteq A_i))$ is identical to the received y. Note that $w_J(x(J))$ for $J \subseteq A_i$ is already available at receiver i from its side information $x(A_i)$. As in the analysis of flat coding in Section 6.9, decoding in the first step is successful w.h.p. if

$$\sum_{J \subseteq [n]: J \nsubseteq A_i} S_J < 1. \tag{6.20}$$

This condition can be interpreted as local time sharing among composite indices, which corresponds to a similar operation of local time sharing among subproblems as in (6.5b) and (6.6b).

Second, assuming the composite indices were recovered correctly as $\hat{w}_J = w_J$, receiver i recovers a subset of messages $x(D_i)$ for some $D_i \subseteq [n] \setminus A_i$ that contains i, based on $x(A_i)$ and the composite indices $\hat{w}_J$, $J \subseteq [n]$. More specifically, it finds a message tuple $\hat{x}(D_i)$ such that

$$w_J(\hat{x}(J \cap D_i), x(J \cap A_i)) = \hat{w}_J, \qquad J \subseteq D_i \cup A_i.$$

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The probability of decoding error in the second step at receiver i is analyzed in Appendix 6.B, which tends to zero as $r \to \infty$ if

$$\sum_{j \in L} R_j < \sum_{J \subseteq D_i \cup A_i : J \cap L \neq \emptyset} S_J \tag{6.21a}$$

for every $L \subseteq [n]$ such that

$$i \in L \subseteq D_i. \tag{6.21b}$$

Combining the rate constraints for all receivers and taking similar steps as before, we can conclude the following.

Proposition 6.11 (Composite coding inner bound)**.** The capacity region of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G includes the rate region

$$\mathscr{R}_{\text{comp}}(G) = \text{co}\left(\bigcup_{\mathbf{D} \in \Delta} \bigcup_{\mathbf{S} \in \Sigma} \bigcap_{i \in [n]} \mathscr{R}_{D_i|A_i}(\mathbf{S}) \right), \tag{6.22}$$

where $\text{co}(\cdot)$ denotes the convex hull, $\Delta = \{\mathbf{D} \colon i \in D_i \subseteq [n] \backslash A_i,\ i \in [n]\}$, $\Sigma = \{\mathbf{S} \colon (6.20) \text{ holds with nonstrict inequalities for } i \in [n]\}$, and the rate region $\mathscr{R}_{D_i|A_i}(\mathbf{S})$ consists of all rate tuples $(R_1, \ldots, R_n)$ satisfying (6.21) with nonstrict inequalities. In particular,

$$\beta(G) \leq \beta_{\text{comp}}(G) := \min_{R : (R, \ldots, R) \in \mathscr{R}_{\text{comp}}(G)} \frac{1}{R}. \tag{6.23}$$

Remark 6.12. It can be easily verified that the composite coding inner bound can be equivalently expressed as

$$\begin{aligned} \mathscr{R}_{\text{comp}}(G) &= \text{co}\left(\bigcup_{\mathbf{S} \in \Sigma} \bigcap_{i \in [n]} \bigcup_{D_i : i \in D_i} \mathscr{R}_{D_i|A_i}(\mathbf{S}) \right) \\ &= \text{co}\left(\bigcap_{i \in [n]} \bigcup_{D_i : i \in D_i} \bigcup_{\mathbf{S} \in \Sigma} \mathscr{R}_{D_i|A_i}(\mathbf{S}) \right) \\ &= \text{co}\left(\bigcup_{\mathbf{D} \in \Delta} \bigcap_{i \in [n]} \bigcup_{\mathbf{S} \in \Sigma} \mathscr{R}_{D_i|A_i}(\mathbf{S}) \right). \end{aligned}$$

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For each fixed $\mathbf{D}$, the rate region $\cup_{\mathbf{S}} \cap_i \mathscr{R}_{D_i|A_i}(\mathbf{S})$ in (6.22) is a convex polytope. Hence, computing this region is equivalent to Fourier–Motzkin elimination of $\mathbf{S}$ from linear inequalities in (6.20) and (6.21).

Remark 6.13. An upper bound on the broadcast rate can be similarly characterized as the solution $\beta_{\text{comp}(\mathbf{D})}(G)$ to the linear program

$$\begin{aligned} \text{minimize} \quad & \max_{i\in[n]} \sum_{J\subseteq[n]:J\not\subseteq A_i} \gamma_J \\ \text{subject to} \quad & \min_{i\in[n]} \min_{i\in L\subseteq D_i} \frac{1}{|L|} \sum_{J\subseteq D_i\cup A_i J\cap L\neq\emptyset} \gamma_J \geq 1, \\ & \gamma_J \geq 0, \quad J\subseteq[n]. \end{aligned}$$

This follows by setting $\gamma_J = \beta_{\text{comp}(\mathbf{D})} S_J$, $J \subseteq [n]$, and $R_i = 1/\beta_{\text{comp}(\mathbf{D})}$, $i \in [n]$, and rearranging the expressions in (6.20) and (6.21). Optimizing over $\mathbf{D}$, we have the upper bound on the broadcast rate as

$$\beta(G) \leq \beta_{\text{comp}}(G) \leq \min_{\mathbf{D}} \beta_{\text{comp}(\mathbf{D})}(G), \tag{6.24}$$

where $\beta_{\text{comp}}(G)$ is defined in (6.23) and derived directly from the composite coding inner bound $\mathscr{R}_{\text{comp}}(G)$. The second inequality in (6.24) is strict in general, as illustrated by the 6-message index coding problem shown in Figure 6.14 (see [10, Example 6.1] for more details).

In computing $\mathscr{R}_{\text{comp}}$ in (6.22), we take the union over all $\mathbf{D} \in \Delta$, since at each receiver i there can be a tradeoff in choosing the decoding set D_i. On the one hand, if we recover more messages at receiver i

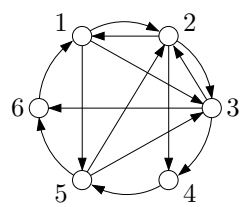


Figure 6.14: The index coding problem $(1|2,6)$,$(2|1,3,5)$,$(3|1,2,5)$,$(4|2,3)$,$(5|1,4)$, $(6|3,5)$ with $\beta_{\text{comp}} = 10/3 < \min_{\mathbf{D}} \beta_{\text{comp}(\mathbf{D})} = 7/2$. For this problem $\beta_{\text{MAIS}} = \beta = 3$, which is achievable by the scalar linear code $(x_1+x_2+x_6, x_1+x_4+x_5, x_3+x_5+x_6)$ over $\mathbb{F}_2$.

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(i.e., choose a larger set D_i in (6.21)), then more composite indices contribute in recovering x_i (with more J on the RHS of (6.21)). On the other hand, recovering more messages results in more constraints on the set of achievable rates (with each $L \subseteq D_i$ corresponding to a distinct inequality in (6.21)). Similar to the case of interference networks [21], the union over different decoding sets can be achieved automatically by maximum likelihood decoding for the message x_i at receiver i, without separately specifying particular decoding sets. Also note that with the union of all decoding sets, the constraint $i \in L$ in (6.21b) becomes irrelevant and the resulting union is identical with or without it.

Time sharing among different $\mathbf{D}$ in (6.22) is embodied by the convex hull operation of the union over the rate regions for different $\mathbf{D} \in \Delta$. Similar to coded time sharing for interference channels [61, Remark 6.1] that improves upon uncoded time sharing, the composite coding scheme can be further improved by allocating the message rates $R_i(\mathbf{D})$ and composite rates $S_J(\mathbf{D})$ over different $\mathbf{D}$ and aggregating these rates.

Theorem 6.4 (Enhanced composite coding inner bound [97])**.** The capacity region of the index coding problem $(i|A_i)$, $i \in [n]$, with side information graph G includes the rate region $\mathscr{R}^*_{\text{comp}}(G)$ on the capacity region that consists of all rate tuples $(R_1, \ldots, R_n)$ satisfying

$$R_i = \sum_{\mathbf{D} \in \Delta} R_i(\mathbf{D}), \quad i \in [n], \tag{6.25a}$$

for some $R_i(\mathbf{D})$, $i \in [n]$, $\mathbf{D} \in \Delta$, and $S_J(\mathbf{D})$, $J \subseteq [n]$, $\mathbf{D} \in \Delta$, such that for each $i \in [n]$,

$$\sum_{j \in L} R_j(\mathbf{D}) \leq \sum_{J \subseteq D_i \cup A_i : J \cap L \neq \emptyset} S_J(\mathbf{D}), \quad i \in L \subseteq D_i, \tag{6.25b}$$

and

$$\sum_{J \subseteq [n] : J \not\subseteq A_i} \sum_{\mathbf{D} \in \Delta} S_J(\mathbf{D}) \leq 1. \tag{6.25c}$$

It can be easily checked that the enhanced composite coding inner bound $\mathscr{R}^*_{\text{comp}}(G)$ is convex. Since $\mathscr{R}^*_{\text{comp}}(G)$ includes $\cup_{\mathbf{S}} \cap_i \mathscr{R}_{D_i|A_i}(\mathbf{S})$ for each $\mathbf{D}$, it follows that

$$\mathscr{R}_{\text{comp}}(G) \subseteq \mathscr{R}^*_{\text{comp}}(G).$$

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This inclusion [97] is strict in general, as illustrated by the 6-message index coding problem in Figure 6.15.

The composite coding scheme performs better than most structured linear codes we discussed earlier.

Proposition 6.12. For any index coding problem G,

$$\mathscr{R}_{\mathrm{R}}(G) \subseteq \mathscr{R}^{*}_{\mathrm{comp}}(G).$$

The proof is relegated to Appendix 6.C. This inclusion is strict, as illustrated by the 5-message index coding problem in Figure 6.9 for which $\beta_{\mathrm{MAIS}} = \beta_{\mathrm{comp}} = \beta^{*}_{\mathrm{comp}} = 3$ while $\beta_{\mathrm{R}} = 7/2$.

Composite coding, however, can be outperformed by linear codes.

Example 6.1. For the 6-message index coding problem in Figure 6.15, $\beta^{*}_{\mathrm{comp}} = 77/23 \simeq 3.3478$ while $\beta_{\mathrm{MAIS}} = \beta = 3$, achieved by a *scalar linear code* over $\mathbb{F}_2$.

Example 6.2. For the 6-message index coding problem in Figure 6.14, $\beta_{\mathrm{comp}} = \beta^{*}_{\mathrm{comp}} = 10/3$, while $\beta_{\mathrm{MAIS}} = \beta = 3$, achieved by the *interlinked cycle covering* scheme [150], which is a special case of scalar linear codes over $\mathbb{F}_2$.

Composite coding is not always outperformed by scalar linear codes, as illustrated by the 5-message index coding problem in Figure 5.2, for which scalar linear codes (in any finite field) achieve $\mathrm{minrank}_{\mathbb{F}} = 3$, but $\beta_{\mathrm{comp}} = \beta = 5/2$. It is not known whether composite coding, which is nonlinear, is always dominated by *vector* linear codes.

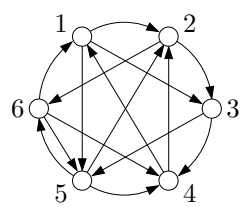


Figure 6.15: The index coding problem $(1|4,6)$,$(2|1,4,5)$,$(3|1,2)$,$(4|3,5)$,$(5|1,3,6)$, $(6|2,5)$ with $\beta^{*}_{\mathrm{comp}} = 77/23 \simeq 3.348 < \beta_{\mathrm{comp}} = 27/8 = 3.375$. Both upper bounds can be computed by respective linear programs. For this problem, $\beta_{\mathrm{MAIS}} = \beta = 3$, which is achievable by the scalar linear code $(x_1 + x_4 + x_5, x_2 + x_5 + x_6, x_3 + x_4 + x_5)$ over $\mathbb{F}_2$.

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Open Problem 6.3. Is there an index coding problem with $\beta^*_{\text{comp}} < \lambda_{\mathbb{F}}$ for every $\mathbb{F}$? How about $\beta^*_{\text{comp}} < \inf_{\mathbb{F}} \lambda_{\mathbb{F}}$?

Computation of composite coding bounds involves many auxiliary rates such as $R_i(\mathbf{D})$ and $S_J(\mathbf{D})$ in (6.25), the number of which grows exponentially in n even for a fixed $\mathbf{D}$. In addition, the number of $\mathbf{D}$ can be computed, starting with $D_i = \{i\}$ and excluding A_i, as

$$|\Delta| = \prod_{i=1}^{n} 2^{n-|A_i|-1} = 2^{n^2-n-\sum_{i=1}^{n}|A_i|},$$

which grows superexponentially in n. Therefore, although the composite coding inner bound is computable in principle, it is rather difficult to compute directly.

Techniques to reduce the computational complexity of the composite coding inner bound were explored in [99]. In particular, if $A_i \subseteq A_j \cup D_j$ for some $j \neq i$, then receiver j already knows or will ultimately know more than receiver i does. Therefore, one can update $D_j \leftarrow D_j \cup \{i\}$ at no cost to the achievable rate region. Starting with $D_i \leftarrow \{i\}$, $i \in [n]$, we can thus build *natural decoding sets* D_i^* by adding new elements to each D_i iteratively. Once equipped with natural decoding sets, we can reduce Δ by focusing on D_i that includes D_i^*. The number of composite indices can be also reduced without affecting the resulting rate region; see [99, Sec. IV].

Remark 6.14. Instead of flat coding, transmission of composite indices can be further layered. The resulting *three-layer composite coding* scheme [100] can strictly outperform the regular two-layer composite coding scheme, at the cost of introducing more auxiliary rates. For the 7-message index coding problem

$$\begin{array}{lll} (1|5), & (2|3,5,6), & (3|4,6,7), \\ (4|1,2,7), & (5|2,3,4,7), & (6|3,4,7), \\ (7|1,2,4), & & \end{array}$$

$\beta^*_{\text{comp}} = 3.25$, but the three-layer composite coding scheme achieves $\beta = 3$ [100, Example 2]. The latter scheme involves only 5 composite rates and 3 doubly composite rates.

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6.A Proof of Proposition 6.5

First note that for every G,

$$\left(\frac{1}{\kappa(G)},\ldots,\frac{1}{\kappa(G)}\right)\in\mathscr{R}_{\mathrm{R}}(G). \tag{6.26}$$

To prove this claim, set

$$\delta_J=\begin{cases}\frac{1}{\kappa(G)} & |J|=1,\\ 0 & \text{otherwise,}\end{cases}$$

and

$$T_{iJ}=\begin{cases}1 & J=\{i\},\\ 0 & \text{otherwise.}\end{cases}$$

Then, the condition in (6.7c) is satisfied automatically since $T_{ij}\in\mathscr{R}_{\mathrm{R}}(G|_J)=[0,1]$ for $J=\{i\}$. In addition, since

$$n-|A_i|\le\kappa(G),\qquad i\in[n], \tag{6.27}$$

the condition in (6.7b) is satisfied as

$$\sum_{J\subseteq[n]:J\not\subseteq A_i}\delta_J=\sum_{j\notin A_i}\delta_{\{j\}}=\sum_{j\notin A_i}\frac{1}{\kappa(G)}=\frac{n-|A_i|}{\kappa(G)}\le 1.$$

Therefore, by the definition of $\mathscr{R}_{\mathrm{R}}(G)$, the rate tuple $(R_1,\ldots,R_n)$ such that

$$R_i=\sum_{J\subseteq[n]:i\in J}\delta_J T_{iJ}=\delta_{\{i\}}T_{i\{i\}}=\frac{1}{\kappa(G)}$$

is contained in $\mathscr{R}_{\mathrm{R}}(G)$.

Now consider any $(R'_1,\ldots,R'_n)$ and $(\delta'_J, J\subseteq[n])$ that satisfy (6.6), i.e., $(R'_1,\ldots,R'_n)\in\mathscr{R}_{\mathrm{FLP}}(G)$. We show that there exist $(\delta_J, J\subsetneq[n])$ and $(T_{iJ}, i\in J\subsetneq[n])$ such that the conditions in (6.7b) and (6.7c) are satisfied, and

$$R'_i\le\sum_{J\subseteq[n]:i\in J}\delta_J T_{iJ}.$$

Then it follows that $(R'_1,\ldots,R'_n)\in\mathscr{R}_{\mathrm{R}}(G)$. We consider two cases.

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Case 1. If $\delta'_{[n]} = 0$, the desired result follows by setting $\delta_J = \delta'_J$ and $T_{iJ} = 1/\kappa(G|_J)$ for each $J \subsetneq [n]$ and each $i \in J$. Indeed, (6.7b) follows by (6.6b), and (6.7c) follows by (6.26). In addition,

$$R'_i \le \sum_{J \subseteq [n]: i \in J} \frac{\delta'_J}{\kappa(G|_J)} = \sum_{J \subsetneq [n]: i \in J} \delta_J T_{iJ}.$$

Hence, $(R'_1, \ldots, R'_n) \in \mathscr{R}_{\mathrm{R}}(G)$.

Case 2. If $\delta'_{[n]} \neq 0$, then set

$$\delta_J = \begin{cases} \delta'_J + \frac{\delta'_{[n]}}{\kappa(G)} & |J| = 1, \\ \delta'_J & \text{otherwise,} \end{cases}$$

and $T_{iJ} = 1/\kappa(G|_J)$. As before, the condition in (6.7c) is satisfied due to (6.26). To check (6.7b), consider

$$\begin{aligned} \sum_{J \subsetneq [n]: J \not\subseteq A_i} \delta_J &= \sum_{J \subsetneq [n]: J \not\subseteq A_i} \delta'_J + \sum_{j \notin A_i} \frac{\delta'_{[n]}}{\kappa(G)} \\ &\le \sum_{J \subsetneq [n]: J \not\subseteq A_i} \delta'_J + \delta'_{[n]} \qquad (6.28) \\ &\le 1, \qquad (6.29) \end{aligned}$$

where (6.28) follows by (6.27), and (6.29) follows by (6.6b). Finally, since $\kappa(G|_{\{i\}}) = 1$, we have

$$\begin{aligned} R'_i &\le \sum_{J \subseteq [n]: i \in J} \frac{\delta'_J}{\kappa(G|_J)} \\ &= \frac{\delta'_{\{i\}}}{\kappa(G|_{\{i\}})} + \frac{\delta'_{[n]}}{\kappa(G)} + \sum_{J \subsetneq [n]: i \in J, J \neq \{i\}} \frac{\delta'_J}{\kappa(G|_J)} \\ &= \frac{\delta_{[n]}}{\kappa(G|_{\{i\}})} + \sum_{J \subsetneq [n]: i \in J, J \neq \{i\}} \frac{\delta_J}{\kappa(G|_J)} \\ &= \sum_{J \subsetneq [n]: i \in J} \delta_J T_{jJ}. \end{aligned}$$

Hence, $(R'_1, \ldots, R'_n) \in \mathscr{R}_{\mathrm{R}}(G)$.

Combining both cases completes the proof that $\mathscr{R}_{\mathrm{FLP}}(G) \subseteq \mathscr{R}_{\mathrm{R}}(G)$.

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6.B Analysis of Error Probability for Composite Coding

Recall that receiver i finds a message tuple $\hat{x}(D_i)$ such that

$$w_J(\hat{x}(J \cap D_i), x(J \cap A_i)) = \hat{w}_J$$

for every $J \subseteq D_i \cup A_i$. A decoding error occurs only if there exists another tuple $\tilde{x}(D_i)$ with $\tilde{x}_i \neq x_i$ such that $w_J(\tilde{x}(J \cap D_i), x(J \cap A_i)) = \hat{w}_J$ for every $J \subseteq D_i \cup A_i$. We partition this error event according to each nonempty subset $L \subseteq D_i$ of erroneous message indices, i.e., $\tilde{x}_j \neq x_j$ iff $j \in L$. Then by the union of events bound,

$$\begin{aligned}
P_e &= \mathsf{P}\Big\{w_J(\tilde{x}(J \cap D_i), x(J \cap A_i)) = \hat{w}_J \text{ for all } J \subseteq D \cup A \\
&\qquad\qquad \text{for some } \tilde{x}(D_i) \text{ such that } \tilde{x}_i \neq x_i\Big\} \\
&= \bigcup_{L \subseteq D_i : i \in L} \mathsf{P}\Big\{w_J(\tilde{x}(J \cap L), x(J \setminus L)) = \hat{w}_J \text{ for all } J \subseteq D_i \cup A_i \\
&\qquad\qquad \text{for some } \tilde{x}(D_i) \text{ such that } \tilde{x}_j \neq x_j \text{ iff } j \in L\Big\} \\
&= \sum_{L \subseteq D_i : i \in L} \sum_{\substack{\tilde{x}(D_i) \\ \tilde{x}_j \neq x_j \text{ iff } j \in L}} \mathsf{P}\left(\bigcap_{J \subseteq D_i \cup A_i} \{w_J(\tilde{x}(J \cap L), x(J \setminus L)) = \hat{w}_J\}\right) \\
&= \sum_{L \subseteq D_i : i \in L} \sum_{\substack{\tilde{x}(D_i) \\ \tilde{x}_j \neq x_j \text{ iff } j \in L}} \mathsf{P}\left(\bigcap_{\substack{J \subseteq D_i \cup A_i \\ J \cap L \neq \emptyset}} \{w_J(\tilde{x}(J \cap L), x(J \setminus L)) = \hat{w}_J\}\right) \\
&\leq \sum_{L \subseteq D_i : i \in L} 2^{\sum_{j \in L} t_j - \sum_{J \subseteq D_i \cup A_i : J \cap L \neq \emptyset} s_J} \qquad (6.30) \\
&\leq \sum_{L \subseteq D_i : i \in L} 2^{\sum_{j \in L} rR_j + 1 - \sum_{J \subseteq D_i \cup A_i : J \cap L \neq \emptyset} rS_J},
\end{aligned}$$

where (6.30) holds since for each L the number of erroneous tuples is $\Pi_{j \in L}\left(2^{t_j} - 1\right)$, and for each erroneous tuple $\tilde{x}(D_i)$ with $\tilde{x}_j \neq x_j$ iff $j \in L$, the probability that it is mapped to the same $\hat{w}_J = w_J(x(J))$ for all J in the random codebook generation is $2^{-\sum s_J}$. Thus, the error probability P_e, averaged over random messages and codebooks, tends to zero as $r \to \infty$, provided that for every $L \subseteq D_i$ with $i \in L$,

$$\sum_{j \in L} R_j < \sum_{J \subseteq D_i \cup A_i : J \cap L \neq \emptyset} S_J.$$

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6.C Proof of Proposition 6.12

We show that the rate region achieved by local time sharing of enhanced composite coding for subproblems is achieved by enhanced composite coding itself. Then by induction on subproblems, i.e., $\mathscr{R}_{\mathrm{R}}(G|_J) \subseteq \mathscr{R}^*_{\mathrm{comp}}(G|_J)$ for every $J \subsetneq [n]$, and using the definition of $\mathscr{R}_{\mathrm{R}}(G)$ in (6.7), it follows that $\mathscr{R}_{\mathrm{R}}(G) \subseteq \mathscr{R}^*_{\mathrm{comp}}(G)$. Details are as follows.

Throughout this proof, we use the rate region $\mathscr{T}(G) \subseteq \mathscr{R}^*_{\mathrm{comp}}(G)$, instead of $\mathscr{R}^*_{\mathrm{comp}}(G)$ itself, which is defined exactly as $\mathscr{R}^*_{\mathrm{comp}}(G)$ except that $i \in L$ is not required in (6.25b), and $\mathscr{T}(G|_K)$ be its restriction to the subproblem $G|_K$. Consider any rate tuple $(R_1, \dots, R_n)$ achieved by local time sharing among all subproblems $\mathscr{T}(G|_K)$, $K \subsetneq [n]$, that is, any rate tuple $(R_1, \dots, R_n)$ satisfying (6.7a) with $\mathscr{T}(G|_K)$ in place of $\mathscr{R}_{\mathrm{R}}(G|_K)$ with some δ_K and $(T_{iK}, i \in K) \in \mathscr{T}(G|_K)$. It suffices to show that this rate tuple $(R_1, \dots, R_n)$ satisfies (6.25a) for some $R_i(\mathbf{D})$ and $S_J(\mathbf{D})$.

First note that each component rate T_{iK} for R_i can be expressed as

$$T_{iK} = \sum_{\tilde{\mathbf{D}}^{(K)} \in \Delta^{(K)}} T_{iK}(\tilde{\mathbf{D}}^{(K)}), \quad i \in K, \tag{6.31a}$$

for some $T_{iK}(\tilde{\mathbf{D}}^{(K)})$, $i \in K$, $\tilde{\mathbf{D}}^{(K)} \in \Delta^{(K)}$, and $U_{JK}(\tilde{\mathbf{D}}^{(K)})$, $J \subseteq K$, $\tilde{\mathbf{D}}^{(K)} \in \Delta^{(K)}$, such that for each $i \in K$,

$$\sum_{j \in L} T_{jK}(\tilde{\mathbf{D}}^{(K)}) \le \sum_{\substack{J \subseteq \tilde{D}_i^{(K)} \cup (A_i \cap K) \\ J \cap L \neq \emptyset}} U_{JK}(\tilde{\mathbf{D}}^{(K)}), \quad L \subseteq \tilde{D}_i^{(K)}, \tag{6.31b}$$

and

$$\sum_{\substack{J \subseteq K \\ J \not\subseteq A_i}} \sum_{\tilde{\mathbf{D}}^{(K)} \in \Delta^{(K)}} U_{JK}(\tilde{\mathbf{D}}^{(K)}) \le 1. \tag{6.31c}$$

Here,

$$\Delta^{(K)} = \{(D_i, i \in K) \colon i \in D_i \subseteq K \setminus A_i\}$$

is the set of decoding sets for the subproblem $G|_K$. Hence, by the

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decomposition as in (6.7a), we can express R_i as

$$\begin{aligned} R_i &= \sum_{K\subsetneq[n]:i\in K} \delta_K\, T_{iK} \\ &= \sum_{K\subsetneq[n]:i\in K} \delta_K \sum_{\tilde{\mathbf{D}}^{(K)}\in\Delta^{(K)}} T_{iK}(\tilde{\mathbf{D}}^{(K)}) \\ &= \sum_{\mathbf{D}\in\Delta} \sum_{K\subsetneq[n]:i\in K} \frac{\delta_K}{N(\mathbf{D}|_K)} T_{iK}(\mathbf{D}|_K) \\ &= \sum_{\mathbf{D}\in\Delta} R_i(\mathbf{D}), \end{aligned}$$

where $\mathbf{D}|_K = (D_i \cap K, i \in K) \in \Delta^{(K)}$ is the restriction of $\mathbf{D} \in \Delta$ on K,

$$N(\tilde{\mathbf{D}}^{(K)}) = |\{\mathbf{D}\in\Delta : \mathbf{D}|_K = \tilde{\mathbf{D}}^{(K)}\}|,$$

$$R_i(\mathbf{D}) = \sum_{K\subsetneq[n]:i\in K} \frac{\delta_K}{N(\mathbf{D}|_K)} T_{iK}(\mathbf{D}|_K). \tag{6.32a}$$

Similarly, let

$$S_J(\mathbf{D}) = \sum_{K\subsetneq[n]:J\subseteq K} \frac{\delta_K}{N(\mathbf{D}|_K)} U_{JK}(\mathbf{D}|_K). \tag{6.32b}$$

Now it suffices to check whether $R_i(\mathbf{D})$ and $S_K(\mathbf{D})$ defined in (6.32) satisfy (6.25b), without the constraint $i \in L$, and (6.25c). First, for each $i \in [n]$, $L \subseteq D_i$, and $\mathbf{D} \in \Delta$,

$$\begin{aligned} \sum_{j\in L} R_j(\mathbf{D}) &= \sum_{j\in L} \sum_{K\subsetneq[n]:j\in K} \frac{\delta_K}{N(\mathbf{D}|_K)} T_{jK}(\mathbf{D}|_K) \\ &= \sum_{K\subsetneq[n]} \sum_{j\in L\cap K} \frac{\delta_K}{N(\mathbf{D}|_K)} T_{jK}(\mathbf{D}|_K) \\ &\le \sum_{K\subsetneq[n]} \sum_{\substack{J\subseteq(D_i\cap K)\cup(A_i\cap K):\\ J\cap(L\cap K)\neq\emptyset}} \frac{\delta_K}{N(\mathbf{D}|_K)} U_{JK}(\mathbf{D}|_K) \qquad (6.33) \\ &= \sum_{J\subseteq D_i\cup A_i: J\cap L\neq\emptyset} \sum_{K\subsetneq[n]:J\subseteq K} \frac{\delta_K}{N(\mathbf{D}|_K)} U_{JK}(\mathbf{D}|_K) \\ &= \sum_{J\subseteq D_i\cup A_i: J\cap L\neq\emptyset} S_J(\mathbf{D}), \qquad (6.34) \end{aligned}$$

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where (6.33) follows by (6.31b), and (6.34) follows by (6.32b). Next, for each $i \in [n]$,

$$\begin{aligned}
\sum_{J\subseteq[n]:J\not\subseteq A_i}\sum_{\mathbf{D}\in\Delta} S_J(\mathbf{D}) &= \sum_{J\subseteq[n]:J\not\subseteq A_i}\sum_{\mathbf{D}\in\Delta}\sum_{K\subsetneq[n]:J\subseteq K}\frac{\delta_K}{N(\mathbf{D}|_K)}U_{JK}(\mathbf{D}|_K)\\
&= \sum_{J\subseteq[n]:J\not\subseteq A_i}\sum_{K\subsetneq[n]:J\subseteq K}\delta_K\sum_{\tilde{\mathbf{D}}^{(K)}\in\Delta^{(K)}}U_{JK}(\tilde{\mathbf{D}}^{(K)})\\
&= \sum_{K\subsetneq[n]:K\not\subseteq A_i}\delta_K\sum_{J\subseteq K:J\not\subseteq A_i}\sum_{\tilde{\mathbf{D}}^{(K)}\in\Delta^{(K)}}U_{JK}(\tilde{\mathbf{D}}^{(K)}) \qquad (6.35)\\
&\le \sum_{K\subsetneq[n]:K\not\subseteq A_i}\delta_K \qquad (6.36)\\
&\le 1, \qquad (6.37)
\end{aligned}$$

where (6.35) follows since $J \not\subseteq A_i$ and $J \subseteq K$ implies that $K \not\subseteq A_i$, (6.36) follows by (6.31c), and (6.37) follows by (6.7b). Therefore, the rate tuple $(R_1, \ldots, R_n)$ we started with is contained in $\mathscr{T}(G)$. Since $\mathscr{R}_{\mathrm{R}}(G|_{\{i\}}) = \mathscr{T}(G|_{\{i\}}) = [0, 1]$, by induction on all subsets $K \subseteq [n]$, this implies that $\mathscr{R}_{\mathrm{R}}(G|_K) \subseteq \mathscr{T}(G|_K)$ for every $K \subseteq [n]$, and in particular, $\mathscr{R}_{\mathrm{R}}(G) \subseteq \mathscr{T}(G) \subseteq \mathscr{R}^*_{\mathrm{comp}}(G)$. This completes the proof.

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7

Criticality

Let e be an edge of the side information graph $G = (V, E)$ of an index coding problem. We denote the graph resulting from removing e from G by G_e, i.e.,

$$V(G_e) = V(G) \text{ and } E(G_e) = E(G) \setminus \{e\}.$$

Given the index coding problem G, the edge $e \in E$ is said to be *critical* if $\mathscr{C}(G_e) \neq \mathscr{C}(G)$, or in words, if the removal of e from G strictly reduces the capacity region. The index coding problem G itself is said to be *critical* if every $e \in E(G)$ is critical. Thus, each critical graph (= index coding problem) cannot be made "simpler" into another one of the same capacity region.

The notion of criticality was introduced by Tahmasbi, Shahrasbi, and Gohari [147]. In this section, we present sufficient and necessary conditions for criticality of an index coding problem.

7.1 A Sufficient Condition

Recall that a cycle in a graph $G = (V, E)$ is a closed path $i_1 \to i_2 \to \cdots \to i_k = i_1$. We say that the cycle $i_1 \to i_2 \to \cdots \to i_k = i_1$ is a *unicycle* or *chordless cycle* if it does not contain another cycle as a proper

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subgraph, or equivalently, the subgraph induced by $\{i_1, i_2, \ldots, i_{k-1}\}$ has no edge other than $i_j \to i_{j+1}$, $j \in [k-1]$. As an example, in Figure 7.1(a), $1 \to 2 \to 3 \to 4 \to 1$ is a cycle, but not a unicycle, but $G|_{\{1,2,3\}} = 1 \to 2 \to 3 \to 1$ is a unicycle. As another example, for the graph in Figure 7.1(b), both $G|_{\{1,2,3\}} = 1 \to 2 \to 3 \to 1$ and $G|_{\{1,3,4\}} = 1 \to 4 \to 3 \to 1$ are unicycles. Note that if the subgraph $G|_J$ is a unicycle, then $G|_{J'}$ cannot be a unicycle for any J' that is a proper subset or superset of J.

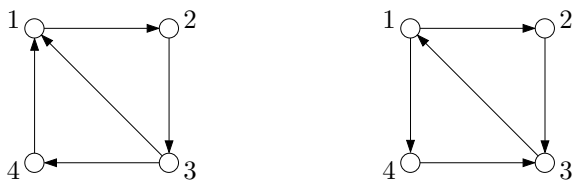


Figure 7.1: (a) $G|_{\{1,2,3\}}$ is a unicycle, but $1 \to 2 \to 3 \to 4 \to 1$ is not a unicycle. (b) $G|_{\{1,2,3\}}$ and $G|_{\{1,3,4\}}$ are both unicycles.

Suppose that $G|_J$ is a unicycle for some $J \subseteq V$. Then the rate tuple $(R_1, \ldots, R_n)$ satisfying

$$R_i = \begin{cases} \frac{1}{|J|-1}, & i \in J, \\ 0, & i \notin J, \end{cases} \tag{7.1}$$

is achievable for index coding problem G, for example, using an MDS code over J. For any edge $e \in G|_J$, however, the vertex-induced subgraph $G_e|_J$ is acyclic (since there is no other cycle in $G|_J$). Therefore, by the MAIS outer bound in (5.2), any rate tuple $(R'_1, \ldots, R'_n) \in \mathscr{C}(G_e)$ must satisfy

$$\sum_{i \in J} R'_i \leq 1. \tag{7.2}$$

The rate tuple in (7.1), however, does not satisfy (7.2) and thus is not in $\mathscr{C}(G_e)$. This implies that removing edge e from G strictly reduces the capacity region, i.e., $\mathscr{C}(G_e) \neq \mathscr{C}(G)$, which establishes the following sufficient condition for the criticality of an index coding problem.

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Theorem 7.1 (Union-of-unicycles condition [11])**.** If every edge of G belongs to some unicycle, then G is critical.

Example 7.1. Consider the index coding problem G with side information graph depicted in Figure 7.1(b). Since every edge of G belongs to some unicycle, Theorem 7.1 shows that G is critical.

The union-of-unicycles condition, however, is not necessary for the criticality of a graph, as illustrated by the following.

Example 7.2. The capacity region of the 5-message index coding problem shown in Figure 7.2 is characterized by

$$R_1 + R_3 \leq 1, \tag{7.3a}$$
$$R_1 + R_4 \leq 1, \tag{7.3b}$$
$$R_2 + R_4 \leq 1, \tag{7.3c}$$
$$R_2 + R_5 \leq 1, \tag{7.3d}$$
$$R_3 + R_4 \leq 1, \tag{7.3e}$$
$$R_3 + R_5 \leq 1. \tag{7.3f}$$

Although the edge $3 \to 5$ does not belong to any unicycle, removing it from the side information graph (see Figure 5.3) reduces the capacity region to the one characterized by (7.3) and

$$R_1 + R_2 + R_3 + R_4 + R_5 \leq 2. \tag{7.4}$$

The union-of-unicycles condition actually captures the "criticality" with respect to the MAIS outer bound.

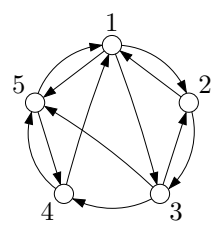


Figure 7.2: The index coding problem $(1|2,4,5),(2|1,3),(3|1,2),(4|3,5),(5|1,3,4)$. Although the edge $3 \to 5$ does not belong to any unicycle, it is critical. The capacity region is achieved by composite coding with or without the edge $3 \to 5$.

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Proposition 7.1. An edge e belongs to some unicycle iff $\mathscr{R}_{\text{MAIS}}(G_e) \neq \mathscr{R}_{\text{MAIS}}(G)$.

Indeed, if e belongs to a unicycle $G|_J$, then $(R_1, \ldots, R_n)$ defined in (7.1) is included in $\mathscr{R}_{\text{MAIS}}(G)$. But it does not satisfy (7.2) and hence it is not included in $\mathscr{R}_{\text{MAIS}}(G_e)$. Hence, $\mathscr{R}_{\text{MAIS}}(G) \neq \mathscr{R}_{\text{MAIS}}(G_e)$. Conversely, if $\mathscr{R}_{\text{MAIS}}(G_e) \neq \mathscr{R}_{\text{MAIS}}(G)$, then for some J, the edge e belongs to a cycle in $G|_J$ and $G_e|_J$ is acyclic. Let $J_{\min}$ be a minimal such subset. Then, $G|_{J_{\min}}$ is a cycle that contains e and does not contain any other cycle. Hence, it is a unicycle.

Similarly, if e belongs to some unicycle, then $\mathscr{R}_{\text{PM}}(G_e) \neq \mathscr{R}_{\text{PM}}(G)$. It is not known, however, whether the converse holds.

Open Problem 7.1. Find the sufficient and necessary condition for $\mathscr{R}_{\text{PM}}(G_e) \neq \mathscr{R}_{\text{PM}}(G)$.

Let G^* be the subgraph of G resulting from removing all edges of G that do not belong to any unicycle, i.e., $V(G^*) = V(G)$ and

$$E(G^*) = \{e \in E(G) \colon e \text{ belongs to some unicycle of } G\}.$$

For example, consider the 6-message side information graph G depicted in Figure 7.3(a). The edges $3 \to 4$, $4 \to 6$, and $6 \to 2$ do not belong to any unicycle. The graph G^* resulting from removing these three edges of G is depicted in Figure 7.3(c).

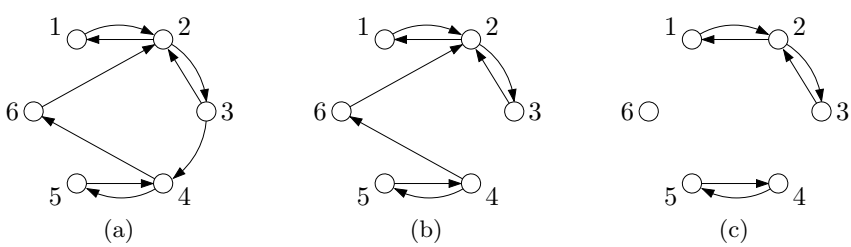


Figure 7.3: (a) The side information graph G in which the edges $3 \to 4$, $4 \to 6$, and $6 \to 2$ do not belong to any unicycle. (b) The subgraph G_e resulting from removing the edge $3 \to 4$. (c) The subgraph G^* resulting from removing all edges of G that do not belong to any unicycle.

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If distinct edges $e, e', e'', \ldots$ do not belong to any unicycle in G, then $e', e'', \ldots$ do not belong to any unicycle in G_e either. Hence by applying Proposition 7.1 edge by edge, we can establish the following.

Proposition 7.2. $\mathscr{R}_{\mathrm{MAIS}}(G^*) = \mathscr{R}_{\mathrm{MAIS}}(G)$.

Note that for any edge e, if the MAIS bound is tight for G_e, then

$$\mathscr{R}_{\mathrm{MAIS}}(G_e) = \mathscr{C}(G_e) \subseteq \mathscr{C}(G) \subseteq \mathscr{R}_{\mathrm{MAIS}}(G).$$

This observation, combined with Propositions 7.1 and 7.2, leads to the following condition under which the capacity region of the index coding problem G is equal to the capacity of the simpler problem G_e.

Proposition 7.3. If e does not belong to any unicycle and the MAIS bound is tight for G_e, then

$$\mathscr{R}_{\mathrm{MAIS}}(G_e) = \mathscr{C}(G_e) = \mathscr{C}(G) = \mathscr{R}_{\mathrm{MAIS}}(G).$$

In particular, if the MAIS bound is tight for G^*, then

$$\mathscr{R}_{\mathrm{MAIS}}(G^*) = \mathscr{C}(G^*) = \mathscr{C}(G) = \mathscr{R}_{\mathrm{MAIS}}(G).$$

Consequently, if the MAIS bound is tight for G^*, then all the edges that do not belong to any unicycle can be removed without reducing the capacity region. As an example, the capacity region of the index coding problem G^* in Figure 7.3(c) is characterized by

$$R_1 + R_3 + R_4 \leq 1, \tag{7.5a}$$
$$R_1 + R_3 + R_5 \leq 1, \tag{7.5b}$$
$$R_2 + R_3 + R_4 + R_6 \leq 1, \tag{7.5c}$$
$$R_2 + R_3 + R_5 + R_6 \leq 1, \tag{7.5d}$$

which is equal to the MAIS bound. Thus, by Proposition 7.3, the capacity region of the index coding problem G in Figure 7.3(a), as well as that of any problem in between G^* and G, such as G_e in Figure 7.3(b), is also characterized by (7.5).

Remark 7.1. The condition in Proposition 7.3 can be weakened for the case of broadcast rate. For any edge e, if $\beta_{\mathrm{MAIS}}(G_e) = \beta(G_e)$, then $\beta_{\mathrm{MAIS}}(G) \leq \beta(G) \leq \beta(G_e) = \beta_{\mathrm{MAIS}}(G_e)$. By Proposition 7.1, if in addition e does not belong to any unicycle, then $\beta_{\mathrm{MAIS}}(G) = \beta_{\mathrm{MAIS}}(G_e)$ and thus $\beta_{\mathrm{MAIS}}(G) = \beta(G) = \beta(G_e) = \beta_{\mathrm{MAIS}}(G_e)$.

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Recall from Section 2.1 that the *transpose* G^T of an index coding problem G satisfies $V(G^T) = V(G)$ and

$$E(G^T) = \{(i, j) : (j, i) \in E(G)\}.$$

In other words, the side information graphs of G and G^T are of the opposite orientations, as illustrated by the pair of 5-message index coding problems in Figure 7.4. Both problems are unions of unicycles and hence are critical. In fact, any index coding problem G is a union of unicycles iff its transpose G^T is a union of unicycles.

Open Problem 7.2. Is the criticality of G equivalent to the criticality of G^T (cf. Open Problem 4.1)?

The correspondence between critical problems and their capacity regions is not one-to-one. For both problems in Figure 7.4, the capacity region is achieved by the flat coding scheme in Section 6.9 and is characterized by the inequalities

$$\begin{aligned} R_1 + R_2 + R_3 + R_4 &\le 1, \\ R_1 + R_2 + R_4 + R_5 &\le 1, \\ R_3 + R_5 &\le 1, \end{aligned}$$

which coincides with the MAIS outer bound. Note that for any index coding problem G, $\mathscr{R}_{\text{MAIS}}(G) = \mathscr{R}_{\text{MAIS}}(G^T)$ in general.

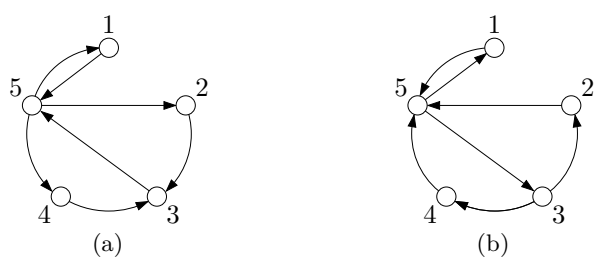


Figure 7.4: (a) The index coding problem $G = (1|5), (2|5), (3|2, 4), (4|5), (5|1, 3)$. (b) The index coding problem $G^T = (1|5), (2|3), (3|5), (4|3), (5|1, 2, 4)$, in which every edge is in the opposite direction of the corresponding edge in G.

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7.2 Necessary Conditions

We now discuss a few necessary conditions for G to be critical, which imply the conditions under which an edge e can be removed without reducing the capacity region. We start with the following condition established by Tahmasbi, Shahrasbi, and Gohari [147].

Proposition 7.4 (Union-of-cycles condition [147])**.** If G is critical, then every edge lies on a cycle.

To prove this, suppose that G can be vertex-partitioned by two subgraphs G_1 and G_2 such that there exists no edge from G_2 to G_1 (see Figure 4.1(b)). The set of edges from G_1 to G_2 is referred to as a *directed cut-set*, removing which will disconnect G_1 and G_2. The necessary condition in Proposition 7.4 can be viewed as a direct application of Remark 4.1, which implies that every edge in a directed cut-set can be removed without affecting the capacity region. Therefore, if G is critical, then by the Farkas lemma [19, Th. 2.2] (which states that each edge in a directed graph either lies on a directed cycle or belongs to a directed cut-set but not both), every edge lies on a (directed) cycle.

The second necessary condition we present is based on the relationship among different side information sets at the receivers. We say that the side information sets $A_1, \ldots, A_n$ of an index coding problem G are *degraded* if there exist $i, j \in V(G)$ such that $j \in A_i$ and $A_j \subseteq A_i$. In this case, the edge $j \to i$ can be removed since x_j can be recovered at receiver i. This observation leads to the following necessary condition.

Proposition 7.5 (Nondegradedness condition [11])**.** If G is critical, then side information sets are nondegraded.

Next, we present another necessary condition by rephrasing Proposition 7.3. Suppose that G is critical, i.e., $\mathscr{C}(G_e) \neq \mathscr{C}(G)$ for every edge e. If the MAIS bound is tight for G_e for some e, then

$$\mathscr{R}_{\mathrm{MAIS}}(G_e) = \mathscr{C}(G_e) \subsetneq \mathscr{C}(G) \subseteq \mathscr{R}_{\mathrm{MAIS}}(G).$$

Now this edge e must belong to a unicycle. Otherwise, by Proposition 7.3, $\mathscr{C}(G_e) = \mathscr{C}(G)$, which contradicts the criticality of G. This establishes the following partial converse to Theorem 7.1.

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Proposition 7.6 (MAIS looseness condition [11])**.** If $G = (V, E)$ is critical, then for every edge $e \in E$ that does not belong to a unicycle, the MAIS bound is loose for G_e, i.e., $\mathscr{R}_{\text{MAIS}}(G_e) \neq \mathscr{C}(G_e)$.

In other words, an edge e is not critical if it does not belong to any unicycle and the MAIS bound is tight for G_e. Recall that the edge $3 \to 5$ in Figure 7.2 is critical and does not belong to any unicycle. As is implied by Proposition 7.6 and verified by (7.4), the MAIS bound is not tight for the graph resulting from removing this edge.

The following summarizes all the necessary conditions for the criticality of a problem that we have discussed so far.

Theorem 7.2. If $G = (V, E)$ is critical, then

1. every edge lies on a cycle,
2. side information sets are nondegraded, and
3. for each $e \in E$ that is not in some unicycle, $\mathscr{R}_{\text{MAIS}}(G_e) \neq \mathscr{C}(G_e)$.

The next three examples demonstrate that these necessary conditions are mutually independent.

Example 7.3. The 6-message side information graph G in Figure 7.5 satisfies the union-of-cycles and nondegradedness conditions, but it does not satisfy the MAIS looseness condition, as the edge $6 \to 1$ does not belong to any unicycle and the MAIS bound is tight after removing this edge. Hence, G is not critical.

Example 7.4. The 6-message side information graph G in Figure 7.6 satisfies the union-of-cycles and MAIS looseness conditions, but not the nondegradedness condition as $A_6 \subseteq A_4$. Hence, G is not critical.

Example 7.5. The 6-message side information graph G in Figure 7.7 satisfies the nondegradedness and MAIS looseness conditions, but not the union-of-cycles condition, as the edge $2 \to 6$ does not belong to any cycle. Hence, G is not critical.

Satisfying all the necessary conditions in Theorem 7.2 at the same time, however, does not imply the criticality.

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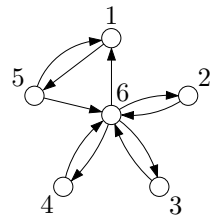


Figure 7.5: The index coding problem $(1|5,6),(2|6),(3|6),(4|6),(5|1),(6|2,3,4,5)$ that satisfies the union-of-cycles and nondegradedness conditions, but not the MAIS looseness condition. The capacity region is the same with or without the edge $6 \to 1$.

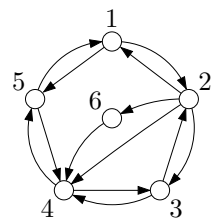


Figure 7.6: The index coding problem $(1|2,5),(2|1,3),(3|2,4),(4|2,3,5,6),(5|1,4),(6|2)$ that satisfies the union-of-cycles and MAIS looseness conditions, but not the nondegradedness condition. The capacity region is the same with or without the edge $6 \to 4$.

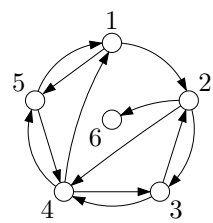


Figure 7.7: The index coding problem $(1|4,5),(2|1,3),(3|2,4),(4|2,3,5),(5|1,4),(6|2)$ that satisfies the nondegradedness and MAIS looseness conditions, but not the union-of-cycles condition. The capacity region is the same with or without the edge $2 \to 6$.

Example 7.6. The 5-message side information graph G in Figure 7.8 satisfies the union-of-cycles and nondegradedness conditions. It also satisfies the MAIS looseness condition, since the edge $1 \to 4$ is the only edge that does not belong to any unicycle and the MAIS bound

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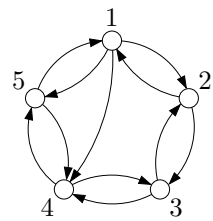


Figure 7.8: The index coding problem $(1|2,5),(2|1,3),(3|2,4),(4|1,3,5),(5|1,4)$ that satisfies all the three necessary conditions in Theorem 7.2, but is not critical. The capacity region is the same with or without the edge $1 \to 4$.

is not tight after removing this edge (see Example 5.2). Nonetheless, the index coding problem G is not critical as the capacity region is characterized by (5.14) and (5.15) with or without the edge $1 \to 4$.

It is not known how the gap between Theorem 7.1 and 7.2 can be closed.

Open Problem 7.3. Find the sufficient and necessary condition for the criticality of G.

Remark 7.2. If G satisfies the union-of-unicycles condition, it trivially satisfies the union-of-cycles condition and the MAIS looseness. As logically expected, the union-of-unicycles condition also implies the nondegradedness condition. Suppose that G has degraded side information sets. Then, there exists an edge $j \to i$ such that $A_j \subseteq A_i$. This edge cannot belong to a unicycle. Indeed, if the edge $j \to i$ does not belong to any cycle, then trivially it does not belong to any unicycle. Otherwise, $j \to i$ lies on a cycle $C = (j \to i \to \cdots \to k \to j)$, which by degradedness must have at least three vertices. Thus, $k \in A_j$ and, by the degradedness assumption, $k \in A_i$. Hence, $i \to \cdots \to k \to i$ is also a cycle, which implies that C cannot be a unicycle.

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8

Index Coding Problems with Known Capacity

In this section, we present several special instances of the index coding problem for which the broadcast rate or the capacity region is known. We refer the reader to [169, 153, 119], and [150] among others for additional classes of index coding instances with known broadcast rate or capacity region.

8.1 Problems with Broadcast Rate of Two

One of the simplest index coding problems is the one with a complete side information graph G, i.e., $j \in A_i$ for all distinct $i, j \in [n]$. In this case, the broadcast rate is one and the rates

$$R_1 = \cdots = R_n = 1$$

are achievable simultaneously.

If side information graph G is not complete, then $j \notin A_i$ for some $i \neq j$. Hence, by the MAIS bound, $2 \leq \beta(G)$. This also implies that there is no index coding instance with $1 < \beta < 2$.

The following result characterizes all instances of the index coding problem with $\beta = 2$, based on the notion of alignment subsets discussed in Section 6.8.

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Proposition 8.1 ([30, 107]). The broadcast rate of the index coding problem with side information graph G is 2 iff G is not complete and can be partitioned into alignment subsets such that for every pair (i, j) in an alignment subset, $i \in A_j$ and $j \in A_i$.

This sufficient and necessary condition for $\beta = 2$ was established independently by Maleki, Cadambe, and Jafar [107] and by Blasiak, Kleinberg, and Lubetzky [30], the latter of whom also presented a polynomial time algorithm to determine whether $\beta = 2$. The proof of sufficiency follows by Proposition 6.8. For the necessity, let $i \overset{k}{\Longleftrightarrow} j$ denote $i, j \notin A_k$ for distinct indices i, j, k. The following lemma can be proved by the polymatroidal outer bound in Theorem 5.2.

Lemma 8.1 ([107]). For any $k+1$ distinct indices $i_0, i_1, \ldots, i_k \in [n]$ and k indices $j_1, \ldots, j_k \in [n]$, if

$$i_0 \overset{j_1}{\Longleftrightarrow} i_1 \overset{j_2}{\Longleftrightarrow} i_2 \overset{j_3}{\Longleftrightarrow} \cdots \overset{j_{k-1}}{\Longleftrightarrow} i_{k-1} \overset{j_k}{\Longleftrightarrow} i_k$$

and

$$i_0 \notin A_{i_k},$$

then

$$R_{i_0} + R_{j_1} + R_{i_1} + R_{j_2} + R_{i_2} + \cdots + R_{j_k} + R_{i_k} \leq 1.$$

Returning to the proof of Proposition 8.1, suppose that i and j belong to the same alignment subset. Then, there must exist a chain of alignment relations connecting j and i as

$$j \overset{j_1}{\Longleftrightarrow} i_1 \overset{j_2}{\Longleftrightarrow} i_2 \overset{j_3}{\Longleftrightarrow} \cdots \overset{j_{k-1}}{\Longleftrightarrow} i_{k-1} \overset{j_k}{\Longleftrightarrow} i.$$

If $j \notin A_i$ (or $i \notin A_j$), then by Lemma 8.1,

$$R_j + R_{j_1} + R_{i_1} + R_{j_2} + R_{i_2} + \cdots + R_{j_k} + R_i \leq 1. \tag{8.1}$$

Since the length of the chain k is at least 1, the summation in (8.1) consists of at least three terms, which implies that the symmetric rate of 1/2 (i.e., the broadcast rate of 2) is not achievable.

Remark 8.1. As mentioned above, there is a dichotomy of the broadcast rate as either $\beta = 1$ or $\beta \geq 2$. Above $\beta = 2$, there exists an index coding instance of the broadcast rate arbitrarily close to 2 (see Proposition 8.6 below).

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8.2 Problems with Very Small Capacity

At the other extreme, if an n-message side information graph G is acyclic, then uncoded transmission is optimal, that is, the broadcast rate $\beta(G) = n$ and the capacity region $\mathscr{C}(G)$ is characterized by

$$R_1 + \cdots + R_n \leq 1.$$

If G contains a cycle, however, then partial clique covering (see Remark 6.5) of that cycle and the remaining single-vertex cliques achieves the upper bound of $n-1$ on the broadcast rate. Consequently, there is no index coding instance with $n-1 < \beta < n$.

The following characterizes all instances of the index coding problem with $\beta = n-1$.

Proposition 8.2 ([118]). The broadcast rate of the n-message index coding problem with side information graph G is $n-1$ iff G contains a cycle and there is a vertex that belongs to all cycles. In other words, $\beta(G) = n-1$ iff $\beta_{\text{MAIS}}(G) = n-1$.

The sufficient condition follows immediately since G contains a cycle (hence $\beta \leq n-1$) and the removal of the common vertex induces an acyclic subgraph of $n-1$ vertices (hence $\beta \geq n-1$ by the MAIS bound). The necessary condition was established by Ong [118], who established a stronger result—if G contains a cycle and the intersection of all cycles is empty, then it is not just that $\beta \neq n-1$, but that $\beta \leq n-2$. The proof of this result relies on the following.

Lemma 8.2 ([118, Lemma 3]). If G has a cycle and the intersection of all cycles is empty, then G contains a subgraph G' (i.e., $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G)$) that is either two disjoint cycles or of the form shown in Figure 8.1.

To establish the upper bound of $n-2$ on the broadcast rate, assume without loss of generality that G is itself a union of two disjoint cycles or a graph of the form in Figure 8.1. Otherwise, any remaining $n-n'$ messages of the index coding problem G that are not in the subproblem G' with n' messages (of either structure) can be communicated in $n-n'$

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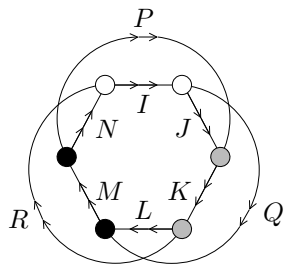


Figure 8.1: The common subgraph structure with a single "inner" circle consisting of six paths I through N and three "outer" paths P, Q, and R. The paths do not share common vertices except the end points. Each pair of white, gray, and black vertices may be identical (no edges between them), but all paths other than I, K, and M must contain at least one edge.

transmissions without coding. Any additional edge (side information) in G' can be ignored and still result in a valid upper bound on the broadcast rate. Hence, the broadcast rate upper bound of $n-2$ is achievable for G if the broadcast rate upper bound of $n'-2$ is achievable for G'.

In the first case, G consists of two cycles with n_1 and $n_2 = n - n_1$ vertices. Then partial clique covering of the two cycles (with partial clique numbers of $n_1 - 1$ and $n_2 - 1$) achieves the upper bound of $(n_1 - 1) + (n_2 - 1) = n - 2$ on the broadcast rate.

In the second case, G has the form in Figure 8.1. Then the *interlinked cycle covering* scheme by Thapa, Ong, and Johnson [150] can achieve the same upper bound of $n-2$. To show this, denote the vertices in one of the 9 paths labeled by $S \in \{I, J, K, L, M, N, P, Q, R\}$ as $s_0 \to s_1 \to \cdots \to s_{-1}$. As some paths meet at end points, we have

$$\begin{aligned} i_0 = n_{-1} = r_{-1}, \quad k_0 = j_{-1} = p_{-1}, \quad m_0 = l_{-1} = q_{-1}, \\ j_0 = q_0 = i_{-1}, \quad l_0 = r_0 = k_{-1}, \quad n_0 = p_0 = m_{-1}. \end{aligned}$$

Note that every vertex except i_0, k_0, m_0 has only one incoming edge. Consider the scalar linear code for G that transmits

$$x_{i_0} + x_{k_0} + x_{m_0}, \tag{8.2a}$$

$$x_s + x(A_s), \qquad s \neq i_0, k_0, m_0. \tag{8.2b}$$

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The total number of transmissions is $1 + (n-3) = n-2$. Clearly, receiver $s \neq i_0, k_0, m_0$ can recover x_s from (8.2b) and $x(A_s)$. Receiver i_0 can compute along the paths N and R

$$\begin{aligned}&(x_{n_0} + x_{n_1}) + (x_{n_1} + x_{n_2}) + \cdots + (x_{n_{-3}} + x_{n_{-2}})\\ &\quad + (x_{r_0} + x_{r_1}) + (x_{r_1} + x_{r_2}) + \cdots + (x_{r_{-3}} + x_{r_{-2}}),\end{aligned}$$

which is simplified by telescoping to $x_{n_0} + x_{r_0} + x_{n_{-2}} + x_{r_{-2}}$. Combining this with (8.2a), and using $x(A_{i_0}) = (x_{n_{-2}}, x_{r_{-2}})$, receiver i_0 can recover x_{i_0}. By symmetry, receivers k_0 and m_0 can similarly recover their messages, and thus the upper bound of $n-2$ is achievable.

Combined with the MAIS bound, this argument also implies the following.

Proposition 8.3 ([118]). The broadcast rate of the n-message index coding problem with side information graph G is $n-2$ if $\beta_{\mathrm{MAIS}}(G) = n-2$.

The necessity of this condition is not known.

Open Problem 8.1. Characterize all n-message index coding instances with $\beta = n-2$.

Remark 8.2. There is a trichotomy that the broadcast rate β must be n, $n-1$, or $\leq n-2$. Below $\beta = n-2$, there are index coding instances with $n-3 < \beta < n-2$ (cf. Remark 8.1). For example, the 5-message index coding problem in Figure 5.2 has $\beta = 5/2 = n - 5/2 = n/2$.

Open Problem 8.2. Is there an n-message index coding instance with $n - 5/2 < \beta < n-2$?

8.3 Perfect Graphs

For an index coding problem with an undirected (bidirectional) side information graph U, the MAIS bound implies $\alpha(U) \leq \beta(U)$ and the clique covering bound implies $\beta(U) \leq \chi(\overline{U})$. When U is perfect (and thus, $\overline{U}$ is perfect by Lemma 2.3),

$$\omega(\overline{U}) = \alpha(U) = \beta(U) = \chi(\overline{U}),$$

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which is achieved by the clique covering scheme [23]. The following generalizes this result.

Proposition 8.4 ([167])**.** The capacity region of an index coding problem with undirected perfect side information graph U is equal to the MAIS outer bound and is achieved by fractional clique covering.

To prove the proposition, note that the polytope associated with U in (2.6) is equivalent to the MAIS outer bound $\mathscr{R}_{\text{MAIS}}$ in Theorem 5.1. The other polytope associated with U in (2.5) is achievable by fractional clique covering, since the rate tuple given by the incidence vector of each clique in U is achievable (see Remark 6.3). Since U is perfect, the two polytopes are equivalent by Lemma 2.3.

8.4 Locally Acyclic Graphs

We say that a side information graph G is *locally acyclic* if for every $i \in [n]$, the induced subgraph $G|_{[n]\setminus A_i}$ is acyclic. For example, the side information graph depicted in Figure 8.2 is locally acyclic since the subgraphs induced by $[3] \setminus A_1 = \{1, 3\}$, $[3] \setminus A_2 = \{2\}$, and $[3] \setminus A_3 = \{2, 3\}$ are respectively acyclic.

The following characterizes the capacity region for this class of side information graphs.

Proposition 8.5. For any index coding problem G, $\mathscr{C}(G) = \mathscr{R}_{\text{flat}}(G)$ iff G is locally acyclic.

The sufficiency follows by comparing the flat coding inner bound $\mathscr{R}_{\text{flat}}$ in Proposition 6.10 and the MAIS outer bound $\mathscr{R}_{\text{MAIS}}$ in Theorem 5.1. For the necessity, suppose that $\mathscr{R}_{\text{flat}}(G)$ is the capacity region

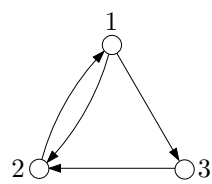


Figure 8.2: The index coding problem $(1|2), (2|1, 3), (3|1)$ with $\beta = 2$, which is achievable by an MDS code.

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and $G|_{[n]\setminus A_i}$ contains a cycle for some $i \in [n]$. Then by partial clique covering, the rate tuple $(\tilde{R}_1, \ldots, \tilde{R}_n)$ such that

$$\tilde{R}_j = \begin{cases} \frac{1}{n-1-|A_i|} & j \in [n] \setminus A_i, \\ 0 & \text{otherwise} \end{cases}$$

is achievable. But

$$\sum_{j \notin A_i} \tilde{R}_j = \frac{n - |A_i|}{n - 1 - |A_i|} > 1$$

and thus $(\tilde{R}_1, \ldots, \tilde{R}_n) \notin \mathscr{R}_{\text{flat}}(G)$, which contradicts the assumption that $\mathscr{R}_{\text{flat}}(G)$ is the capacity region.

As a corollary to Proposition 8.5, the broadcast rate of any index coding problem with locally acyclic side information graph is

$$\beta(G) = \kappa(G) = n - \text{minindeg}(G) = n - \min_i |A_i|, \tag{8.3}$$

which is achieved, for example, by an MDS code. Local acyclicity, however, is not necessary for (8.3) to hold. For example, the 4-message index coding problem depicted in Figure 8.3 is not locally acyclic since the subgraph induced by $[4] \setminus A_1 = \{1, 2, 3\}$ contains a cycle, yet $\kappa(G) = \beta_{\text{MAIS}}(G) = \beta(G) = 3$.

Open Problem 8.3. Characterize all n-message index coding problems G for which (8.3) holds.

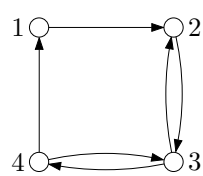


Figure 8.3: The index coding problem $(1|4), (2|1, 3), (3|2, 4), (4|3)$ with $\beta = 3$, which is achievable by an MDS code.

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8.5 Symmetric Neighboring Side Information

In Proposition 6.9, we established an upper bound on the broadcast rate for index coding problems that have symmetric neighboring side information, i.e.,

$$A_i = \{i \ominus d, \ldots, i \ominus 1, i \oplus 1, \ldots, i \oplus u\}, \quad i \in [n], \tag{8.4}$$

where $\oplus$ and $\ominus$ denote addition and subtraction modulo n, respectively, and $0 \leq d \leq u \leq u+d \leq n-2$ or $(u,d) = (n-1,0)$. The polymatroidal outer bound in Section 5.2 implies the following bound on achievable rates for these problems.

Lemma 8.3 ([107]). If $(R_1, \ldots, R_n)$ is achievable for an index coding problem with side information sets in (8.4), then

$$\sum_{i=1}^{n} R_i \leq \frac{(d+1)n}{n-(u-d)}. \tag{8.5}$$

Proposition 6.9 and Lemma 8.3 characterize the broadcast rate of this class of index coding problems as follows.

Proposition 8.6 ([107]). For any index coding problem G with symmetric neighboring side information sets in (8.4),

$$\beta(G) = \frac{n-(u-d)}{d+1}. \tag{8.6}$$

The broadcast rate in (8.6) is also achievable by the composite coding scheme [8]. It is unknown, however, the composite coding inner bound in Proposition 6.11 and the polymatroidal outer bound in Theorem 5.2 coincide and characterize the capacity region.

Open Problem 8.4. Characterize the capacity region for index coding problems with symmetric neighboring side information sets.

Remark 8.3. By taking different values of (u,d), the broadcast rates of all n-message index coding problems with symmetric neighboring side information sets can be enumerated as

$$\mathfrak{B}_n = \{1\} \cup \bigcup_{k=1}^{\lfloor \frac{n}{2} \rfloor} \left\{ \frac{2k}{k}, \frac{2k+1}{k}, \ldots, \frac{n}{k} \right\}. \tag{8.7}$$

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For example,

$$\mathfrak{B}_5 = \left\{1, 2, \frac{5}{2}, 3, 4, 5\right\},$$
$$\mathfrak{B}_6 = \left\{1, 2, \frac{5}{2}, 3, 4, 5, 6\right\},$$
$$\mathfrak{B}_7 = \left\{1, 2, \frac{7}{3}, \frac{5}{2}, 3, \frac{7}{2}, 4, 5, 6, 7\right\}.$$

These values cover the broadcast rates of many other index coding problems, but not all of them. For example, the 6-message index coding problem in Figure 8.4 that combines the 5-message problem with broadcast rate of 5/2 in Figure 5.2 and a separate single-message problem has $\beta = 5/2 + 1 = 7/2 \notin \mathfrak{B}_6$.

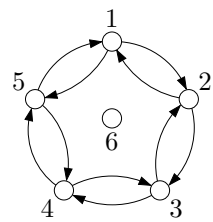


Figure 8.4: The index coding problem $(1|2,5), (2|1,3), (3|2,4), (4|3,5), (5|1,4), (6|-)$ with $\beta = \beta(G|_{\{1,2,3,4,5\}}) + \beta(G|_{\{6\}}) = 5/2 + 1 = 7/2$.

8.6 Problems with Small Number of Messages

As mentioned earlier in Section 1.2, the number of nonisomorphic instances of the index coding problem with n messages grows superexponentially in n. In this section, we discuss the capacity regions of all index coding problems with $n \leq 5$ and the broadcast rates of all index coding problems with $n \leq 6$. The software used to compute these results is available at a GitHub repository [16].

For $n = 1$, there is one index coding instance $(1|-)$, the capacity region of which is $\mathscr{C}_1 = [0, 1]$ and can be expressed by the inequality

$$1 \cdot R_1 \leq 1.$$

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Trivially, this instance is critical with $\beta = 1$.

For $n = 2$, there are 3 nonisomorphic index coding instances

$$(1|-),(2|-), \qquad (1|-),(2|1), \qquad (1|2),(2|1)$$

as shown in Figure 4.3. By Remark 4.1 (or more generally, by Theorem 4.1), the capacity region $\mathscr{C}_2^{(a)}$ of the first two is characterized by

$$\begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{R} \leq \mathbf{1}.$$

By Remark 4.2 or Theorem 4.1, the capacity region $\mathscr{C}_2^{(c)}$ of the third is characterized by

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{R} \leq \mathbf{1}.$$

The first and third problems are critical and the second one is not. The corresponding broadcast rates are $2, 2$, and 1.

For $n = 3$, there are 16 nonisomorphic index coding instances as shown in Figure 8.5. For problems (a) through (f), among which only (a) is critical, the broadcast rate is 3 and the capacity region $\mathscr{C}_3^{(a)}$ is the polytope characterized by

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \mathbf{R} \leq 1.$$

For problems (g) through (l), among which (g) is critical, the broadcast rate is 2 and the capacity region $\mathscr{C}_3^{(g)}$ is characterized by

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{R} \leq \mathbf{1}.$$

For problem (m), which is critical, the broadcast rate is 2 and the capacity region $\mathscr{C}_3^{(m)}$ is characterized by

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{R} \leq \mathbf{1}.$$

For problems (n) and (o), among which (n) is critical, the broadcast rate is 2 and the capacity region $\mathscr{C}_3^{(n)}$ is characterized by

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{R} \leq 1.$$

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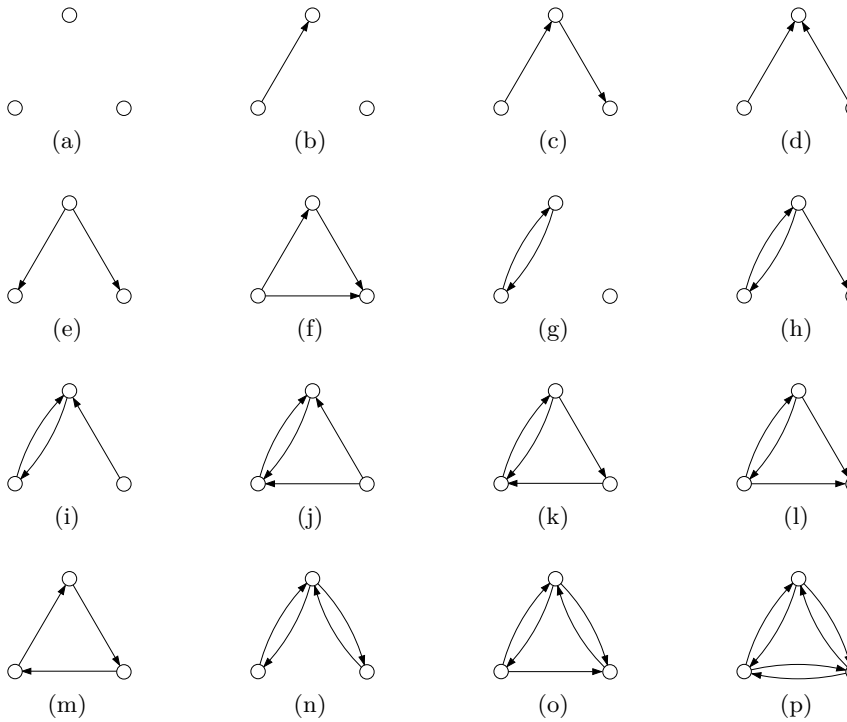


Figure 8.5: Sixteen nonisomorphic 3-message index coding instances.

Finally, for problem (p), which is critical, the broadcast rate is 1 and the capacity region $\mathscr{C}_3^{(p)}$ is characterized by

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{R} \leq 1.$$

For each of the 16 instances, the capacity region is achieved by the fractional (local) partial clique covering scheme and coincides with the MAIS outer bound. The criticality of each instance can be checked by comparing the union-of-unicycles condition in Theorem 7.1 and the three necessary conditions in Theorem 7.2, which agree for all sixteen 3-message index coding instances (in fact, only the MAIS looseness condition in Proposition 7.6 is needed). Among the five critical instances,

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(a), (g), (n), and (p) can be decomposed into smaller problems by Theorem 4.1 as

$$\begin{aligned}
\mathscr{C}_3^{(a)} &= \mathscr{C}_2^{(a)} \circ (\mathscr{C}_1, \mathscr{C}_2^{(a)}),\\
\mathscr{C}_3^{(g)} &= \mathscr{C}_2^{(a)} \circ (\mathscr{C}_1, \mathscr{C}_2^{(c)}),\\
\mathscr{C}_3^{(n)} &= \mathscr{C}_2^{(c)} \circ (\mathscr{C}_1, \mathscr{C}_2^{(a)}),\\
\mathscr{C}_3^{(p)} &= \mathscr{C}_2^{(c)} \circ (\mathscr{C}_1, \mathscr{C}_2^{(c)}).
\end{aligned}$$

Hence, problem (m) is the only bona-fide 3-message index coding problem, for which the flat coding inner bound and the MAIS outer bound on the capacity region coincide (cf. Proposition 8.5).

For $n = 4$, there are 218 nonisomorphic index coding instances. For each instance, the fractional local partial clique covering inner bound coincides with the MAIS outer bound on the capacity region. In addition, the broadcast rate of each instance is achievable by a scalar linear code over $\mathbb{F}_2$. There are 18 critical instances, which are unions of unicycles and have distinct capacity regions, as listed in Table 8.1. Among these, 4 problems (o), (p), (q), and (r), shown in Figure 8.6, are not decomposable into smaller problems.

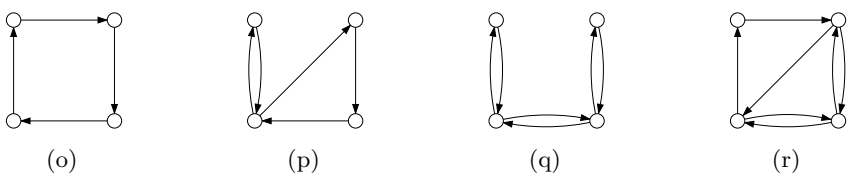


Figure 8.6: Four 4-message nonisomorphic, critical, indecomposable index coding problems.

For $n = 5$, there are 9,608 nonisomorphic index coding instances. For each instance, the composite coding inner bound coincides with the polymatroidal outer bound on the capacity region [8]. In addition, the broadcast rate is achievable by a scalar linear code over $\mathbb{F}_2$ for all but 28 instances. For those few exceptions, a vector linear code over $\mathbb{F}_2$ with $t = 2$ suffices [118]. There are 136 instances that satisfy the union-of-unicycles sufficient condition in Theorem 7.1 and 175 instances that

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Table 8.1: Eighteen 4-message nonisomorphic critical index coding instances and their capacity regions.

	Problem	Capacity region
(a)	$(1\|-),(2\|-),(3\|-),(4\|-)$	$\mathscr{C}_2^{(a)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(a)})$
(b)	$(1\|-),(2\|3),(3\|2),(4\|-)$	$\mathscr{C}_2^{(a)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(g)})$
(c)	$(1\|-),(2\|4),(3\|2),(4\|3)$	$\mathscr{C}_2^{(a)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(m)})$
(d)	$(1\|-),(2\|3,4),(3\|2),(4\|2)$	$\mathscr{C}_2^{(a)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(n)})$
(e)	$(1\|-),(2\|3,4),(3\|2,4),(4\|2,3)$	$\mathscr{C}_2^{(a)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(p)})$
(f)	$(1\|2),(2\|1),(3\|4),(4\|3)$	$\mathscr{C}_2^{(a)} \circ (\mathscr{C}_2^{(c)}, \mathscr{C}_2^{(c)})$
(g)	$(1\|2,3,4),(2\|1),(3\|1),(4\|1)$	$\mathscr{C}_2^{(c)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(a)})$
(h)	$(1\|2,3,4),(2\|1,3),(3\|1,2),(4\|1)$	$\mathscr{C}_2^{(c)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(g)})$
(i)	$(1\|2,3,4),(2\|1,4),(3\|1,2),(4\|1,3)$	$\mathscr{C}_2^{(c)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(m)})$
(j)	$(1\|2,3,4),(2\|1,3,4),(3\|1,2),(4\|1,2)$	$\mathscr{C}_2^{(c)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(n)})$
(k)	$(1\|2,3,4),(2\|1,3,4),(3\|1,2,4),(4\|1,2,3)$	$\mathscr{C}_2^{(c)} \circ (\mathscr{C}_1, \mathscr{C}_3^{(p)})$
(l)	$(1\|3,4),(2\|3,4),(3\|1,2),(4\|1,2)$	$\mathscr{C}_2^{(c)} \circ (\mathscr{C}_2^{(a)}, \mathscr{C}_2^{(a)})$
(m)	$(1\|3,4),(2\|1),(3\|2),(4\|2)$	$\mathscr{C}_3^{(m)} \circ (\mathscr{C}_1, \mathscr{C}_1, \mathscr{C}_2^{(a)})$
(n)	$(1\|3,4),(2\|1),(3\|2,4),(4\|2,3)$	$\mathscr{C}_3^{(m)} \circ (\mathscr{C}_1, \mathscr{C}_1, \mathscr{C}_2^{(c)})$
(o)	$(1\|4),(2\|1),(3\|2),(4\|3)$	$\begin{bmatrix}1&1&1&0\\1&1&0&1\\1&0&1&1\\0&1&1&1\end{bmatrix}\mathbf{R} \leq \mathbf{1}$
(p)	$(1\|4),(2\|4),(3\|2),(4\|1,3)$	$\begin{bmatrix}1&1&1&0\\0&1&0&1\\0&0&1&1\end{bmatrix}\mathbf{R} \leq \mathbf{1}$
(q)	$(1\|4),(2\|3),(3\|2,4),(4\|1,3)$	$\begin{bmatrix}1&1&0&0\\1&0&1&0\\0&1&0&1\end{bmatrix}\mathbf{R} \leq \mathbf{1}$
(r)	$(1\|4),(2\|1,3),(3\|2,4),(4\|2,3)$	$\begin{bmatrix}1&1&0&0\\1&0&1&0\\1&0&0&1\\0&1&0&1\end{bmatrix}\mathbf{R} \leq \mathbf{1}$

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satisfy the necessary conditions in Theorem 7.2. Both conditions fail to capture the criticality and there are in fact 141 critical instances (recall Open Problem 7.3). The five instances that are critical yet not unions of unicycles are shown in Figure 8.7. The first two problems (a) and (b) are transposes of each other, while each of the next three problems (c), (d), and (e) is isomorphic to its transpose (recall Open Problem 7.2). Including these 5 instances, there are 68 critical instances that are not decomposable into smaller problems. As mentioned earlier in Section 7.1, some instances share the same capacity region, and there are 118 distinct capacity regions (up to relabeling). In Figure 8.7, problems (a), (b), and (c), and problems (d) and (e), respectively, have the same capacity regions. The capacity regions of all 141 critical instances coincide with the corresponding MAIS inner bounds, except the two indecomposable problems in Figures 5.2 and 5.3. It can be verified that $\beta = 3$ iff $\beta_{\text{MAIS}} = 3$ (recall Open Problem 8.1).

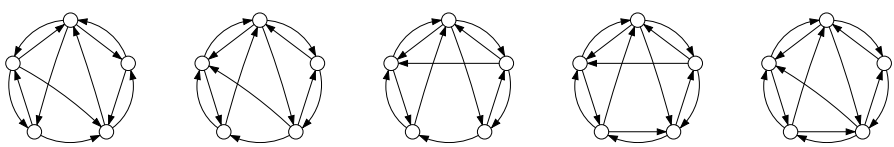


Figure 8.7: Five 5-message nonisomorphic, critical, indecomposable index coding problems that are not unions of unicycles.

For $n = 6$, there are 1,540,944 nonisomorphic instances. For simplicity, we focus on the broadcast rate. For each instance, the broadcast rate is equal to the polymatroidal lower bound, which coincides with either composite coding or scalar linear coding [10]; see Example 6.1. In order to isolate the problems that are critical and indecomposable, we adapt the notion of criticality for the capacity region to the one for the broadcast rate, namely, an index coding problem G is *β-critical* if $\beta(G_e) > \beta(G)$ for every edge e. Note that β-criticality implies ($\mathscr{C}$-)criticality. Hence, the necessary conditions for the $\mathscr{C}$-criticality discussed in Section 7.2 are also necessary for the β-criticality. In light of Remark 7.1, the MAIS looseness condition in Proposition 7.6 can be

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relaxed as the condition $\beta_{\text{MAIS}}(G_e) \neq \beta(G_e)$ for every edge $e \in E$ that does not belong to a unicycle. In summary, we have the following necessary conditions for the β-criticality.

Proposition 8.7. If $G = (V, E)$ is β-critical, then

1. every edge lies on a cycle,
2. side information sets are nondegraded, and
3. for each $e \in E$ that is not in some unicycle, $\beta_{\text{MAIS}}(G_e) \neq \beta(G_e)$.

These conditions are mutually independent (cf. Examples 7.3–7.5) and there are 9,747 instances that satisfy all three of them. A closer examination reveals that there are only 259 β-critical instances with 6 messages. Among these, 152 instances are not decomposable into smaller problems. For all but 13 of these 152 indecomposable instances, the composite coding upper bound coincides with the polymatroidal lower bound on the broadcast rate. For the remaining 13 instances, the broadcast rate is 3 and scalar linear codes over $\mathbb{F}_2$ achieve it. It can be verified that $\beta = 4$ iff $\beta_{\text{MAIS}} = 4$ (Open Problem 8.1), and that $\beta(G^T) = \beta(G)$ (Open Problem 4.1) for all 6-message index coding instances.

We conclude this section by summarizing the collective behavior of all index coding instances with $n \leq 6$. Let $\mathfrak{B}_n^*$ be the set of all values of the broadcast rate an n-message index coding problem can take (cf. $\mathfrak{B}_n$ in Section 8.5). We have

$$\begin{aligned}
\mathfrak{B}_1^* &= \{1\},\\
\mathfrak{B}_2^* &= \{1, 2\},\\
\mathfrak{B}_3^* &= \{1, 2, 3\},\\
\mathfrak{B}_4^* &= \{1, 2, 3, 4\},\\
\mathfrak{B}_5^* &= \left\{1, 2, \frac{5}{2}, 3, 4, 5\right\},\\
\mathfrak{B}_6^* &= \left\{1, 2, \frac{5}{2}, 3, \frac{13}{4}, \frac{10}{3}, \frac{7}{2}, 4, 5, 6\right\}.
\end{aligned}$$

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This brings up the following challenging question that spans beyond all the simple examples we have seen so far (see Remarks 8.1 and 8.2 as well as Open Problem 8.2).

Open Problem 8.5. Characterize the set $\mathfrak{B}_n^*$ of all values of the broadcast rate an n-message index coding problem can take.

Simpler (but still challenging) problems are as follows.

Open Problem 8.6. Is $\mathfrak{B}_n^* \subseteq \mathbb{Q}$?

Open Problem 8.7. How does $|\mathfrak{B}_n^*|$ scale in n?

Similar problems can be posed for capacity regions, starting with $n = 6$.

Open Problem 8.8. Characterize the capacity regions of all 6-message index coding problems.

We can also compute the expected broadcast rate $\bar{\beta}_n$ of the random n-message index coding problem whose side information graph is the directed Erdös–Rényi random graph $G(n, 1/2)$ (i.e., $\bar{\beta}_n$ is the uniform average of the broadcast rates of all $2^{n(n-1)}$ index coding instances including isomorphic ones):

$$\begin{aligned}
\bar{\beta}_1 &= 1,\\
\bar{\beta}_2 &= 1.75,\\
\bar{\beta}_3 &= 2.375,\\
\bar{\beta}_4 &= 2.9075,\\
\bar{\beta}_5 &= 3.3794,\\
\bar{\beta}_6 &= 3.7960.
\end{aligned}$$

Open Problem 8.9. How does $\bar{\beta}_n$ scale in n?

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9

Capacity Approximation

With no computable characterization of the broadcast rate (or capacity) of a general index coding problem, the next step forward seems to be approximating it. Keeping in mind that the broadcast rate β for an n-message index coding problem is $O(n)$, we present in this section capacity approximation results of sublinear multiplicative factors. Our main focus will be several classes of index coding problems for which the multiplicative factor can be tightened to $O(n^{1-\epsilon})$ for some $\epsilon > 0$.

9.1 Ramsey Numbers

We first review some results in Ramsey theory which we will use in the rest of the section to approximate the broadcast rate. Given a class $\mathcal{G}$ of undirected graphs and two positive integers i and j, the *Ramsey number* $R_{\mathcal{G}}(i, j)$ is defined as the smallest positive integer such that every graph in $\mathcal{G}$ with at least $R_{\mathcal{G}}(i, j)$ vertices has a clique of size i or an independent set of size j. If $\mathcal{G}$ is the class of all undirected finite simple graphs, then the Ramsey number is denoted by $R(i, j)$. In general, determining Ramsey numbers for most classes $\mathcal{G}$ is quite difficult, but they are easily computed for very small values of i and j.

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Lemma 9.1 (Belmonte, Heggernes, Hof, Rafiey, and Saei [24])**.** For any nonempty graph class $\mathcal{G}$ of undirected graphs,

$$R_{\mathcal{G}}(1, j) = R_{\mathcal{G}}(i, 1) = 1, \quad i, j \geq 1.$$

If $\mathcal{G}$ is the set of all edgeless graphs, i.e., all $U = (V, E)$ with $E = \emptyset$, then

$$R_{\mathcal{G}}(2, j) = j, \quad j \geq 1.$$

Similarly, if $\mathcal{G}$ is the set of all complete graphs, i.e., all $U = (V, E)$ with $E = \{\{i, j\} : i \neq j \in V\}$, then

$$R_{\mathcal{G}}(i, 2) = i, \quad i \geq 1.$$

The following upper bound on the Ramsey number is well-known.

Lemma 9.2 (Erdős and Szekeres [68])**.** For any $i, j \geq 1$,

$$R(i, j) \leq \binom{i + j - 2}{i - 1}.$$

For some classes of graphs, this upper bound can be tightened.

A graph is said to be *planar* if it can be drawn on a plane without edges crossing each other. Figure 9.1 shows examples of a 4-vertex planar graph and a 5-vertex nonplanar graph. Note that an edgeless graph is planar. Let $\mathcal{P}$ be the class of undirected planar graphs. The Ramsey number for this class is completely determined; see Table 9.1.

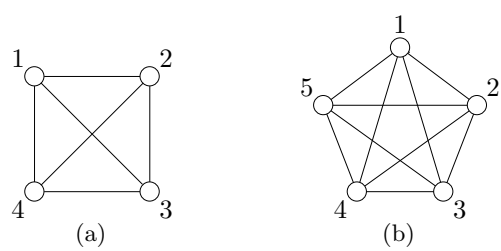


Figure 9.1: (a) A 4-vertex planar graph (the edge $\{1, 3\}$ can be drawn such that it does not cross $\{2, 4\}$). (b) A 5-vertex nonplanar graph.

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Table 9.1: Ramsey numbers for planar graphs

(i, j)	$R_{\mathcal{P}}(i, j)$	Reference
$i = 1,\ j \geq 1$	1	Lemma 9.1
$i = 2,\ j \geq 1$	j	Lemma 9.1
$i = 3,\ j \geq 1$	$3j - 3$	[144]
$i \geq 4,\ j \geq 1,\ (i, j) \neq (4, 2)$	$4j - 3$	[144]
$(4, 2)$	4	[144]

The *line graph* of an undirected graph U is obtained by associating a vertex with each edge of the graph U and connecting two vertices with an edge iff the corresponding edges of U have a vertex in common. Figure 9.2 shows a graph and its line graph. Edgeless graphs and complete graphs belong to the class of line graphs. Table 9.2 lists Ramsey numbers for the class $\mathcal{L}$ of line graphs (of some undirected graphs).

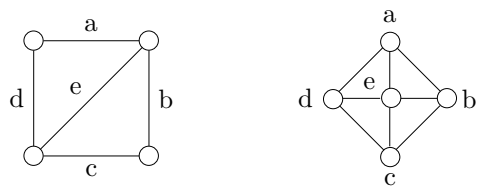


Figure 9.2: (a) A graph with 5 edges. (b) The corresponding 5-vertex line graph.

Table 9.2: Ramsey numbers for line graphs

(i, j)	$R_{\mathcal{L}}(i, j)$	Reference
$i = 1,\ j \geq 1$	1	Lemma 9.1
$i = 2,\ j \geq 1$	j	Lemma 9.1
$i \geq 1,\ j = 1$	1	Lemma 9.1
$i \geq 1,\ j = 2$	i	Lemma 9.1
$i = 3,\ j \geq 3$	$\lfloor (5j - 3)/2 \rfloor$	[109]
$i \geq 4,\ j \geq 3$	$\leq i(j - 1) + 2^*$	[24]

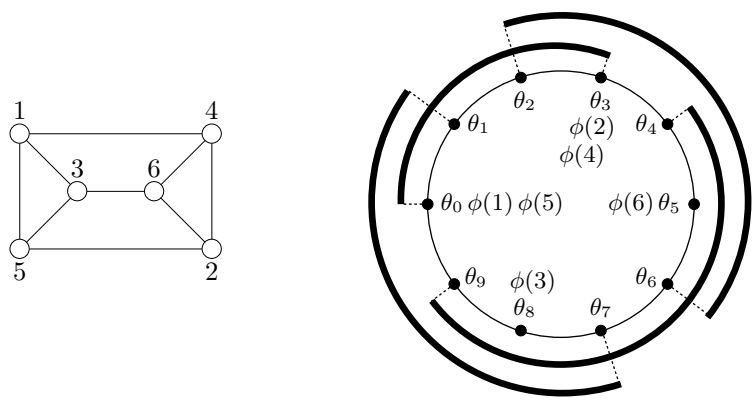
* The exact value is known [24].

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An undirected graph $U = (V, E)$ is said to be a *fuzzy circular interval* graph [47] if

- there exists a set F of closed intervals $\{e^{j\theta} : a \leq \theta \leq b\}$ of the unit circle $\mathbb{T} = \{e^{j\theta} : \theta \in [0, 2\pi)\}$ such that (1) no interval includes another and (2) no point of $\mathbb{T}$ is an endpoint of more than one interval in F, and
- there exists a mapping $\phi : V \to \mathbb{T}$ such that (1) if $\{i, j\} \in E$, then $\phi(i)$ and $\phi(j)$ belong to a common interval of F, and (2) if $\{i, j\} \notin E$, then either there is no interval in F that contains both $\phi(i)$ and $\phi(j)$, or there is exactly one interval in F whose endpoints are $\phi(i)$ and $\phi(j)$.

Note that an edgeless graph is a fuzzy circular interval graph ($F = \emptyset$) and a complete graph is a fuzzy circular interval graph ($F = \mathbb{T}$). Figure 9.3 shows a more interesting example of $\overline{C}_6$. Let $\mathcal{F}$ be the class of fuzzy circular interval graphs. The Ramsey number for this class is given in Table 9.3.



(a) (b)

Figure 9.3: (a) The complement $\overline{C}_6$ of the cycle graph C_6. (b) The fuzzy circular interval model of $\overline{C}_6$, where the intervals $[\theta_0, \theta_3], [\theta_2, \theta_6], [\theta_4, \theta_9], [\theta_7, \theta_1]$ are shown in thick arcs and $\phi(1) = \phi(5) = \theta_0$, $\phi(2) = \phi(4) = \theta_3$, $\phi(3) = \theta_8$, and $\phi(6) = \theta_5$. It is easy to check that F and ϕ satisfy the aforementioned conditions. For example, the edge $\{3, 6\}$ exists and the interval $[\theta_4, \theta_9]$ contains $\phi(3)$ and $\phi(6)$. The edge $\{1, 2\}$ does not exist and there is exactly one interval $[\theta_0, \theta_3]$ with endpoints $\phi(1)$ and $\phi(2)$.

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Table 9.3: Ramsey numbers for fuzzy circular interval graphs

(i, j)	$R_{\mathcal{F}}(i, j)$	Reference
$i = 1,\ j \geq 1$	1	Lemma 9.1
$i = 2,\ j \geq 1$	j	Lemma 9.1
$i \geq 1,\ j = 1$	1	Lemma 9.1
$i \geq 1,\ j = 2$	i	Lemma 9.1
$i \geq 3,\ j \geq 3$	$(i-1)j$	[24]

We summarize these results as a simple bilinear upper bound on the Ramsey number for the classes of planar, line, and fuzzy circular interval graphs.

Lemma 9.3. For $\mathcal{G} = \mathcal{P}, \mathcal{L}$, or $\mathcal{F}$,

$$R_{\mathcal{G}}(i, j) \leq ij, \quad i, j \geq 1.$$

9.2 Approximate Capacity for Some Classes of Problems

Blasiak, Kleinberg, and Lubetzky [30] established the following approximation result for index coding instances with undirected (bidirectional) side information graphs based on the clique covering scheme discussed in Section 6.2.

Proposition 9.1 ([30])**.** For any n-message index coding problem with bidirectional (undirected) side information graph U,

$$\frac{\chi(\overline{U})}{\beta(U)} = O\left(\frac{n}{\log n}\right).$$

In other words, the clique covering scheme approximately achieves the broadcast rate within a multiplicative factor of $O(n/\log n)$.

To prove the proposition, we consider two cases. First, if $\chi(\overline{U}) < 4n/\log n$, then trivially

$$\frac{\chi(\overline{U})}{\beta(U)} \leq \chi(\overline{U}) < \frac{4n}{\log n}.$$

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Second, for the other case of $\chi(\overline{U}) \geq 4n/\log n$, we rely on the following lemma relating the independence number of an undirected graph with the chromatic number of its complement via Ramsey numbers.

Lemma 9.4 (Alon and Kahale [4]). Let $U = (V, E)$ be an undirected graph with $|V| = n$. Let $\alpha_i(m) = \max\{j : R(i, j) \leq m\}$. If $\chi(\overline{U}) \geq (n/i) + m$, then an independent set of size $\alpha_i(m)$ can be found in U.

Continuing with the proof of Proposition 9.1, let $i = (\log n)/2$ and $m = n/i$. Then for n sufficiently large,

$$\begin{aligned}\alpha(U) &\geq \max\left\{j : R(\tfrac{1}{2}\log n, j) \leq \frac{n}{i}\right\} &(9.1)\\ &\geq \max\left\{j : \binom{\frac{1}{2}\log n + j - 2}{\frac{1}{2}\log n - 1} \leq \frac{n}{i}\right\} &(9.2)\\ &\geq \frac{1}{2}\log n,\end{aligned}$$

where (9.1) follows by Lemma 9.4, and (9.2) follows by Lemma 9.2. Hence, by Remark 5.2, we have

$$\frac{\chi(\overline{U})}{\beta(U)} \leq \frac{n}{\beta(U)} \leq \frac{n}{\alpha(U)} \leq O\left(\frac{n}{\log n}\right).$$

This completes the proof of Proposition 9.1.

Next, we consider the class of *unidirectional* graphs. In this case, by Lemma 2.1, for any unidirectional graph G on n vertices and any tournament G' resulting from adding edges with arbitrary direction between the vertices that are not connected in G, we have

$$\log n \leq \beta(G') \leq \beta(G).$$

Since uncoded transmission trivially achieves the upper bound of n on the broadcast rate, we have the following.

Proposition 9.2 ([14]). For any n-message index coding problem with unidirectional side information graph G,

$$\frac{n}{\beta(G)} \leq \frac{n}{\log n}.$$

In words, uncoded transmission approximately achieves the broadcast rate within a multiplicative factor of $O(n/\log n)$.

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For general *directed* graphs, we have the following result by Blasiak, Kleinberg, and Lubetzky, the proof of which can be found in [30, Sec. V.B].

Proposition 9.3 ([30])**.** For any index coding problem with n messages with side information graph G,

$$\frac{\chi_f(U(\overline{G}))}{\beta(G)} = O\left(\frac{n \log\log n}{\log n}\right).$$

In words, the fractional clique covering scheme approximately achieves the broadcast rate within a multiplicative factor of $O(n \log\log n / \log n)$.

Beyond Propositions 9.1 through 9.3, there is no known approximation for the broadcast rate of a general (undirected/bidirectional, unidirectional, or directed) index coding problem.

Open Problem 9.1. Improve the multiplicative factors in Propositions 9.1 through 9.3.

A stronger question is as follows.

Open Problem 9.2. Approximate the broadcast rate of an n-message index coding problem from above within a multiplicative factor of $O(n^{1-\epsilon})$ for some $\epsilon > 0$.

For now it is not even clear whether such a tight approximation is feasible at all (cf. Open Problem 3.2).

Open Problem 9.3. Does there exist a finite, terminating algorithm that approximates the broadcast rate β of an arbitrary n-message index coding problem by producing lower and upper bounds of the form

$$\Omega(n^{\epsilon}) \leq \beta \leq O(n^{\epsilon'})$$

for some $0 \leq \epsilon \leq \epsilon' \leq 1$ such that $\epsilon' - \epsilon < 1$?

The approximation results in Propositions 9.1 through 9.3 continue to hold for the linear broadcast rate. Questions similar to Open Problems 9.1 through 9.3 can be asked for the linear broadcast rate $\lambda_{\mathbb{F}}$ and its minimum over all $\mathbb{F}$.

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As attacking Open Problem 9.2 in general is very difficult, we limit ourselves to studying the conditions under which a tighter approximation exists. We first adapt the technique used in the proof of Proposition 9.1 to present a condition under which the broadcast rate of an index coding problem can be approximated within a factor of $O(n^{1-\epsilon})$ for some $\epsilon > 0$. The proof is relegated to Appendix 9.A.

Theorem 9.1 ([14]). Let $\mathcal{G}$ be a class of undirected graphs for which $R_{\mathcal{G}}(i,j) \leq ci^a j^b$ holds for some constants a, b, and c. For any index coding problem with side information graph $G \in \mathcal{G}$ such that $|V(G)| = n$, the clique covering scheme approximates the broadcast rate within a multiplicative factor of $2^{(a+1)/(a+b+1)} c^{1/(a+b+1)} n^{(a+b)/(a+b+1)}$.

As stated in Lemma 9.3, planar graphs, line graphs and fuzzy circular interval graphs are three classes that satisfy the condition of Theorem 9.1 with $a = b = c = 1$.

Corollary 9.1. If G is a planar graph or a line graph or a fuzzy circular interval graph with n vertices, the clique covering scheme approximates the broadcast rate within a multiplicative factor of $(2n)^{2/3}$.

For (directed) planar graphs, we can establish a tighter approximation with simple lower and upper bounds. We recall the celebrated four-color theorem.

Lemma 9.5 (Four-color theorem [6]). Every planar graph U is four-colorable, i.e., $\chi(U) \leq 4$.

If the undirected graph $U(G)$ associated with the n-vertex side information graph G is planar, then

$$\begin{aligned} \frac{n}{4} &\leq \frac{n}{\chi(U(G))} && (9.3)\\ &\leq \frac{n}{\chi_f(U(G))} && (9.4)\\ &\leq \alpha(U(G)) && (9.5)\\ &\leq \beta(U(G)) && (9.6)\\ &\leq \beta(G) && (9.7)\\ &\leq n, \end{aligned}$$

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where (9.3) follows by the four-color theorem, (9.4) by (2.2), (9.5) by Lemma 2.2, (9.6) by Remark 5.2 and (9.7) holds since adding side information decreases the broadcast rate. Similarly, if $U(\overline{G})$ is planar,

$$1 \leq \beta(G) \leq \chi(U(\overline{G})) \leq 4.$$

The following summarizes these results.

Theorem 9.2 ([14]). For any index coding problem with side information graph G such that $U(G)$ $[U(\overline{G})]$ is planar, uncoded transmission [clique covering] approximates the broadcast rate within a multiplicative factor of four.

Remark 9.1. The approximation results in Theorems 9.1 and 9.2 hold for the linear broadcast rate as well. For bidirectional *outerplanar* graphs [43], which form a special subclass of planar graphs, the clique covering scheme actually achieves the scalar linear capacity [25].

Remark 9.2. For any index coding problem with planar side information graph G, and (1) G is bidirectional, (2) G is unidirectional, (3) $U(G)$ is triangle-free, (4) G is bidirectional and triangle-free, (5) the length of the smallest directed cycle of G is 4, or (6) the length of the smallest directed cycle of G is at least 5, then Theorem 9.2 can be further strengthened to the factor of 2, 3, 3, 3/2, 12/5, and 2, respectively [112, 166].

Sparse graphs form another class for which we have a partial solution to Open Problem 9.2. We recall the celebrated Turán's theorem in two forms.

Lemma 9.6 (Turán's theorem [152]). If an undirected graph $U = (V, E)$ is K_{m+1}-free, then

$$|E| \leq \left(1 - \frac{1}{m}\right)\frac{|V|^2}{2}.$$

Equivalently, for any undirected graph $U = (V, E)$,

$$\alpha(U) \geq \frac{|V|^2}{2|E| + |V|}.$$

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Suppose that the side information graph $G = (V, E)$ is sparse with $|V| = n$ and $|E| = O(n^{1+\epsilon})$ for some $\epsilon < 1$. Then by Turán's theorem,

$$\Omega(n^{1-\epsilon}) = \alpha(U(G)) \leq \beta(G) \leq n.$$

An argument similar to the proof of Theorem 9.2 establishes the following result by Yi and Caire [166] and by Agarwal, Flodin, and Mazumdar [1].

Theorem 9.3 ([166, 1])**.** For any n-message index coding problem with side information graph $G = (V, E)$, if $|E| = O(n^{1+\epsilon})$, then uncoded transmission approximately achieves the broadcast rate within a multiplicative factor of $O(n^{\epsilon})$.

Remark 9.3. Tahmasbi, Shahrasbi, and Gohari [147] used a tight version of Turán's theorem (see, for example, [155, Problem 5.2.18]) to find the minimum number of edges necessary for a given broadcast rate, which also implies Theorem 9.3.

Remark 9.4. Agarwal, Flodin, and Mazumdar [1] also established tight approximation results for several other classes of index coding problems, including *unit disk graphs*.

In all class-specific improvements of Proposition 9.3 we have discussed thus far, we approximated the broadcast rate from above using (fractional) clique covering and the trivial lower bound of $\beta \geq 1$ or from below using the independence number and the trivial upper bound of $\beta \leq n$ (uncoded transmission).

Open Problem 9.4. Find a class of n-message index coding problems for which

$$\Omega(n^{\epsilon}) \leq \beta \leq O(n^{\epsilon'})$$

for some $0 < \epsilon \leq \epsilon' < 1$.

Remark 9.5. Although approximating β itself is rather difficult for a general n-message index coding problem, approximating $n - \beta$ is much easier. This *complementary index coding* rate, namely, the amount of relative improvement of the optimal index code over uncoded transmission, was originally studied by Chaudhry, Asad, Sprintson, and

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Langberg [44], who showed that $n-\beta$ as well as $n-\lambda_{\mathbb{F}}$ can be approximated within a multiplicative factor of $O(\log n \log\log n)$. Mazumdar [111] strengthened this result and also established a factor-of-two approximation of both $n-\beta$ and $n-\lambda_{\mathbb{F}}$ for bidirectional graphs. We will see the complementary index coding rate again in Section 11.4 when we discuss the performance limit of distributed storage networks.

9.A Proof of Theorem 9.1

Let $U \in \mathcal{G}$ and k be a positive integer to be specified later. We consider two cases. First, if $\chi(\overline{U}) < 2n/k$, then trivially

$$\frac{\chi(\overline{U})}{\beta(U)} < \frac{2n}{k}.$$

Second, for the other case of $\chi(\overline{U}) \geq 2n/k$, by Lemma 9.4 with $i = k$ and $m = \lfloor n/k \rfloor$,

$$\begin{aligned}\left(\frac{n}{ck^{a+1}}\right)^{1/b} &= \max\{j\colon ck^a j^b \leq n/k\}\\ &\leq \max\{j\colon R_{\mathcal{G}}(k,j) \leq n/k\}\\ &\leq \alpha(U)\\ &\leq \beta(U)\\ &\leq \chi(\overline{U})\\ &\leq n.\end{aligned}$$

Thus,

$$\frac{\chi(\overline{U})}{\beta(U)} \leq \frac{n}{\beta(U)} \leq n^{1-1/b}(ck^{a+1})^{1/b}.$$

As k increases, the upper bound on $\chi(\overline{U})/\beta(U)$ decreases in the first case, and increases in the second case. Hence, to minimize the upper bound on the multiplicative gap between $\chi(\overline{U})$ and $\beta(U)$, we choose $k = 2^{b/(a+b+1)}(n/c)^{1/(a+b+1)}$, which makes the upper bounds in both cases equal to the desired multiplicative gap. This completes the proof of the theorem.

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10

Index Coding and Network Coding

Multiple-unicast network coding [71, 70, 164, 80] is a long-lasting and central open problem in information theory. As we saw earlier in Section 1 (especially in Figure 1.4), index coding is a rather special case of multiple-unicast network coding. In this section, we take the opposite direction and associate a general multiple-unicast network coding problem with a corresponding index coding problem. This association allows a nontrivial translation of a code for the index coding problem to a code for the network coding problem, and vice versa. We elucidate this relationship and discuss to what extent it can be leveraged to solve one problem by the solution to the other.

10.1 Multiple Unicast Network Coding

An instance of the multiple-unicast network coding problem is defined by a directed acyclic graph $G = (V, E)$, a set of k distinct source nodes $S = \{s_1, \ldots, s_k\} \subseteq V$, and a set of k distinct destination nodes $D = \{d_1, \ldots, d_k\} \subseteq V$. We assume without loss of generality that each source s_i has no incoming edges, that each destination d_i has no outgoing edges, and that there exists a directed path from s_i to d_i

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for each $i \in [k]$. Destination d_i wishes to receive message x_i that is available at source s_i, $i \in [k]$. Each source node s_i encodes the message x_i into sequences carried by its outgoing edges and each intermediate node encodes the sequences carried by its incoming edges into those carried by its outgoing edges.

For any node $v \in V$, let $\mathrm{In}(v) = \{(u, v) \in E\}$ be the set of its incoming edges. Overloading this notation, for an edge e that is outgoing from a source $s \in S$ let $\mathrm{In}(e) = \{s\}$, and for any other edge $e = (u, v)$ let $\mathrm{In}(e) = \{(t, u) \in E\}$. As an example, for the graph in Figure 10.1(a), $\mathrm{In}(d_2) = \{e_1, e_6\}$, $\mathrm{In}(e_2) = \{s_1\}$, and $\mathrm{In}(e_5) = \{e_2, e_3\}$. Under this notation, a $((t_i, i \in [k]), (t_e, e \in E))$ *network code* can be formally defined by

- k messages, where the i-th message x_i takes values in $\{0,1\}^{t_i}$,
- $|E|$ encoders, where the encoder $f_e : \Pi_{a \in \mathrm{In}(e)} \{0,1\}^{t_a} \to \{0,1\}^{t_e}$ at edge $e \in E$ maps the sequences carried by the inputs to edge e to the sequence carried by edge e, and
- k decoders, where the decoder $g_i : \Pi_{e \in \mathrm{In}(d_i)} \{0,1\}^{t_e} \to \{0,1\}^{t_i}$ at destination $d_i \in D$ maps the sequences carried by the inputs to destination d_i back to x_i.

Due to the acyclic structure of the network graph G, the sequence carried by edge e can be written as a function of the message tuple $\mathbf{x}_S = (x_1, \ldots, x_k)$ alone by recursively applying the *local encoding functions* f_e, f_a, $a \in \mathrm{In}(e)$, and so on. We refer to the resulting function $f'_e(\mathbf{x}_S)$ as the *global encoding function* at edge e.

A tuple $(\mathbf{R}, \mathbf{C}) = ((R_i, i \in [k]), (C_e, e \in E))$ of nonnegative real numbers is said to be an *achievable rate–capacity tuple* if there exist a $((t_i, i \in [k]), (t_e, e \in E))$ network code and an integer r such that

$$R_i \leq \frac{t_i}{r}, \quad i \in [k],$$

and

$$C_e \geq \frac{t_e}{r}, \quad e \in E.$$

Note that if $(\mathbf{R}, \mathbf{C})$ is achievable, then $(\alpha\mathbf{R}, \alpha\mathbf{C})$ is also achievable for every rational $\alpha > 0$. For a given $\mathbf{C} \in \mathbb{R}_{\geq 0}^{|E|}$, the *capacity region* $\mathscr{C}_{\mathrm{N}}(\mathbf{C})$ is defined as the closure of all $\mathbf{R}$ such that $(\mathbf{R}, \mathbf{C})$ is achievable.

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10.2 Relationship Between Index Coding and Network Coding

Figure 1.4 illustrated that for any index coding problem, there exists an equivalent network coding problem such that the encoding function for the network coding problem on the bottleneck edge on the top defines the encoding problem for the index coding problem. In particular, the capacity region of the index coding problem is identical to the capacity region of the network coding problem for $C_e = 1$, $e \in E$.

In this section, we show that for any network coding problem, there exists a corresponding index coding problem such that a code for one yields a code for the other. This relationship was established by El-Rouayheb, Sprintson, and Georghiades for the case of linear codes [62] and by Effros, El-Rouayheb, and Langberg for the general case [60]. Here we provide a slightly refined account of this relationship.

Let N be an instance of the network coding problem on a directed acyclic graph $G = (V, E)$ with k source nodes $s_1, \ldots, s_k$, k destination nodes $d_1, \ldots, d_k$, and k messages $x_1, \ldots, x_k$. Given N, we define an instance $I(N)$ of the index coding problem corresponding to this network coding instance by a set of $k + |E|$ messages $(\mathbf{x}_S, \mathbf{x}_E) = ((x_i, i \in [k]), (x_e, e \in E))$, and sets of side information $A_i = \text{In}(d_i)$, $i \in [k]$, and $A_e = \text{In}(e)$, $e \in E$. For example, the butterfly network, a famous multiple-unicast network coding instance, and the side information graph of its corresponding index coding instance are depicted in Figure 10.1. The butterfly network has two source nodes s_1 and s_2, and seven edges. Thus, the corresponding index coding instance has nine messages x_1, x_2, and $x_{e_i}, i \in [7]$.

A network coding instance N and the corresponding index coding instance $I(N)$ have the following relationship.

Theorem 10.1. Let N be a k-unicast network coding instance over the graph $G = (V, E)$ and let $I(N)$ be its corresponding index coding instance with $k + |E|$ messages. Then a $((t_i : i \in [k]), (t_e : e \in E))$ network code exists for N iff a $((t_i : i \in [k]), (t_e : e \in E), \sum_{e \in E} t_e)$ index code exists for $I(N)$.

To prove the necessity (only if), let f_e and f'_e be the local and global encoding functions of the $((t_i : i \in [k]), (t_e : e \in E))$ network code for

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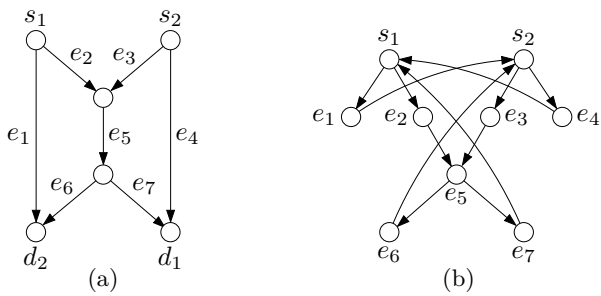


Figure 10.1: (a) The butterfly network. (b) The side information graph of the corresponding index coding instance.

edge $e \in E$, respectively, and g_i be the decoding function of destination $d_i \in D$. Let $|E| = m$ and $\mathbf{x}_E = (x_{e_1}, \ldots, x_{e_m})$. We construct an index code for the index coding instance $I(N)$ with encoding function

$$\phi(\mathbf{x}_S, \mathbf{x}_E) = (x_{e_1} + f'_{e_1}(\mathbf{x}_S), \ldots, x_{e_m} + f'_{e_m}(\mathbf{x}_S)) \in \{0,1\}^{\sum_{e\in E} t_e}. \tag{10.1}$$

We now argue that every receiver can recover its desired message using $y = \phi(\mathbf{x}_S, \mathbf{x}_E)$ and its side information. The receiver for message x_e, $e \in E$, has $(x_a : a \in \text{In}(e))$ as side information and can perform the following steps to recover x_e: (1) For each $a \in \text{In}(e)$, use $x_a + f'_a(\mathbf{x}_S)$ to compute $f'_a(\mathbf{x}_S)$; (2) compute $f'_e(\mathbf{x}_S) = f_e(f'_a(\mathbf{x}_S) : a \in \text{In}(e))$; and (3) use $x_e + f'_e(\mathbf{x}_S)$ to compute x_e. The receiver for $x_i, i \in [k]$, has $(x_e : e \in \text{In}(d_i))$ and can perform the following steps: (1) For each $e \in \text{In}(d_i)$, use $x_e + f'_e(\mathbf{x}_S)$ to compute $f'_e(\mathbf{x}_S)$; and (2) compute the decoding function $g_i(f'_e(\mathbf{x}_S) : e \in \text{In}(d_i))$, which is equal to x_i. Therefore, the encoder in (10.1) and aforementioned decoders define a $((t_i : i \in [k]), (t_e : e \in E), \sum_{e\in E} t_e)$ index code for $I(N)$.

To prove the sufficiency (if), let ϕ and ψ_i, $i \in [k]$, ψ_e, $e \in E$, be the encoding and decoding functions of a $((t_i : i \in [k]), (t_e : e \in E), \sum_{e\in E} t_e)$ index code for $I(N)$, respectively. Let $r = \sum_{e\in E} t_e$ and partition the set of message tuples by $P_y = \{(\mathbf{x}_S, \mathbf{x}_E) : \phi(\mathbf{x}_S, \mathbf{x}_E) = y\}$, $y \in \{0,1\}^r$. Before we proceed with the proof we present two lemmas that are proved in Appendices 10.A and 10.B.

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Lemma 10.1. For any $\mathbf{x}_S \in \Pi_{i\in[k]}\{0,1\}^{t_i}$, if $\mathbf{x}_E \neq \mathbf{x}'_E$, then

$$\phi(\mathbf{x}_S, \mathbf{x}_E) \neq \phi(\mathbf{x}_S, \mathbf{x}'_E). \tag{10.2}$$

Lemma 10.2. For every $y \in \{0,1\}^r$, $|P_y| = 2^{\sum_{i=1}^k t_i}$.

Thus, in a partition P_y, each value of $\mathbf{x}_S$ appears exactly once with a unique value for $\mathbf{x}_E$. This uniqueness allows us to define encoding functions for the network coding instance N as follows.

Take *any* fixed y. For each $\mathbf{x}_S$, let $\mathbf{x}_E(\mathbf{x}_S)$ be the unique vector $\mathbf{x}_E$ such that $(\mathbf{x}_S, \mathbf{x}_E) \in P_y$, and let $x_e(\mathbf{x}_S)$ be the component of $\mathbf{x}_E(\mathbf{x}_S)$ that corresponds to edge e. Since the index code is valid,

$$\psi_e(\phi(\mathbf{x}_S, \mathbf{x}_E), (x_a, a \in \mathrm{In}(e))) = x_e, \quad e \in E,$$

and

$$\psi_i(\phi(\mathbf{x}_S, \mathbf{x}_E), (x_a, a \in \mathrm{In}(d_i))) = x_i, \quad i \in [k].$$

In particular, plugging in $\mathbf{x}_E = \mathbf{x}_E(\mathbf{x}_S)$,

$$\psi_e(y, (x_a(\mathbf{x}_S), a \in \mathrm{In}(e))) = x_e(\mathbf{x}_S), \quad e \in E, \tag{10.3a}$$

and

$$\psi_i(y, (x_a(\mathbf{x}_S), a \in \mathrm{In}(d_i))) = x_i, \quad i \in [k]. \tag{10.3b}$$

Now we define the local encoding functions for the network coding instance N by

$$f_e(x_a, a \in \mathrm{In}(e)) = \psi_e(y, (x_a, a \in \mathrm{In}(e))), \quad e \in E. \tag{10.4}$$

Then, by (10.3a) and induction, the global encoding function at edge e satisfies

$$f'_e(\mathbf{x}_S) = x_e(\mathbf{x}_S), \tag{10.5}$$

since the global encoding functions at the edges $a \in \mathrm{In}(e)$ satisfy $f'_a(\mathbf{x}_S) = x_a(\mathbf{x}_S)$. We also define the k decoding functions at the destinations as

$$g_i(x_a, a \in \mathrm{In}(d_i)) = \psi_i(y, (x_a, a \in \mathrm{In}(d_i))), \quad i \in [k]. \tag{10.6}$$

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Then, by (10.3b) and (10.5), for the given message tuple $\mathbf{x}_S$, the decoding outputs at the destinations satisfy

$$\psi_i(y, (x_a(\mathbf{x}_S), a \in \mathrm{In}(d_i))) = x_i, \quad i = [k].$$

Therefore, the encoding functions in (10.4) and the decoding functions in (10.6) define a valid $((t_i : i \in [k]), (t_e : e \in E))$ network code. This completes the proof of Theorem 10.1.

Theorem 10.1 implies the following relationship between the achievable rate tuples for a network coding instance and its corresponding index coding instance.

Corollary 10.1. Let N be a k-unicast network coding instance over the graph $G = (V, E)$ and let $I(N)$ be its corresponding index coding instance with $k+|E|$ messages. Suppose that $\mathbf{C} = (C_e : e \in E)$ is rational. Then, $(\mathbf{R}, \mathbf{C})$ is achievable for N iff $(\mathbf{R}/C_E, \mathbf{C}/C_E)$ is achievable for $I(N)$, where $C_E = \sum_{e \in E} C_e$.

To prove the necessity, suppose that $(\mathbf{R}, \mathbf{C})$ is achievable for N. Then, there exists a $((t_i : i \in [k]), (t_e : e \in E))$ network code and an integer r such that

$$R_i \leq \frac{t_i}{r}, \quad i \in [k], \tag{10.7a}$$

and

$$C_e \geq \frac{t_e}{r}, \quad e \in E. \tag{10.7b}$$

Since $\mathbf{C}$ is rational, we have

$$C_e = \frac{a_e}{b_e}, \quad e \in E,$$

where $a_e, b_e \in \mathbb{N}$, $e \in E$. Let r' be divisible by r as well as b_e, $e \in E$. Define $t'_i = (r'/r)t_i$, $i \in [k]$, and $t'_e = r'C_e \geq (r'/r)t_e$, $e \in E$. Then, by repeating the network code satisfying (10.7a) and (10.7b) (r'/r) times and by appropriate zero padding (to be exact, adding $r'C_e - (r'/r)t_e$ zeros for x'_e, $e \in E$), we can construct a $((t'_i : i \in [k]), (t'_e : e \in E))$ network code such that

$$R_i \leq \frac{t'_i}{r'}, \quad i \in [k],$$

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and

$$C_e = \frac{t'_e}{r'}, \quad e \in E.$$

By Theorem 10.1, there exists a $((t'_i : i \in [k]), (t'_e : e \in E), \sum_{e\in E} t'_e)$ index code for $I(N)$, the rates of which are

$$R'_i = \frac{t'_i}{\sum_{e\in E} t'_e} \geq \frac{R_i}{C_E}, \quad i \in [k],$$

and

$$R'_e = \frac{t'_e}{\sum_{e\in E} t'_e} = \frac{C_e}{C_E}, \quad e \in E.$$

Thus, the rate tuple $(\mathbf{R}/C_E, \mathbf{C}/C_E)$ is also achievable for $I(N)$.

To prove the sufficiency, suppose that the rate tuple $(\mathbf{R}/C_E, \mathbf{C}/C_E)$ is achievable for the index coding instance $I(N)$. Then, there exists a $((t_i : i \in [k]), (t_e : e \in E), r)$ index code such that

$$\frac{R_i}{C_E} \leq \frac{t_i}{r}, \quad i \in [k], \tag{10.8a}$$

and

$$\frac{C_e}{C_E} \leq \frac{t_e}{r}, \quad e \in E. \tag{10.8b}$$

On the one hand, by summing over $e \in E$, (10.8b) implies that

$$r \leq \sum_{e\in E} t_e. \tag{10.9}$$

On the other hand, by Lemma 10.1, for any $\mathbf{x}_S$ the function $\phi(\mathbf{x}_S, \mathbf{x}_E)$ is one-to-one, which implies that

$$2^{\sum_{e\in E} t_e} \leq 2^r. \tag{10.10}$$

Combining (10.9) and (10.10), we have

$$\sum_{e\in E} t_e = r. \tag{10.11}$$

Thus, by Theorem 10.1, there exists a $((t_i : i \in [k]), (t_e : e \in E))$ network code for N. Furthermore, by (10.8b) and (10.11),

$$\frac{C_e}{C_E} = \frac{t_e}{r}, \quad e \in E.$$

Hence, $(\mathbf{R}/C_E, \mathbf{C}/C_E)$, as well as $(\mathbf{R}, \mathbf{C})$, is achievable for the network coding instance N.

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Remark 10.1. The relationships between a network coding instance N and its corresponding index coding instance $I(N)$ in Theorem 10.1 and Corollary 10.1 also hold for the cases with vanishing probability of error [62, 60].

Sometimes the exact code-level relationship between N and $I(N)$ presented in Theorem 10.1 is referred to, perhaps for the sake of brevity, as the "equivalence" between the two classes of the problems. It should be noted, however, that in the network coding problem N the goal is to achieve a larger message rate tuple $\mathbf{R}$ for a smaller link capacity tuple $\mathbf{C}$, but that in the corresponding index coding problem $I(N)$ the goal is simply to achieve a larger message rate tuple $(\mathbf{R}_S, \mathbf{R}_E)$. These incompatible goals are tangent to each other only under the special condition $\mathbf{R}_E = \mathbf{C}/C_E$, which can be made valid if $\mathbf{C}$ is rational.

We illustrate the resulting relationship between the rate regions in Corollary 10.1 in Figure 10.2, when the rates are projected to the two dimensions (R_S, C_E) or (R_S, R_E), where $R_S = \sum_{i=1}^{k} R_i$. The rate–capacity region $\mathscr{R}_N$ (namely, the closure of all achievable $(\mathbf{R}_S, \mathbf{C})$) of the network coding instance is a convex cone (or triangle in two dimensions), while the capacity region $\mathscr{C}_I$ (namely, the closure of all achievable $(\mathbf{R}_S, \mathbf{R}_E)$) of the index coding instance is a bounded convex set. Corollary 10.1 implies that at $R_E = C_E = 1$, both regions contain the same set of *achievable* rates R_S (i.e., the same interior before taking the closure operations).

The rate-level relationship in Corollary 10.1, combined with the fact that $\mathrm{cl}(\mathrm{Proj}(A)) \subseteq \mathrm{Proj}(\mathrm{cl}(A))$ for any set A, implies the following.

Corollary 10.2. Let N be a k-unicast network coding instance over the graph $G = (V, E)$ and let $I(N)$ be its corresponding index coding instance with $k + |E|$ messages. For a given (rational) link capacity tuple $\mathbf{C}$, the capacity region $\mathscr{C}_N(\mathbf{C})$ of the network coding instance N satisfies

$$\begin{aligned} \mathscr{C}_N(\mathbf{C}) &= \mathrm{cl}(\{\mathbf{R}\colon (\mathbf{R}, \mathbf{C}) \text{ is achievable for } N\}) \\ &\subseteq \{\mathbf{R}\colon (\mathbf{R}/C_E, \mathbf{C}/C_E) \in \mathscr{C}_I\}, \end{aligned} \tag{10.12}$$

where $\mathscr{C}_I$ is the capacity region of the index coding instance $I(N)$.

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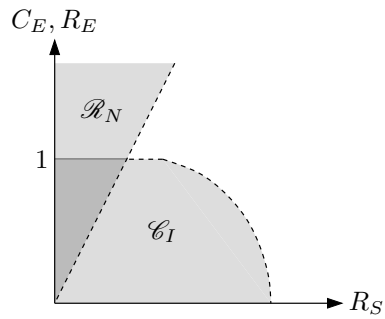


Figure 10.2: The two-dimensional projections of the achievable rate–capacity region $\mathscr{R}_N$ of the network coding instance N and the capacity region $\mathscr{C}_I$ of the corresponding index coding instance $I(N)$. The interiors of the regions are achievable. On the boundary, the rates are marked by solid lines/curves if achievable and by dashed lines otherwise. At $C_E = R_E = 1$, the sets of achievable R_S coincide for both regions.

Thus, the index coding capacity region can be projected to yield an outer bound on the network coding capacity region. It is not known if the inclusion holds in the other direction as well.

Open Problem 10.1. Is the inclusion in (10.12) strict in general?

10.A Proof of Lemma 10.1

Suppose that the edges in E are labeled in the order such that $\mathrm{In}(e_j) \subseteq \{e_1, \ldots, e_{j-1}\}$. Note that such a labeling is possible since G is acyclic. Suppose that $\mathbf{x}_E = (x_{e_1}, \ldots, x_{e_m}) \neq (x'_{e_1}, \ldots, x'_{e_m}) = \mathbf{x}'_E$. Then we have one of the following two cases.

Case 1. If e_j is outgoing from some source node $s \in S$ and $x_{e_j} \neq x'_{e_j}$, then we must have $\phi(\mathbf{x}_S, \mathbf{x}_E) \neq \phi(\mathbf{x}_S, \mathbf{x}'_E)$, since otherwise the receiver for message x_{e_j} cannot distinguish its desired message x_{e_j} from x'_{e_j} based on $\phi(\mathbf{x}_S, \mathbf{x}_E)$ and their common side information x_s.

Case 2. Otherwise, let $j \in [m]$ be the first coordinate at which $\mathbf{x}_E$ and $\mathbf{x}'_E$ differ, i.e., $(x_{e_1}, \ldots, x_{e_{j-1}}) = (x'_{e_1}, \ldots, x'_{e_{j-1}})$ and $x_j \neq x'_j$. Since $\mathrm{In}(e_j) \subseteq \{e_1, \ldots, e_{j-1}\}$ by assumption, once again we must have $\phi(\mathbf{x}_S, \mathbf{x}_E) \neq \phi(\mathbf{x}_S, \mathbf{x}'_E)$.

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10.B Proof of Lemma 10.2

Lemma 10.1 implies that there exist at most $2^{\sum_{i=1}^{k} t_i}$ message tuples in each P_y, i.e.,

$$|P_y| \leq 2^{\sum_{i=1}^{k} t_i}, \quad y \in [2^r]. \tag{10.13}$$

But combining all $(\mathbf{x}_S, \mathbf{x}_E)$ tuples over all $y \in [2^r]$, we must have

$$\sum_{y \in [2^r]} |P_y| = |\{(\mathbf{x}_S, \mathbf{x}_E)\}| = (2^{\sum_{i=1}^{k} t_i})(2^{\sum_{e \in E} t_e}) = (2^{\sum_{i=1}^{k} t_i})2^r,$$

which implies that (10.13) must hold with equality, i.e.,

$$|P_y| = 2^{\sum_{i=1}^{k} t_i}, \quad y \in [2^r].$$

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11

Index Coding, Distributed Storage, and Guessing Games

As the amount of data continues to expand, it is becoming increasingly important to store them reliably in a distributed storage system of interconnected servers. The index coding problem is closely related to the locally recoverable distributed storage problem, which studies fundamental limits and coding schemes for such distributed storage networks. In an unexpected twist, the locally recoverable distributed storage problem turns out to be equivalent to the recreational mathematics problem of guessing games on directed graphs. In this section, we introduce the locally recoverable distributed storage problem and the problem of guessing games on graphs, and develop the relationship between these two problems and the index coding problem.

11.1 Locally Recoverable Distributed Storage Problem

In the locally recoverable distributed storage problem, which hereafter will be referred to as the distributed storage problem, a set of servers collectively store data so that if a single server fails, its contents can be efficiently reconstructed from the contents of the other servers (see, for example, [131, 123, 36]). The goal is to design a distributed storage

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code that maximizes the amount of data that can be stored while satisfying the single-failure recovery constraint. Mazumdar [111] considered a distributed storage system in which each server has only access to a subset of the other servers and model the topology of the system by a directed graph. The same model was also considered independently and concurrently by Shanmugam and Dimakis [132].

Suppose that some data y is stored across n servers in the system and that it can be recovered exactly by accessing all servers. Let x_i denote the content of server $i \in [n]$. Each server has access to the contents of a subset of other servers, $x(A_i)$, $A_i \subseteq [n] \setminus \{i\}$. The set A_i is referred to as the *recovery set* of server i. The goal is to find the maximum amount of data that can be stored in the network so that if any single server fails, its content can be still recovered from the contents of its recovery set.

As before, we can formulate the problem more precisely by defining a $(t_1, \ldots, t_n, r)$ *distributed storage code* that consists of

- a message set $[2^r]$,
- a *one-to-one* encoding function $x^n : [2^r] \to \prod_{i=1}^{n} \{0,1\}^{t_i}$ that assigns a distinct codeword $x^n(y)$ to each $y \in [2^r]$, (the set $\mathcal{C} = \{x^n(1), \ldots, x^n(2^r)\}$ is referred to as the codebook), and
- n recovery functions, where the recovery function at server $i \in [n]$, $f_i : \prod_{j \in A_i} \{0,1\}^{t_j} \to \{0,1\}^{t_i}$ maps the contents $x(A_i)$ of the recovery set to x_i.

Thus, for every $x^n \in \mathcal{C}$,

$$f_i(x(A_i)) = x_i, \quad i \in [n].$$

A rate tuple $(R_1, \ldots, R_n)$ is said to be *achievable* for the distributed storage problem if there exists a $(t_1, \ldots, t_n, r)$ distributed storage code such that

$$R_i \geq \frac{t_i}{r}, \quad i \in [n].$$

The *optimal rate region* $\mathscr{R}$ of the distributed storage problem is defined as the closure of the set of achievable rate tuples. For any nonnegative

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real tuple $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_n)$, the *optimal* $\boldsymbol{\mu}$*-directed rate* $R(\boldsymbol{\mu})$ of the distributed storage problem is defined as

$$R(\boldsymbol{\mu}) = \min\{R\colon R\boldsymbol{\mu} \in \mathscr{R}\}. \tag{11.1}$$

The optimal $\mathbf{1}$-directed rate of the distributed storage problem is referred to as the *optimal symmetric rate*

$$R_{\mathrm{sym}} = R(\mathbf{1}) = \min\{R\colon (R, \ldots, R) \in \mathscr{R}\}.$$

The reciprocal of the optimal symmetric rate is sometimes referred to as the *storage capacity* σ [111].

Remark 11.1. The optimal rate region can be written in terms of optimal $\boldsymbol{\mu}$-directed rates (cf. Remark 1.2) as

$$\mathscr{R} = \bigcup_{\boldsymbol{\mu}} \{(R_1, \ldots, R_n)\colon R_i \geq R(\boldsymbol{\mu})\mu_i,\ i \in [n]\}. \tag{11.2}$$

Note that if $\boldsymbol{\mu} = c\boldsymbol{\mu}'$ for some constant c, then $R(\boldsymbol{\mu})\boldsymbol{\mu} = R(\boldsymbol{\mu}')\boldsymbol{\mu}'$. Thus, it suffices to take the union in (11.2) only over normalized vectors, e.g., over $\boldsymbol{\mu}$ such that $\sum_{i=1}^{n} \mu_i = n$.

For any nonnegative real vector $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_n)$, the *optimal* $\boldsymbol{\nu}$*-weighted sum-rate* $\overline{R}(\boldsymbol{\nu})$ of the distributed storage problem is defined as

$$\overline{R}(\boldsymbol{\nu}) = \min\left\{\boldsymbol{\nu}^T\mathbf{R} = \sum_{i=1}^{n} \nu_i R_i\colon (R_1, \ldots, R_n) \in \mathscr{R}\right\}.$$

The optimal $\mathbf{1}$-weighted sum-rate $\overline{R}(\mathbf{1})$ is referred to simply as the *optimal sum-rate*

$$R_{\mathrm{sum}} = \overline{R}(\mathbf{1}) = \min\{\mathbf{1}^T\mathbf{R}\colon (R_1, \ldots, R_n) \in \mathscr{R}\}.$$

Remark 11.2. The optimal rate region can be also written in terms of optimal $\boldsymbol{\nu}$-weighted rates as

$$\mathscr{R} = \bigcup_{\boldsymbol{\nu}} \{\mathbf{R} = (R_1, \ldots, R_n)\colon \boldsymbol{\nu}^T\mathbf{R} \geq \overline{R}(\boldsymbol{\nu})\}. \tag{11.3}$$

Since $\overline{R}(c\boldsymbol{\nu}) = c\overline{R}(\boldsymbol{\nu})$, it suffices to take the union in (11.3) only over $\boldsymbol{\nu}$ such that $\sum_{i=1}^{n} \nu_i = n$.

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Any instance of the distributed storage problem is fully represented by a directed graph $G = (V, E)$ referred to as the *storage recovery graph* in which each vertex represents a server and there exists a directed edge $j \to i$ iff server j is in the recovery set of server i, i.e, $j \in A_i$. We identify an instance of the distributed storage problem with its storage recovery graph G and often write "distributed storage problem G" and use notations like $\mathscr{R}(G), R_{\text{sym}}(G)$, and $R_{\text{sum}}(G)$.

Given a storage recovery graph G and a tuple $\mathbf{t} = (t_1, \ldots, t_n)$ of positive integers, consider the confusion graph $\Gamma = \Gamma_{\mathbf{t}}(G)$ as defined in Section 3.1, with vertices representing the potential contents of the servers. Suppose that two vertices $\mathbf{x}$ and $\mathbf{z}$ are not confusable. Then the storage network can store a single bit of information, represented by $x^n(0) = \mathbf{x}$ and $x^n(1) = \mathbf{z}$, since the content of each server (say, x_i or z_i) can be determined by the contents of its recovery set. To store more information, we consider the maximal independent set of the confusion graph Γ. Since no two content tuples among these $\alpha(\Gamma)$ tuples are confusable, each can represent a distinct information satisfying the single-server failure recovery condition. Therefore, we can store $r = \lfloor \log \alpha(\Gamma) \rfloor$ bits of information in the distributed storage network. This proves the existence of a $(t_1, \ldots, t_n, \lfloor \log \alpha(\Gamma_{\mathbf{t}}(G)) \rfloor)$ distributed storage code.

Conversely, consider any $(\mathbf{t}, r)$ distributed storage code, which has at least 2^r distinct n-tuples that satisfy the required function relationship. By definition, these n-tuples form an independent set of the confusion graph $\Gamma = \Gamma_{\mathbf{t}}(G)$. Thus, $\alpha(\Gamma) \geq 2^r$, or equivalently, $r \leq \lfloor \log \alpha(\Gamma) \rfloor$. Therefore, any achievable rate tuple $(R_1, \ldots, R_n)$ must satisfy

$$R_i \geq \frac{t_i}{\lfloor \log \alpha(\Gamma_{\mathbf{t}}(G)) \rfloor}, \quad i \in [n],$$

for some $\mathbf{t} = (t_1, \ldots, t_n)$. This establishes the following.

Proposition 11.1 ([111, 13])**.** The optimal rate region $\mathscr{R}$ of the distributed storage problem G with n servers is the closure of all rate tuples $(R_1, \ldots, R_n)$ such that

$$R_i \geq \frac{t_i}{\log \alpha(\Gamma_{\mathbf{t}}(G))}, \quad i \in [n],$$

for some $\mathbf{t} = (t_1, \ldots, t_n)$.

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11.2 Guessing Games on Directed Graphs

There are many popular "hat problems" in recreational mathematics. Here is one. There are n players sitting on a circle, each wearing a hat. The colors of the hats are randomly chosen to be red or blue. Each player can observe the hats of all other players except her own. The players are to make respective guesses of their own hat colors simultaneously without any communication. As a team, they win this game if all n guesses are correct. Once the game starts, they are given no information beside the colors of the other players, but they can premeditate a strategy to maximize the probability of winning. What is the maximum winning probability? What is the optimal strategy?

Instead of revealing the answers, we move directly to a more general version of this riddle, which is referred to as a *guessing game* on a directed graph and was introduced by Riis [125]. Given a directed graph $G = (V, E)$ with $V = [n]$, associate player $i \in [n]$ to vertex i and randomly assign a value $x_i \in \{0, 1\}^{t_i}$ independently from the other players. Suppose that player i can observe $x(A_i)$ assigned to the players in her neighbor set $A_i \subseteq [n] \setminus \{i\}$. The players are to make respective guesses of their own values simultaneously and win the game if all of them are correct. No communication is allowed between the players, but they can agree on a strategy beforehand. The goal is to find the maximum winning probability and the strategy that achieves this maximum. Note that the setting presented here is slightly more general than Riss's in that the range of the values assigned to the players can be different.

As an example, consider the guessing game on a complete graph with n vertices and $t_i = 1$, $i \in [n]$. Then the problem is equivalent to the earlier hat problem. If everybody makes an independent guess on their own, then the team will win with probability $1/2^n$. Surprisingly, they can do much better. Suppose that each player guesses her own value assuming that the parity of all the values is even. Since every player can observe the values of all other players, the players win iff their assumption is correct, which happens with probability 1/2! This strategy makes a significant improvement over the random guessing strategy and is optimal, as the probability that a single player guesses her value correctly is also 1/2.

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Now we formalize the problem with the following definition. A $(t_1, \ldots, t_n, W)$ *guessing strategy* consists of

- n *guessing functions*, where the guessing function of player $i \in [n]$, $h_i : \prod_{j \in A_i} \{0,1\}^{t_j} \to \{0,1\}^{t_i}$, maps the values of the neighbors $x(A_i)$ to x_i, and
- a *winning set* W of n-tuples that can be guessed correctly using these functions, i.e.,

$$W := \bigcap_{i=1}^{n} \Big\{ x^n \in \prod_{j=1}^{n} \{0,1\}^{t_j} : h_i(x(A_i)) = x_i \Big\}.$$

For the example of the complete graph with n vertices and $t_i = 1$, $i \in [n]$, the optimal guessing strategy corresponds to

$$h_i(x(A_i)) = \sum_{j \in A_i} x_j, \quad i \in [n],$$

and

$$W = \Big\{ \mathbf{x} : \sum_{i=1}^{n} x_i = 0 \Big\},$$

where the summations are modulo 2.

Remark 11.3. This definition of guessing strategy does not include any random guessing strategy. It can be easily verified, however, that randomization does not help, and we thus limit our attention to deterministic guessing strategies as in our definition.

Remark 11.4. For any pair of distinct tuples $\mathbf{x}, \mathbf{z}$ in a winning set W of a general guessing strategy, either $x_i = z_i$ or $x(A_i) \neq z(A_i)$ for every $i \in [n]$. Conversely, any set W satisfying this property is a valid winning set for some guessing functions defined appropriately to satisfy $h_i(x(A_i)) = x_i$, $i \in [n]$, for every $\mathbf{x} \in W$.

When the players adopt a $(\mathbf{t}, W)$ strategy, everyone guesses her value correctly and the team win with the probability

$$P_{\text{win}} = \frac{|W|}{\prod_{i=1}^{n} 2^{t_i}}.$$

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The performance of a given $(\mathbf{t}, W)$ guessing strategy can be measured by the notion of *guessing number*

$$k(G, \mathbf{t}) = \log_s \frac{P_{\text{win}}}{P_{\text{rand}}} = \frac{n \log |W|}{\sum_{i=1}^{n} t_i} \in [0, n], \tag{11.4}$$

where $s = 2^{(1/n)\sum_{i=1}^{n} t_i}$ and $P_{\text{rand}} = \prod_{i=1}^{n} (1/2)^{t_i}$ is the winning probability of random guesses (corresponding to $|W| = 1$ after derandomization). For the case in which $t_i = t, i \in [n]$, we write $k(G, t, \ldots, t)$ as $k(G, t) = (1/t) \log |W|$. The *optimal guessing number* $k(G)$ of a directed graph G is defined as

$$k(G) = \sup_{\mathbf{t}} \sup k(G, \mathbf{t}), \tag{11.5}$$

where the second supremum is over all $(\mathbf{t}, W)$ guessing strategies.

The performance of a strategy can be measured alternatively by the *complementary guessing number*

$$k'(G, \mathbf{t}) = \log_s \frac{1}{P_{\text{win}}} = \frac{n(\sum_{i=1}^{n} t_i - \log |W|)}{\sum_{i=1}^{n} t_i} \in [0, n]. \tag{11.6}$$

The *optimal complementary guessing number* is defined similarly as

$$k'(G) = \inf_{\mathbf{t}} \inf k'(G, \mathbf{t}), \tag{11.7}$$

where the second infimum is over all $(\mathbf{t}, W)$ guessing strategies.

Remark 11.5. For any $(\mathbf{t}, W)$ guessing strategy, we have $k'(G, \mathbf{t}) = n - k(G, \mathbf{t})$, and thus

$$k'(G) = n - k(G). \tag{11.8}$$

As in index coding and distributed storage problems, the confusion graph $\Gamma = \Gamma_{\mathbf{t}}(G)$ defined in Section 3.1 for a given directed graph G and an integer tuple $\mathbf{t}$ plays a pivotal role in characterizing the optimal guessing number of the guessing game on G. By Remark 11.4, any independent set of Γ defines a valid guessing strategy. Hence we can use an argument similar to the proof of Proposition 11.1 to establish the following.

Proposition 11.2. For the guessing game on the directed graph G on n vertices,

$$k(G) = \sup_{\mathbf{t}} \frac{n \log \alpha(\Gamma_{\mathbf{t}}(G))}{\sum_{i=1}^{n} t_i}.$$

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11.3 Equivalence of Distributed Storage and Guessing Games

We now consider the distributed storage problem and the guessing game defined on the *same* directed graph G on n vertices. For any integer tuple $\mathbf{t} = (t_1, \ldots, t_n)$, let $s(\mathbf{t}) = 2^{(1/n)\sum_{i=1}^{n} t_i}$. Then the optimal sum-rate of the distributed storage problem G satisfies

$$\begin{aligned} R_{\text{sum}}(G) &= \min_{\mathbf{R}\in\mathscr{R}} \sum_{i=1}^{n} R_i \\ &= \inf_{\mathbf{t}} \frac{\sum_{i=1}^{n} t_i}{\log \alpha(\Gamma_{\mathbf{t}}(G))} \qquad (11.9) \\ &= \frac{n}{\sup_{\mathbf{t}} \frac{\log \alpha(\Gamma_{\mathbf{t}}(G))}{\log s(\mathbf{t})}} \\ &= \frac{n}{k(G)}, \qquad (11.10) \end{aligned}$$

where (11.9) follows by Proposition 11.1, and (11.10) follows by Proposition 11.2. Hence, the optimal guessing number of the guessing game on G is inversely related to the optimal sum-rate of the distributed storage problem G.

Theorem 11.1 ([111, 13])**.** For any directed graph G on n vertices,

$$k(G) = \frac{n}{R_{\text{sum}}(G)}.$$

In fact, the guessing game is equivalent to the distributed storage problem in the following stronger sense, which can be used to prove Theorem 11.1 without involving confusion graphs.

Theorem 11.2. Given any directed graph G on n vertices, a $(\mathbf{t}, W)$ guessing strategy exists iff a $(\mathbf{t}, r)$ distributed storage code exists with $r = \lfloor \log |W| \rfloor$.

To prove this, we show that the winning set W and the codebook $\mathcal{C} = \{x^n(y) : y \in [2^r]\}$ can be used interchangeably for the two problems. First, consider a $(\mathbf{t}, W)$ guessing strategy. We can then construct a $(\mathbf{t}, r)$ distributed storage code by using the winning set of n-tuples W as the codebook $\mathcal{C}$ to store the data in $r = \lfloor \log |W| \rfloor$ bits and by using

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the guessing functions $h_i, i \in [n]$, as the recovery functions. Conversely, consider a $(\mathbf{t}, r)$ distributed storage code. We can then construct a $(\mathbf{t}, W)$ guessing strategy by setting $W = \mathcal{C}$ with $|W| = 2^r$ and by using the recovery function f_i as the guessing function h_i of player $i \in [n]$, so that every tuple in W can be guessed correctly. This completes the proof of Theorem 11.2.

11.4 Complementarity of Index Coding and Distributed Storage

Let G be a directed graph on n vertices. For any integer tuple $\mathbf{t}$, the confusion graph $\Gamma = \Gamma_{\mathbf{t}}(G)$ is vertex transitive. Hence, by Remark 2.1,

$$\log \chi_f(\Gamma) = \sum_{i=1}^{n} t_i - \log \alpha(\Gamma). \tag{11.11}$$

The left and right hand sides of (11.11) appeared in Theorem 3.1 (see Remark 3.1 as well) and Proposition 11.1, respectively, which hints at a complementary relationship between the capacity region $\mathscr{C}(G)$ of the index coding problem with side information graph G and the optimal rate region $\mathscr{R}(G)$ of the distributed storage problem with storage recovery graph G. This relationship can be made more precise by the following observation on the $\boldsymbol{\mu}$-directed rates.

Lemma 11.1 ([13]). For any directed graph G on n vertices and any $\boldsymbol{\mu} \in \mathbb{Q}^n_{\geq 0}$,

$$C(\boldsymbol{\mu}) = \sup_{r: r\boldsymbol{\mu} \in \mathbb{Z}^n_{\geq 0}} \frac{r}{\log \chi_f(\Gamma_{r\boldsymbol{\mu}}(G))}, \tag{11.12a}$$

$$R(\boldsymbol{\mu}) = \inf_{r: r\boldsymbol{\mu} \in \mathbb{Z}^n_{\geq 0}} \frac{r}{\log \alpha(\Gamma_{r\boldsymbol{\mu}}(G))}. \tag{11.12b}$$

Now, by this lemma and (11.11), for any $\boldsymbol{\mu} \in \mathbb{Q}^n_{\geq 0}$,

$$\begin{aligned} C(\boldsymbol{\mu}) &= \sup_{r: r\boldsymbol{\mu} \in \mathbb{Z}^n_{\geq 0}} \frac{r}{\log \chi_f(\Gamma_{r\boldsymbol{\mu}}(G))} \\ &= \sup_{r: r\boldsymbol{\mu} \in \mathbb{Z}^n_{\geq 0}} \frac{r}{r \sum_{i=1}^n \mu_i - \log \alpha(\Gamma_{r\boldsymbol{\mu}}(G))} \\ &= \left(\sum_{i=1}^{n} \mu_i - \frac{1}{R(\boldsymbol{\mu})} \right)^{-1}. \end{aligned}$$

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Since $\mathbb{Q}$ is dense in $\mathbb{R}$, by the continuity of the functions $C(\boldsymbol{\mu})$ and $R(\boldsymbol{\mu})$, we can establish the following complementary relationship between the $\boldsymbol{\mu}$-directed capacity $C(\boldsymbol{\mu})$ and the $\boldsymbol{\mu}$-directed optimal rate $R(\boldsymbol{\mu})$ for any nonnegative real tuple $\boldsymbol{\mu}$.

Theorem 11.3 ([13]). For any directed graph G on n vertices and any $\boldsymbol{\mu} \in \mathbb{R}^n_{\geq 0}$,

$$\frac{1}{C(\boldsymbol{\mu})} = \sum_{i=1}^{n} \mu_i - \frac{1}{R(\boldsymbol{\mu})}. \tag{11.13}$$

By Remarks 1.2 and 11.1, Theorem 11.3 establishes the complementarity between the capacity region $\mathscr{C}(G)$ of the index coding problem and the optimal rate region $\mathscr{R}(G)$ of the distributed storage problem. Given $\mathscr{C}(G)$, Theorem 11.3 completely determines $\mathscr{R}(G)$ and vice versa. This relationship implies as an important special case ($\boldsymbol{\mu} = \mathbf{1}$) the complementarity between the broadcast rate β of the index coding problem and the storage capacity σ (the reciprocal of the optimal symmetric rate) of the distributed storage problem established by Mazumdar [111], and by Shanmugam and Dimakis [132].

Corollary 11.1 ([111, 132]). For any n-message index coding problem and n-server distributed storage problem with the same side information or storage recovery graph G, we have

$$\beta(G) = n - \sigma(G). \tag{11.14}$$

Remark 11.6. Mazumdar [111] established a stronger and direct code-level complementary relationship between index coding and distributed storage problems that bypasses the identity in (11.11).

Remark 11.7. Similar to the linear broadcast rate $\lambda_{\mathbb{F}}$, we can define scalar and vector linear codes as well as the linear storage capacity $M_{\mathbb{F}}$ for the distributed storage problem. Shanmugam and Dimakis [132] established the complementarity between linear index and storage codes, which implies that

$$\lambda_{\mathbb{F}}(G) = n - M_{\mathbb{F}}(G).$$

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Remark 11.8. The complementarity in Corollary 11.1, combined with the approximation results for the complementary index coding rate $n-\beta$ discussed in Remark 9.5, shows that the storage capacity for a general n-server distributed storage problem can be approximated within $O(\log n \log\log n)$ [132]. Tighter approximation results on the storage capacity for special classes of storage recovery graphs were established in [111, 112, 1].

Equation (11.11) can also be used to relate the sum-capacity

$$C_{\text{sum}} = \max_{\mathbf{R}\in\mathscr{C}} \sum_{i=1}^{n} R_i$$

of the index coding problem with the optimal sum-rate R_{sum} of the distributed storage problem.

Proposition 11.3 ([13]). For any n-message index coding problem and n-server distributed storage problem with the same side information or storage recovery graph G, we have

$$\frac{1}{C_{\text{sum}}(G)} = 1 - \frac{1}{R_{\text{sum}}(G)}. \tag{11.15}$$

To prove Proposition 11.3, consider

$$\begin{aligned}
\frac{1}{C_{\text{sum}}(G)} &= \frac{1}{\max_{\mathbf{R}\in\mathscr{C}(G)} \sum_{i=1}^{n} R_i} \\
&= \frac{1}{\sup_{\mathbf{t}} \frac{\sum_{i=1}^{n} t_i}{\log \chi_f(\Gamma_{\mathbf{t}}(G))}} && (11.16)\\
&= \inf_{\mathbf{t}} \frac{\sum_{i=1}^{n} t_i - \log \alpha(\Gamma_{\mathbf{t}}(G))}{\sum_{i=1}^{n} t_i} && (11.17)\\
&= 1 - \frac{1}{\min_{\mathbf{R}\in\mathscr{R}(G)} \sum_{i=1}^{n} R_i} && (11.18)\\
&= 1 - \frac{1}{R_{\text{sum}}(G)},
\end{aligned}$$

where (11.16) follows by Theorem 3.1, (11.17) follows by (11.11), and (11.18) follows by Proposition 11.1.

Combining Proposition 11.3, Theorem 11.1, and Remark 11.5, we can also establish the following relationship between the optimal complementary guessing number and the index coding sum-capacity.

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Corollary 11.2. For any directed graph G on n vertices,

$$k'(G) = \frac{n}{C_{\text{sum}}(G)}.$$

In summary, the index coding problem and the locally recoverable distributed storage problem are *equivalent* in the sense that the capacity region $\mathscr{C}(G)$ of the index coding problem G completely characterizes the optimal rate region $\mathscr{R}(G)$ for the distributed storage problem G and vice versa. The optimal guessing number $k(G)$ and its complement $k'(G)$ of the guessing game on the same graph G are the projections of $\mathscr{R}(G)$ and $\mathscr{C}(G)$ in a specific direction. The relationship between the three problems is summarized in Figure 11.1.

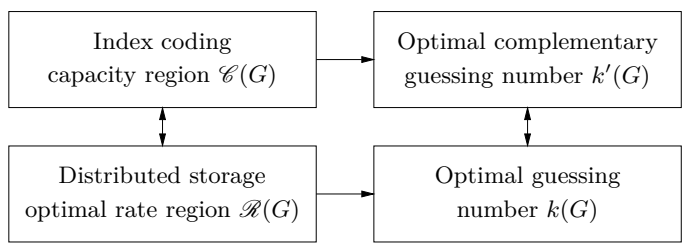


Figure 11.1: The relationships between the index coding capacity region, the distributed storage optimal rate region, the optimal guessing number, and the optimal complementary guessing number defined on the same directed graph. Here a directed arrow from problem A to problem B indicates that the solution to A implies the solution to B.

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12

Further Reading

The literature on index coding is extensive and rapidly growing; see, for example, the dissertations of El Rouayheb [64] and Blasiak [31], as well as a recent survey [34]. In this section, we provide a few pointers to the topics we have not covered thus far.

12.1 Beyond Multiple Unicast

Throughout this monograph, we considered *multiple-unicast* demands of the messages, whereby each of multiple messages is to be recovered at exactly one receiver. In the more general *multiple-groupcast* setting, there can be more than one receiver that demands the same message. As an example, consider the 3-message multiple-groupcast index coding problem $(1|2), (1|2,3), (2|3), (3|2)$ with four receivers $1a$, $1b$, 2, and 3. Both receivers $1a$ and $1b$ demand the same message x_1, but with different sets of side information x_2 and (x_2, x_3), respectively.

An instance of the multiple-groupcast index coding problem can be represented naturally by identifying each receiver with a *directed hyperedge*; for instance, $(1, \{2\})$, $(1, \{2,3\})$, $(2, \{3\})$, and $(3, \{2\})$ in the above example. This hypergraph extension of index coding was introduced

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in [3], and most performance bounds and coding schemes we discussed in Sections 5 and 6 can be readily adapted to such hypergraphs; see, for example, [29] for the generalization of the polymatroidal bound for hypergraphs. The broadcast rate of all multiple-groupcast index coding problems with up to 3 receivers was characterized in [153].

In some applications, such as content distribution networks, each receiver may be interested in receiving *any* message it does not already have as side information. In this *pliable index coding* problem [32, 142], such flexibility can reduce the amount of communication—sometimes significantly. As an example, for the 3-message pliable index coding problem $(*|2), (*|3), (*|1,2)$, the broadcast rate is 1, achieved by transmitting $x_2 + x_3$, so that x_3 is communicated to receivers 1 and 3, while x_2 is communicated to receiver 2. In comparison, the regular index coding problem $(1|2), (2|3), (3|1,2)$ that is consistent with the same side information sets requires the broadcast rate of 2. As a more trivial example, if no receiver has any side information about n messages, pliable index coding allows the transmission of x_1 alone ($\beta = 1$), while regular index coding requires the uncoded transmission of $(x_1, \ldots, x_n)$ ($\beta = n$).

Sometimes side information at a receiver may not be exactly a set of other messages, but some functions of the messages (i.e., codewords) that were overheard from past transmissions. This problem of *index coding with coded side information* was studied in [51], where both the side information at the receivers as well as the demands themselves can be linear combinations of the messages; see also [94, 33].

As yet another variation, side information may not be shared synchronously between the server and receivers. In the *blind index coding* problem [85], the server does not know which side information is available at each receiver, but knows the amount of (coded) side information. In the problem of *index coding with erroneous side information* [87], the receivers may not have accurate side information.

12.2 Noisy Broadcast

One great simplification in the index coding problem is that the broadcast link is noiseless. For some practical applications, however, this

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has to be relaxed. One of the simplest instances of such relaxation is the *complementary broadcast channel* problem [160, 162, 91] depicted in Figure 12.1. This problem generalizes the 2-message index coding problem $(1|2), (2|1)$ by replacing the noiseless broadcast link with a noisy broadcast channel $p(v_1, v_2|u)$ with input U and outputs V_1 and V_2 . As before, the messages are $X_i \in \{0,1\}^{t_i}$, $i = 1, 2$, and the rates of communication over r uses of the broadcast channel with codeword U^r and its noisy observations V_1^r and V_2^r are t_1/r and t_2/r. Thus, a (t_1, t_2, r) code for this problem is defined by the encoder $U^r = \phi(X_1, X_2)$ and two decoders $\hat{X}_1 = \psi_1(V_1^r, X_2)$ and $\hat{X}_2 = \psi_2(V_2^r, X_1)$. The definitions of achievable rate pair and capacity region are the same as in the noiseless index coding problem with vanishing average probability of error presented in Appendix 1.A. The capacity region for this problem can be characterized as the closed convex set consisting of all rate pairs (R_1, R_2) satisfying

$$R_1 \leq I(U; V_1),$$
$$R_2 \leq I(U; V_2)$$

for some $p(u)$.

The communication problem over the broadcast channel with arbitrary receiver side information, which may well be called the *noisy index coding* problem, is very difficult in general, even for a small number of messages. For example, the capacity region of the standard two-receiver broadcast channel [49] (corresponding to $(1|-), (2|-)$) is still

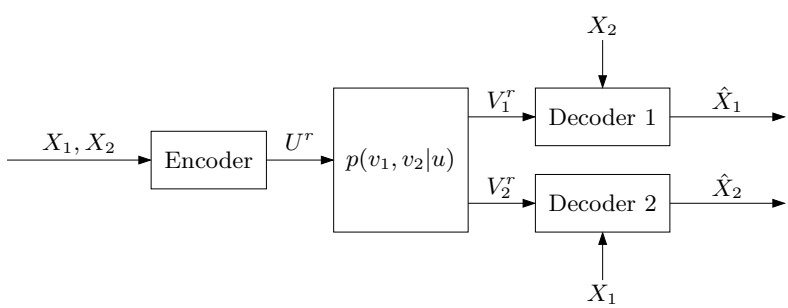


Figure 12.1: The complementary broadcast channel in which each receiver has the other message as side information.

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open. For several interesting special cases such as Gaussian broadcast channels, however, the capacity region can be characterized for a couple of receivers [168, 141, 17].

Several coding techniques have been developed for noisy index coding. Algebraic codes and their performance bounds for index coding were studied in [53]; see also [33]. Other coding techniques were developed based on modulation [115] and lattices [116, 81].

12.3 Distributed Servers

In some applications, the data is geographically distributed and stored across many locations. The *distributed index coding* problem, depicted in Figure 12.2, is an extension of the index coding problem to such communication scenarios. This problem was first studied in [119] and generalized in [127] to $2^n - 1$ servers each encoding some nonempty subset of messages $x(J)$, $J \subseteq [n]$, to its own codeword under some link capacity constraint.

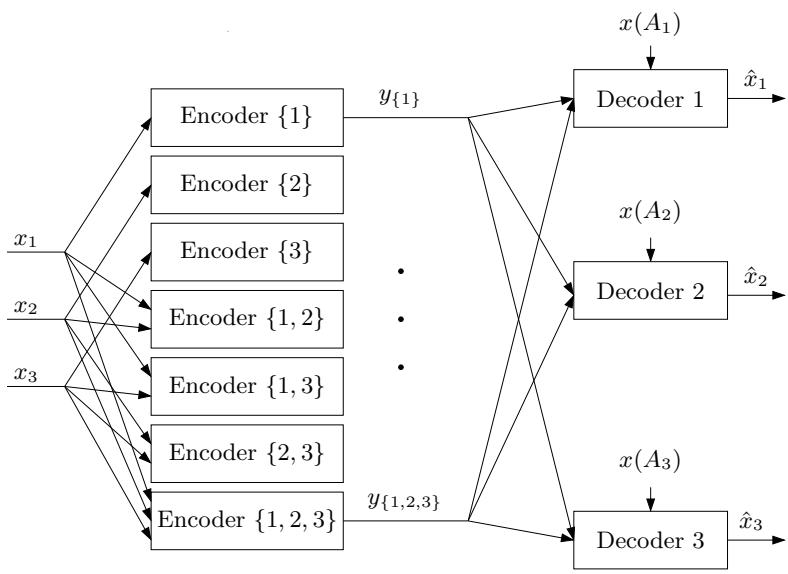


Figure 12.2: The distributed index coding problem with three messages.

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Several performance bounds and coding techniques for the centralized index coding problem we discussed in Sections 5 and 6 can be generalized to the distributed case [149, 95, 97, 98]. In particular, the (enhanced) composite coding scheme developed in Section 6.10 can be extended naturally to multiple servers by matching composite indices to the servers that can generate them [97] and by sending the composite indices cooperatively across those servers [95]; see [98] for details. When juxtaposed with a general outer bound that combines polymatroidal axioms similar to those used in the polymatroidal bound in Section 5.2 with functional dependence separation techniques [90, 148], this distributed composite coding scheme achieves the sum-capacity for all index coding problems with up to four messages under equal link capacities was established in [98].

12.4 Security and Privacy

Secure communication against eavesdroppers is an important issue in many communication networks; see, for example, the seminal paper [38] on secure network coding and its subsequent developments [37, 39, 140, 65]. Consider an index coding problem in which an eavesdropper has partial knowledge of the messages and the broadcast codeword and the server should prevent the eavesdropper from learning any information about some or all messages. This *secure index coding* problem was first studied in [52] in the context of scalar linear codes, and subsequently developed for general nonlinear codes in [121]. The connection between secure network coding and secure index coding (which parallels the relationship between nonsecure versions discussed in Section 10) was developed in [120].

One of the canonical formulations for secure index coding is the case in which the eavesdropper has full access to the broadcast codeword and some subset of messages and it must not learn any information about any single message (beyond the side information it already has). For this problem, the flat and composite coding schemes discussed in Section 6 can be readily extended to achieve reliable communication and *weak* secrecy (namely, vanishing rate of leaked information) simultaneously [101]. When compared to the outer bound that augments the

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polymatroidal bound in Section 5.2 with additional inequalities based on the secrecy constraints, the secure composite coding scheme achieves the capacity region for all problems up to four messages [101].

Another canonical formulation is the *perfectly secure index coding* problem, in which the eavesdropper has no additional side information, but it must not learn any information whatsoever about the messages from the broadcast codeword (namely, zero mutual information as in Shannon's secrecy system [136]). This *perfect secrecy* constraint requires a shared key among the server and receivers. The minimum key length was studied and characterized approximately in [113].

Sometimes there is no adversarial eavesdropper in the system, but it is desirable to reveal each message only to its designated receiver. This *private index coding* problem were studied in [114] with linear codes. Another type of privacy was studied in [86], in which the demand and side information availability of each receiver should be also kept private from other receivers. Index coding in this context is intimately related to *private information retrieval* [45, 20]; see [146] for this intriguing connection.

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